Reverse Foundations of Physics

Separating Physical Postulates, Mathematical Carriers, and Logical Strength

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Working draft, 14 August 2026

Abstract

What part of a physical theory is physics, what part is mathematics, and what part is inherited from the logic and existence principles in which the mathematics is developed? Familiar foundational programmes usually vary one of these coordinates while holding the others fixed. Reverse mathematics calibrates the axioms needed for a theorem. Reverse physics seeks physical postulates sufficient to recover a law. Operational reconstructions recover quantum structure from informational principles. Constructive, computable, choice-free, topos-theoretic, and finite approaches each replace selected pieces of the mathematical architecture. None of those methods alone answers the combined question.

We formulate a programme of *reverse foundations of physics* around the typed derivability judgement

$$L + S + M + \mathrm{Enc}(P) \ \vdash \ O.$$

Here L is the logic, S the set, type, or existence theory, M the mathematical carrier and analytic machinery, P the physical postulates after an explicit encoding, and O one sharply delimited obligation. The programme asks which ingredients are sufficient, necessary, equivalent, independent, or avoidable by reformulation. Its practical research atlas has three coarse axes—six foundational regimes, six carrier families, and sixteen physical obligations—giving 576 navigational coordinates. The cube is not an ontology and its coordinates are not presumed independent.

Five case studies show why the typing matters. An explicit mode labelling can avoid basis-selection uses of Choice without making physics "imply ZF." A Krein fundamental symmetry permits explicit normalized states but does not select a physical state. A represented scalar wave evolution is formalizable over RCA_0, while causal Green support remains a distinct theorem. Exact finite graph causality is not continuum Lorentzian causality. And a bounded finite BV contraction can be checked in PRA without lowering any infinite-dimensional quantum claim. These are sufficiency and separation results, not a universal weakest-foundation theorem. We close with a certificate-oriented research protocol and a set of tractable open problems at the interfaces between logic, analysis, and physics.

1 The missing question

A physical theory is normally presented as if its assumptions came in one list. In practice that list mixes several kinds of commitment. A symmetry principle, a causal condition, and a composition rule are physical postulates. A Hilbert space, a distribution space, a Krein product, or a C^*-algebra is a mathematical carrier. Completeness, compactness, selection of bases, spectral resolution, and

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existence of Green operators are mathematical theorems. Classical logic, set existence, actual infinity, and forms of Choice live one level further down. A calculation can use all four layers without saying where one ends and another begins.

This creates two opposite errors. The first is to treat the conventional mathematical formulation as physical content: quantum mechanics is said to *assume Hilbert space*, even when an operational reconstruction derives a Hilbert representation from other principles. The second is to treat the entire standard mathematical background as unavoidable: a proof uses a basis or compactness theorem, and one concludes that the physical law requires the Axiom of Choice. In both directions the inference skips the representation and the base theory.

Reverse mathematics supplies the right discipline for the mathematical side. Its central question is which set-existence axioms are needed to prove a theorem of ordinary mathematics [18]. Reverse physics applies a parallel strategy to physical laws: begin with accepted laws or theory structure and seek physical starting assumptions sufficient to recover them [7]. Operational reconstructions of quantum theory do something closely related inside a chosen formal arena [11, 8]. Yet the results do not compose automatically. A reconstruction may derive finite-dimensional complex Hilbert structure while relying on classical real analysis. A reverse- mathematical result may calibrate an abstract differential equation while saying nothing about which physical representation should encode its data. Computable analysis may construct an operator under a representation without classifying its proof-theoretic strength. A topos formulation may change the internal logic while leaving the bridge to experimental probabilities open.

The question of this paper is therefore not merely "which physical assumptions imply a theory?" nor merely "which axioms prove its mathematics?" It is:

> For a precisely represented physical obligation, which combination of physical postulates, carrier structure, existence principles, and logic is sufficient; which parts can be recovered from the obligation; and which can be removed by changing representation?

We use *reverse foundations of physics* as a working name for this combined programme. The phrase is descriptive, not a priority claim. The adjacent literatures are extensive, and the present literature survey is scoped rather than exhaustive. Our claim is narrower: the sources reviewed here do not provide a single typed atlas in which physical obligations, carrier choices, and logical or set-theoretic strength are all varied and audited together.

The programme grew from conformal gravity because that theory is a demanding stress test. Gauge symmetry, indefinite pairings, fourth-order evolution, Green operators, local BRST cohomology, state selection, renormalized products, and the quantum master equation all occur, but they occur at different dependency levels. The method is not specific to Weyl gravity. Its natural domain includes classical PDE, quantum mechanics, algebraic quantum theory, lattice systems, operational reconstructions, and any theory in which a change of mathematical foundations may change what "existence" or "state" means.

Contributions. This paper makes four contributions. First, it gives a typed formal object for the combined analysis. Second, it organizes a three-axis research atlas and explains what its coverage counts do and do not mean. Third, it presents five independently checkable case studies in which familiar foundational inferences split into distinct claims. Fourth, it proposes an evidence and certificate protocol designed to turn a suggestive matrix cell into a result that can survive changes of proof, representation, and implementation.

2 A typed derivability problem

2.1 Profiles, obligations, and encodings

The basic object is not a theory name but a derivability profile.

Definition 2.1 (Assumption profile)**.** An assumption profile is a tuple

$$\mathcal{A} = (L, S, M, \mathrm{Enc}(P), R),$$

where L specifies the logic and inference rules, S specifies the existence theory, M specifies the mathematical carrier and theorems made available over $L + S$, P is a list of physical postulates, Enc is a formal encoding of those postulates, and R records the representation of inputs and outputs.

The representation R deserves its own coordinate. A real number may be a Dedekind cut, a rapidly converging Cauchy name, or a point of an internal real object. A field may be a smooth section, a distribution, a Sobolev class, a finite lattice function, or a compatible family of finite data. Those representations can change both the content of an existence claim and the strength of its proof. We display R inside the profile but often absorb it into M and Enc, giving the compact judgement

$$L + S + M + \mathrm{Enc}(P) \ \vdash_R \ O. \tag{1}$$

Definition 2.2 (Obligation)**.** An obligation O is one theorem-level job with a declared result kind and scope: for example, construction of one normalized algebraic state; uniqueness of an energy solution in a represented carrier; existence of retarded and advanced Green maps with a support theorem; classification of local counterterms; or restoration of a local quantum master equation.

The obligation must be smaller than a theory. "Construct quantum field theory" is not an obligation. "Construct a BRST-compatible Hadamard state for the full metric BV complex" is an obligation, albeit a very difficult one. This granularity prevents a finite-mode calculation from being promoted to a continuum field theorem and prevents state existence from being promoted to a state-selection principle.

2.2 Seven relations, not one arrow

An unlabelled implication arrow hides too much. We use the following relation types.

Relation	Meaning
`USED_BY_DISPLAYED_PROOF`	One inspected proof invokes the item. This is neither necessity nor an optimal upper bound.
`SUFFICIENT_OVER_BASE`	A derivation of the obligation is available over the declared base, encoding, and representation.
`NECESSARY_OVER_BASE`	A reversal, separating model, or counterexample proves that the item cannot be omitted over the declared base.
`EQUIVALENT_OVER_BASE`	Both sufficiency and necessity have been proved over the same base and encoding.
`AVOIDED_BY_REFORMULATION`	Another explicit representation reaches the same declared obligation without the item.
`INDEPENDENT_OVER_BASE`	Models or realizations on both sides establish independence over the named base.
`UNKNOWN`	No stronger relation is certified.

These types separate three questions that are often collapsed:

1. Did a proof use a principle?
2. Is the principle sufficient or necessary for the theorem over a named base?
3. Does the physical obligation require the theorem in that representation and generality?

Only the second question is a reverse-mathematical calibration. The third is where reverse mathematics and reverse physics meet.

2.3 Why physical assumptions do not simply imply Choice

The phrase "physical assumption P implies the Axiom of Choice" is usually ill-typed. A physical postulate and a set-theoretic sentence do not even inhabit a shared formal language until an encoding and base theory have been fixed.

Proposition 2.3 (Encoding discipline)**.** *Let P be a physical postulate stated in an experimental or operational vocabulary and let C be a choice principle stated in a foundational language. A claim $P \Rightarrow C$ has no formal derivability content until one supplies a common theory $L + S$, an encoding* $\mathrm{Enc}(P)$*, and a target translation of C. Any proved implication then belongs to that encoding and base; it is not automatically an implication from an experiment to an unrestricted assertion about all sets.*

This does not make such reversals impossible. A physical principle may be encoded as a totality, selection, compactness, or continuity statement, and that statement may turn out to be equivalent to a familiar subsystem or choice fragment. The proposition says what must be provided before the claim is meaningful.

Physical structure can nevertheless reduce mathematical commitments in a more direct way. If the physics supplies a canonical grading and explicit mode labels, a proof need not choose a basis. If it supplies a preferred state or boundary condition, a selection problem may disappear. The correct relation in such cases is often `AVOIDED_BY_REFORMULATION`, not "physics proves Choice" or "physics disproves Choice."

3 The research atlas

The formal profile has many coordinates. For navigation we project it onto three axes: foundational regime, carrier, and physical obligation. The current projection has $6 \times 6 \times 16 = 576$ coordinates. This number is large enough to expose blind spots and small enough to inspect. It is not a claim that the underlying space of theories is finite, that the three axes are independent, or that every coordinate is coherent.

3.1 Six foundational regimes

Regime	Intended question
Classical standard	What is known in ordinary classical ZFC-style analysis, where background Choice is often not audited?
Weak arithmetic	How much proof-theoretic strength is needed in systems such as PRA, RCA_0, WKL_0, or ACA_0?
ZF with weakened Choice	Which constructions survive in classical set theory when full Choice or Countable Choice is absent or isolated?
Constructive/computable	What witnesses, algorithms, moduli, or represented operators can be constructed without treating classical existence as enough?
Topos/internal	What becomes of the theory when objects and truth are interpreted internally, often with Heyting rather than Boolean logic?
Finite/discrete restriction	What survives on finite carriers, exact finite algebras, lattices, or proposals that restrict actual infinity?

These regimes overlap but are not synonyms. ZF without Choice retains classical logic and actual infinity. Bishop-style constructive mathematics changes the meaning of existence. Computable analysis fixes representations and algorithms but is not by itself a subsystem classification. A topos has an internal logic relative to a selected category. A finite cutoff inside ZFC is not finitism.

3.2 Six carrier families

Carrier	What it is meant to track
Finite exact algebra	Finite matrices, integer or rational complexes, and explicit finite-dimensional algebraic data.
Hilbert/operator	Positive Hilbert spaces, completions, operator domains, spectral data, and one-parameter groups.
Krein/indefinite	Indefinite pairings, fundamental symmetries, positive companion topologies, and their operator theory.
Algebraic C^*-system	Observable algebras, states, GNS representations, nets, and algebra-first dynamics.
Smooth/PDE/ distributional	Manifolds, sections, Sobolev and distribution spaces, differential operators, and Green theory.
Localic/synthetic/ internal	Locales, internal algebra objects, formal manifolds, and synthetic smooth structures.

The rows are deliberately not mutually exclusive descriptions of a complete theory. A Krein space normally carries a positive companion Hilbert topology; an algebraic quantum theory may be represented on a Hilbert space; and a PDE may be encoded in a finite exact approximation. The axis asks which carrier is doing the work in the obligation under inspection.

3.3 Sixteen obligations in four groups

The obligation axis is the most important because it blocks silent promotion. For readability the sixteen values fall into four groups.

Group	Obligations
Kinematics and states	kinematics/observables; state existence; state representation; probability rule; physical state selection.
Dynamics and causality	generator/spectral dynamics; evolution and well-posedness; causal propagation and advanced/retarded Green maps.
Gauge and interaction	gauge/BV/cohomology; interaction construction; counterterm classification; anomaly classification.
Quantum completion and comparison	renormalized products; QME restoration; residual quantum transfer; reconstruction and continuum or empirical limits.

This ordering encodes no claim that every theory must pass through the stages linearly. It does show why "we have a state" does not answer which state is physical, why a unitary group does not supply a retarded propagator, and why a local anomaly classification does not restore a quantum master equation.

3.4 Coverage is metadata, not truth

In the current evidence snapshot, all 576 coordinates are emitted and assessed. Of these, 90 are marked as literature results, 127 as local results, 160 as having only pieces, 30 as selected priority gaps, and 169 as reviewed open gaps; zero remain not mapped. A reviewed open gap is a formulated research question with a foundation requirement, carrier requirement, and typed missing certificate, but no direct local or literature result. The evidence interface contains 83 source or certificate records.

Those numbers measure the state of a corpus and classification scheme. They do not measure how much humanity knows, and a reviewed gap is not evidence of impossibility or literature absence. A literature-result label says that a source establishes a claim with a recorded boundary; it does not say the entire cell is solved. A local-result label says that a repository certificate reaches the declared obligation; it does not make the result representation-independent. A priority gap is a selected invitation to research, not a no-go theorem. A reviewed open gap has been made precise enough to investigate, but is not promoted to a programme priority or scientific result.

The counts are reproducible from `FOUNDATIONAL_INTERSECTION_CUBE_V14`, `FOUNDATIONAL_FULL_SURFACE_GAP_AUDIT_V1`, and `FOUNDATIONAL_MATRIX_EXPLORER_SITE_V2`. The cube explicitly records that complete assessment is not literature completeness, does not make every coordinate jointly realizable, proves no weakest base or reversal for the coded wave result, and establishes no new Lorentzian Weyl claim.

4 How adjacent programmes fit—and fail to collapse

4.1 Reverse mathematics

Reverse mathematics most directly calibrates the $L + S \vdash T$ part of Eq. (1). Its strength is the insistence on a named weak base and, in a full reversal, proofs in both directions. Simpson's analysis of the Cauchy–Peano theorem is an instructive physics-adjacent example: over RCA_0, the theorem is equivalent to WKL_0, while the Ascoli lemma and Osgood's maximum-solution theorem have the strength of ACA_0 [17]. "Differential equations require analysis" is thereby replaced with several exact statements whose strength depends on the theorem.

The limitation for our purpose is equally clear. Reverse mathematics begins after a statement and its representation have been selected. It does not by itself decide whether the physical problem requires Cauchy–Peano in that generality, whether a canonical physical flow avoids the compactness step, or whether a different carrier expresses the same observable content.

4.2 Reverse physics and operational reconstruction

Reverse physics begins from laws and seeks physical assumptions sufficient to rederive them [7]. This is the closest methodological relative of the present programme. Our additional demand is to audit the mathematical and logical substrate of every displayed derivation.

Quantum reconstructions show why this matters. In Hardy's five-axiom derivation, continuous reversible transformations are load-bearing: deleting continuity yields classical probability theory within the stated framework [11]. Chiribella, D'Ariano, and Perinotti define a broad operational-probabilistic class using five axioms and use purification to single out quantum theory [8]. These works turn some textbook mathematical postulates into derived representation theorems. They do not, without a separate audit, classify the proof-theoretic cost of the continuum, compactness, convex separation, or infinite-dimensional limit.

Accordingly, "Hilbert space is derived" and "Hilbert-space quantum field theory is constructible over a weak or choice-free base" are very different claims. The former can be a finite-dimensional operational theorem; the latter includes completion, unbounded operators, distribution theory, locality, states, and renormalization.

4.3 Choice-free functional analysis

It is common to summarize the situation as "functional analysis needs Choice." Recent results make that too coarse. Blackadar, Farah, and Karagila study Hilbert spaces and operators in ZF without Countable Choice, showing both robust canonical constructions and unfamiliar pathologies [6]. Blackadar and Farah develop substantial separable C^*-algebra theory in ZF, including representation theorems, spectral mapping for polynomials, and continuous functional calculus for commuting normal elements, while also exhibiting failures outside protected settings [5].

The lesson is not that Choice is irrelevant. It is that the cost is attached to a theorem, a class of objects, and a presentation. Canonically presented separable structures may avoid selections that arise for arbitrary families. The right unit of analysis is therefore not "Hilbert space" or "C^*-algebra" but the exact construction used in a physical obligation.

4.4 Constructive, computable, and proof-mined analysis

Constructive analysis requires existence statements to carry witness content [4]. Computable analysis makes that demand relative to explicit representations of spaces and operators [20]. Weihrauch and Zhong, for example, give an algorithm for distributional fundamental solutions of constant-coefficient differential operators under canonical representations [21]. That result is highly relevant to the carrier and computability axes, but a fundamental solution is not automatically a retarded or advanced solution and the theorem does not by itself supply Lorentzian causal support.

Proof mining asks a different question again: what quantitative or computational information can be extracted from an apparently nonconstructive proof? Pischke's metatheorem for nonlinear semigroups generated by accretive operators illustrates its reach into PDE and evolution, including extracted rates [15]. Proof mining should not be relabelled as a Big-Five reverse-mathematical classification, nor should a computable realizer be silently treated as a constructive proof in every foundational system. The three methodologies can inform one another only after their input and output types are recorded.

4.5 Topos and localic reformulations

Constructive Gelfand duality replaces point-set spectra with point-free spectra in a form valid constructively [9]. Heunen, Landsman, and Spitters place a noncommutative algebra in a topos of its commutative contexts, where the internal algebra is commutative, its spectrum is a locale, and the internal propositional logic is intuitionistic [12]. This is a genuine change of mathematical architecture, not merely deletion of Choice from a classical proof.

It also illustrates a recurring boundary. Internal state objects, locale valuations, and internal dynamics do not automatically supply a physical state-selection rule, a Lorentzian Green operator, an interacting BV complex, or empirical equivalence with the external formulation. Object logic and metalogic must be recorded separately. A non-Boolean algebra of quantum propositions is not, just by itself, an intuitionistic metatheory.

4.6 Finite mathematics and finite physics

Finite structures offer exact computation and sometimes eliminate analytic existence questions. But "finite" has several types. Gibbons, Hoffman, and Wootters use a finite field to label a discrete phase space when the quantum state-space dimension is a prime power [10]. The carrier of state vectors remains an N-dimensional complex Hilbert space. A finite field phase space is therefore not the same thing as a finite-field amplitude theory, a finite-mode truncation, or a rejection of actual infinity.

The distinctions become decisive when a finite regulator is used to motivate a continuum claim. Exact solvability at each finite size supplies a family of finite theorems. A continuum limit requires its own representation, uniform estimates, convergence theorem, and physical comparison. Finitism is a foundational proposal; a lattice cutoff inside ordinary mathematics is an approximation scheme.

5 Five certified separations

The following examples are not offered as a complete theory. They are small enough to verify independently and together show what the combined programme changes in practice.

5.1 Krein structure changes the pairing, not all Hilbert mathematics

A Krein space carries a nondegenerate indefinite product together with a fundamental decomposition or fundamental symmetry. If $J = J^* = J^{-1}$, the companion product

$$(x, y)_J = [x, Jy]$$

is positive and supplies Hilbert topology. Replacing a positive physical pairing by a Krein pairing therefore does not remove completion, topology, or operator-domain questions. It redistributes them.

In the reduced conformal-gravity mode carrier audited in `FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1`, energy, family, chirality, and a finite multiplicity coordinate name every mode. The fundamental symmetry is coordinate multiplication by the sign of the family. At every fixed cutoff, the list and the involution are primitive-recursive finite data; PRA suffices to verify self-adjointness and $J^2 = 1$. At the complete one-particle level, the carrier is an explicitly indexed $\ell^2(I)$, and the occupation number Fock carrier is $\ell^2(\mathrm{Occ}(I))$. The certificate uses ZF results for those canonical completions and no Countable Choice.

The foundational work is done by the representation. Named modes replace an unspecified basis search and a choice of bases across arbitrary families. The certified relation is `AVOIDED_BY_REFORMULATION` for those particular selection steps and `SUFFICIENT_OVER_BASE` for ZF. It is not a proof that arbitrary Krein spaces have choice-free fundamental decompositions or that ZF is the weakest possible base.

This case also corrects the slogan "Krein throws out Hilbert space." The distinguished physical pairing becomes indefinite, but a positive companion Hilbert topology remains part of the displayed structure. What changes are the adjoint relations, positivity problem, null directions, and physical interpretation; what survives must still be audited theorem by theorem.

5.2 State existence is not state selection

Once the coordinate Krein carrier is available, constructing *a* state is easy. A named J-eigenvector e_i defines an explicit normalized vector state on the companion Hilbert algebra,

$$\omega_i(A) = (e_i, Ae_i)_J,$$

and the rank-one density operator $|e_i\rangle\langle e_i|$ is explicit. No choice principle is required for this witness. The same holds for named occupation vectors in the Fock carrier. These constructions are certified by `FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1` with dependency tags `LOCAL-ALGEBRAIC` and `REDUCED-MODE`.

But J does not choose a unique coordinate. Indeed, the certificate proves a scoped no-selection result: no density operator is invariant under the full group of sign-preserving finite-coordinate permutations on the infinite positive and negative sectors. The reason is elementary. Invariance makes all diagonal weights within an infinite sector equal; trace-class normalization then forces each equal weight to vanish, contradicting unit trace. This excludes normal density states with that full invariance. It does not exclude singular states, and it does not prove that the permutation group is a required physical symmetry.

Thus three cells that are often merged must be separate:

$$\text{state existence} \quad \not\Rightarrow \quad \text{canonical state selection} \quad \not\Rightarrow \quad \text{Born interpretation.}$$

Extra physical information—dynamics, boundary conditions, thermality, Hadamard regularity, detector preparation, or an incoming condition—must do the selecting. The algebra and its set-theoretic foundation cannot supply that information merely by existing.

A related audit, `FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1`, reaches the same boundary from an algebra-first direction. An explicitly presented separable compact- operator unitization, concrete states, and a corner GNS representation can be constructed in ZF. A semifinite trace is not a normalized state, and conditioning on a detector projection imports that projection as additional physical data. Algebra, state existence, representation, and physical selection are four obligations.

5.3 Weak wave evolution is not causal Green theory

The certificate `FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1` treats a mean-zero scalar wave on a coded circle. Dense states are pairs of rational periodic step functions whose polygonal primitives encode left- and right-moving profiles. Completed states and real times are supplied as fast Cauchy names with fixed rates. Rational translation is exact and preserves the energy metric. A finite-code time modulus and a primitive-recursive diagonal construction extend the translation group to named real times.

For this representation, RCA_0 suffices for existence, uniqueness, the group law, and energy conservation of the continuous isometric extension. Finite rational-code identities are already checkable in PRA. This is a genuine weak-base upper bound, and it demonstrates how much can be obtained when convergence rates are data rather than objects to be extracted from bare convergence.

The boundary is as important as the theorem. The result proves neither that RCA_0 is necessary nor that the upper bound is representation-invariant. It constructs no spacetime distribution, no advanced or retarded Green map, and no finite-propagation theorem. A unitary or isometric evolution group solves a Cauchy obligation; causal support is a separate spacetime theorem.

Two subsequent certificates make the intervening reconstruction steps explicit. First, `FOUNDATIONAL_CODED_WAVE_OBSERVABLE_RECONSTRUCTION_V1` fixes a rational periodic detector h and constructs finite rational dyadic-time interpolants A_m for the smeared chiral observable O_h. If $K = \mathrm{Lip}(h)(\|a\|_1 + \|b\|_1)$ and $\ell(K)$ is the least natural number with $K \leq 2^{\ell(K)}$, then

$$N(k) = k + \ell(K) + 1, \qquad \sup_{|t|\leq T} |A_{N(k)}(t) - O_h(t)| < 2^{-k}$$

for every positive rational T. The finite arithmetic is PRA-checkable and the coded uniform-limit statement is proved over RCA_0. This reconstructs one bounded observable, not the full wave state.

Second, `FOUNDATIONAL_CODED_LOCAL_WEAK_WAVE_TEST_CLASS_V1` takes the common five-cell rational partition of the declared fixtures and builds ten compact-time characteristic-strip polynomial tests. Their labelled chiral measurement matrix is diagonal of exact rank ten. For $r = a(x - t)$, $l = b(x + t)$, and $u = f(x - t) + g(x + t)$, every displayed test coefficient satisfies

$$\iint r(\phi_t + \phi_x) = 0, \qquad \iint l(\phi_t - \phi_x) = 0, \qquad \iint u(\phi_{tt} - \phi_{xx}) = 0.$$

The first two identities are exact characteristic boundary terms; the third follows by one integration by parts. Rational linear combinations therefore give a finite localized weak test class. The separation retains the labelled right/left carrier, and the result does not cover every smooth compactly supported test, an unlabelled scalar-field quotient, strict causal support, or a Green operator. It closes a finite coefficient-wise gate while leaving the full distributional and causal gates open.

The next certificate makes the representation choice explicit rather than silently identifying that finite span with the classical test-function space. Let rational periodic compact-time C^1 piecewise-polynomial codes on the unit cylinder carry the squared H^2 norm, and define a completed test to be a supplied fast Cauchy name q_i with $\|q_i - q_j\|^2_{H^2} \leq 4^{-i}$ for $i \leq j$. Then `FOUNDATIONAL_CODED_WEAK_WAVE_H2_TEST_COMPLETION_V1` proves over RCA_0 that the coded energy solution defines a continuous spacetime functional on this completion and that all three weak residuals extend from exact zero on the rational codes to exact zero on every named test. For example,

$$|W(u;\delta\phi)|^2 \leq 4E\|\delta\phi\|^2_{H^2}, \qquad N_W(k) = k + \ell(\lceil 4E \rceil).$$

The rate is part of the test name. This covers a smooth periodic compact-time test when such a name is supplied; it does not uniformly manufacture names from bare extensional smooth functions. That boundary matters because the unrestricted space of compactly supported smooth tests has a nonmetrizable LF topology, while represented smooth and bump functions require derivative and support advice [16]. Classical direct approximation by piecewise polynomials in Sobolev norms supplies useful comparison context [19], but is not imported here as a uniform RCA_0 name translator. The theorem proves uniqueness only inside the represented energy solution image, not among arbitrary distributions, and supplies no causal support or Green operator.

Three certificates now close that declared bridge without erasing its representation boundary. First, `FOUNDATIONAL_FIXED_SUPPORT_SMOOTH_TO_H2_TRANSLATOR_V1` takes as input a rational support interval $[a, b]$, a rational collar δ, and rational approximants p_m whose derivatives through order two converge uniformly at the supplied rate 2^{-m}. With the rational cubic smoothstep $h(r) = 3r^2 - 2r^3$, it constructs a global-C^1 cutoff χ and

$$q_n = \chi p_{n+s}, \qquad \|q_n - \phi\|_{H^2}^2 \leq 4^{-n}.$$

Here s is the least integer for which 4^s majorizes the exact sum of the six product-rule constants. The construction is uniform over RCA_0 and uses no choice operation because the support and rate are fields of the input name. It does not manufacture those fields from a bare extensional function.

Second, `FOUNDATIONAL_SUPPORT_INDEXED_TEST_SPACE_COMPARISON_V1` uses the nested rational supports $K_j = [2^{-(j+2)}, 1 - 2^{-(j+2)}] \times (\mathbb{R}/\mathbb{Z})$. A conventional represented test name consisting of a smooth name and a discrete support bound is exactly intertranslatable with a tagged pair (j, p) in the support-indexed union. The fixed-stage translators are compatible after passage to a common later stage, so the coded weak identity holds for every test carrying such a name. This proves a name-level equivalence, not equality with the full locally convex LF topology; the H2 completion also contains nonsmooth elements and the embedding is not surjective.

Third, `FOUNDATIONAL_SCALAR_MINKOWSKI_GREEN_CHOICE_AUDIT_V1` supplies a controlled causal benchmark. For $P = \partial_t^2 - \partial_x^2$ on flat $1+1$-dimensional Minkowski spacetime it constructs the canonical maps

$$(G_{\mathrm{ret}} f)(t, x) = \frac{1}{2} \int_{-\infty}^{t} \int_{x-(t-s)}^{x+(t-s)} f(s, y)\, dy\, ds,$$

with the time-reversed formula for G_{adv}. Exact rational null-coordinate codes certify $PG_{\pm} f = f$, $G_{\pm} P\phi = \phi$ on the declared domains, causal support, and adjoint duality. An explicit energy estimate extends the maps to supplied fast source names over RCA_0. This is a genuine `LORENTZIAN-CAUSAL` result, but only for the displayed scalar flat operator. It supplies no variable-coefficient, curved-spacetime, Weyl/BV, or quantum Green construction. In particular, it is not a Weyl/BV propagator or quantum causal construction.

The literature makes the missing bridge visible. Normally hyperbolic operators on globally hyperbolic spacetimes have advanced and retarded Green operators with support control in classical analysis [2]. Under represented hypotheses, parts of PDE and fundamental-solution theory are computable [22, 21]. The scalar certificate above is an explicit weak-base avoidance result for one canonical formula. It is not a direct reverse-mathematical reversal, a Bishop-style global Green theorem, or a choice-free ZF audit for the full variable-coefficient Lorentzian construction. "Classically known," "computably represented," and "calibrated over a weak base" remain three different statuses.

5.4 Exact finite causality is not continuum causality

On a finite nearest-neighbour graph, a discrete wave recurrence can be solved by exact rational arithmetic. The retarded response at time step n is a polynomial in the finite propagation operator of degree at most n. Consequently its matrix entry vanishes when graph distance exceeds n. The advanced kernel follows by time reversal. The construction and graph-step support theorem are certified in `FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1`.

This is a clean causal theorem, but its type is `LOCAL-ALGEBRAIC/REDUCED-MODE`, not `LORENTZIAN-CAUSAL`. It does not provide a continuum Green operator, a light-cone support theorem, stability under refinement, a regulator- independent limit, or a metric BV propagator. Each of those requires new mathematical data.

The example gives a useful rule for finite programmes:

$$\begin{aligned}&\text{exact finite identity} + \text{uniform estimate}\\&+ \text{represented convergence} + \text{limit identification}\\&\Longrightarrow \text{candidate continuum theorem.}\end{aligned}$$

Only the first term is supplied by exact finite causality. This does not make the finite theorem unimportant. It tells us exactly which part of continuum causality is algebraic and which part is analytic.

5.5 A positive finite Euclidean state is not yet a Krein reconstruction

The Bateman–Turok positive Euclidean lattice provides a second, physically motivated finite case. On a connected periodic graph, write $\Omega_x = \exp(\lambda\phi_x) > 0$, remove the constant mode by $\sum_x \phi_x = 0$, and use the positive action

$$S_{E,L} = \frac{1}{2\lambda^2}\sum_x \left(\frac{(\Delta_L\Omega)_x}{\Omega_x}\right)^2.$$

The certified coercive bound makes the finite-dimensional partition function finite. Hence a normalized Euclidean Gibbs state exists and ordinary event probabilities are defined by integration against its positive density. The interaction is exact rather than a numerical inference: on the two-site mean-zero direction $\phi = (t, -t)$, its series contains $\frac{28}{3}\lambda^2 t^4$ for $\lambda \neq 0$. Certificate `FOUNDATIONAL_BT_EUCLIDEAN_LATTICE_IMPORT_V1` therefore supplies direct local records for kinematics, state existence, state representation, a Euclidean statistical probability rule, and interaction construction in the `FINITE-DISCRETE/SMOOTH-DISTRIBUTIONAL` cell family.

The numerical statement is deliberately weaker. Hamiltonian Monte Carlo and an independently implemented local Metropolis chain agree at the declared four-standard-error gate, but not at a two-standard-error precision gate; the $L = 4$ to $L = 6$ interaction-proxy step remains unresolved. This is tagged `EUCLIDEAN-SPECTRAL` and recorded as coarse numerical reproduction, not empirical validation or a continuum theorem.

There is also a precise carrier boundary. The lattice uses $\Omega > 0$, whereas the two-field BT/Krein path integral ranges over all real Ω. They cannot be identified as the same full nonperturbative configuration space and measure. This scoped non-identity does not rule out a conditional perturbative, reflection-positive, or analytic-continuation bridge. Such a bridge would still have to supply the missing uniform estimates, represented convergence, limit identification, and Lorentzian observable map. The reconstruction cell therefore remains a priority gap.

The first reconstruction decision is now exact at the free endpoint. On the periodic 6^4 lattice with link reflection $\theta(t, \mathbf{x}) = (1 - t, \mathbf{x})$, let A_t be the spatial slice average and take the shift-invariant positive-time observable $F = -A_1 + 2A_2 - A_3$. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_FREE_RECONSTRUCTION_OBSTRUCTION_V1` computes

$$\langle(\theta F)F\rangle_{0,6} = -\frac{1}{1296}.$$

Thus the positive density fails ordinary Osterwalder–Schrader reflection positivity on this finite free lattice. Fixed-volume coercivity supplies a common Gaussian dominating function, so continuity preserves the strict negative sign on some unquantified open coupling interval around zero. This does not decide $\lambda = 0.4$, and it does not obstruct a specifically Krein-compatible reconstruction.

The same certificate also isolates the first viable free continuum topology. For trigonometric interpolation on the unit four-torus,

$$\sup_L \mathbb{E}\|\Phi_L\|_{H^{-1}}^2 \le \frac{15}{32}, \qquad \mathbb{E}\|\Phi_L\|_{L^2}^2 \ge \frac{H_{\lfloor (L-1)/2 \rfloor}}{4\pi^4}.$$

The L^2 second moment therefore diverges logarithmically, while a negative Sobolev estimate survives. This is a free-family uniform estimate, not interacting tightness, represented convergence, or limit identification.

A bounded interacting preflight, `REVERSE_PHYSICS_BT_EUCLIDEAN_OS_WITNESS_PREFLIGHT_V1`, evaluates the same reflected product at $\lambda = 0.4$ with four independent HMC and four independent local-Metropolis replicas. All eight chain means are negative. Equal-replica pooling gives -6.25 standard errors for HMC and -2.53 for the slower-mixing local chain; the two algorithm means differ by 0.64 combined standard errors. This is two-sampler numerical support for a negative sign of that linear witness, not its exact sign certificate.

The finite-volume ordinary-OS question at this coupling is nevertheless now decided by a different exact witness. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_LAMBDA04_OS_KERNEL_OBSTRUCTION_V1` chooses two spatially constant, half-sum-zero configurations with $\lambda\phi$ in integer multiples of $\log 2$. Every Boltzmann exponent entering their two-by-two reflected density kernel is therefore rational, and the exact action gap is

$$S_{pp} + S_{qq} - 2S_{pq} = \frac{19361025}{512} > 0.$$

It follows that the kernel determinant is strictly negative. Smooth compact bumps around the two configurations convert this point-kernel direction into an admissible positive-time cylinder function with negative OS form. Thus ordinary reflection positivity is obstructed at $\lambda = 0.4$ on the declared 6^4 lattice; the numerical preflight is supporting only. This does not decide every nonzero coupling, construct an interacting uniform estimate, or obstruct an indefinite-metric reconstruction.

The first natural route to that interacting estimate is itself now exactly obstructed. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_UNIFORM_CONVEXITY_OBSTRUCTION_V1` uses the spatially constant mean-zero family $k(a) = (-a, 0, 0, -a, a, a)$ and direction $v = (-1, -1, 1, 1, 1, -1)$. The ratio of the nonlinear directional Hessian to the free bilaplacian form is

$$\frac{2^a + 1}{2^{2a+1}} \longrightarrow 0.$$

Hence no positive field-independent bilaplacian strong-convexity constant exists, even at fixed volume, and the standard global Brascamp–Lieb comparison cannot prove the desired bound. This does not establish nonconvexity or failure of the interacting H^{-1} estimate; it directs the next analysis toward Schwinger–Dyson, annealed-Hessian, or multiscale control.

The direct mode identity has also been decided at the pointwise level. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_SCHWINGER_DYSON_MODE_OBSTRUCTION_V1` proves, for every mean-zero lattice mode h,

$$\mathbb{E}[\langle h, \phi \rangle \langle h, \nabla S_\lambda \rangle] = \|h\|_2^2.$$

For a Laplacian eigenmode this splits into its free covariance term and an interaction remainder. A lowest-mode configuration on the 6^4 lattice makes that remainder strictly negative, so a pointwise sign cannot yield free covariance domination. The same certificate proves the all-volume deterministic estimate

$$S_\lambda(\phi) \ge \frac{\lambda^2}{2L^4} \left[\sum_{\{x,y\}} |\phi_y - \phi_x|^2 \right]^2 \ge 128\lambda^2 |\widehat{\Phi}_L(e_\mu)|^4.$$

The second inequality concerns any lowest axial coefficient. It is uniform action-sublevel control, not a normalized Gibbs moment bound: an annealed remainder or marginal-partition estimate is still missing.

The deterministic control can be sharpened to a bilaplacian radial envelope. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_BILAPLACIAN_REFERENCE_BRIDGE_V1` proves, with $B_\phi = \sum_x (\Delta\phi_x)^2$ on a q-regular periodic graph,

$$S_\lambda(\phi) \geq \frac{B_\phi}{4N} + \frac{\lambda^2 B_\phi^2}{16q^2 N}.$$

The normalized radial reference measure defined by the right-hand side has the volume-uniform bound

$$\sup_L \mathbb{E}_\nu \|\Phi_L\|_{H^{-1}}^2 \leq \frac{q\sqrt{15}}{4|\lambda|} = 5\sqrt{15} \quad (q = 8,\ \lambda = 0.4).$$

This is not yet a bound for the actual BT measure. Pointwise domination does not compare normalized moments, and an exact 6^4 witness gives a directional Hessian below $-31/3$ for the actual action minus the reference action. Thus convex-perturbation transfer is obstructed; a genuinely nonconvex, annealed, coarea, or direct-marginal bridge remains open.

The Hessian obstruction persists even if curvature is requested only in one lowest mode. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_LOW_MODE_UV_SCHUR_OBSTRUCTION_V1` decomposes the earlier degenerating direction exactly as

$$v = -\frac{2}{3}h + \frac{1}{3}g,$$

where $(-\Delta_6)h = h$ and $(-\Delta_6)g = 4g$. At the center family $k(a) = (-a, 0, 0, -a, a, a)$, $x = 2^a$, eliminating the alternating mode g gives a lowest-mode Hessian Schur complement

$$0 < \kappa(x) \leq \frac{72}{x}, \qquad \lim_{x\to\infty} x\kappa(x) = 18.$$

Thus no field-independent pointwise Hessian estimate controls even this single infrared mode. The same centers have action $A(x) \sim x^4$, with $\kappa(x)A(x)^{1/4} \to 18$. This leaves action-weighted or annealed curvature control viable along that family, but does not make the quarter power a global exponent.

The successor certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_ACTION_WEIGHT_VIRIAL_OBSTRUCTION_V1` decides that exponent negatively. At the period-three center

$$k(a) = a(-2, 1, 1, -2, 1, 1), \qquad x = 2^{3a},$$

the chosen lowest mode couples, by the exact shift-and-reflection symmetry decomposition of the full mean-zero six-cycle sector, only to the alternating mode. Its full Schur curvature and center action obey

$$\kappa(x) \sim \frac{36}{x}, \qquad A(x) \sim 4x^2.$$

Consequently

$$\kappa(x)A(x)^p \longrightarrow 0 \quad (p < 1/2), \qquad \kappa(x)A(x)^{1/2} \longrightarrow 72.$$

In particular the quarter-power candidate is obstructed. The first exponent not ruled out by this family is one half. With full volume $N = 6^4$, the scale-compatible version uses action density:

$$C(x)=\frac{\kappa(x)}{\kappa(0)}\sqrt{1+\frac{A_{\mathrm{total}}(x)}{N}},\qquad \lim_{x\to\infty} C(x)^2=6.$$

This is a surviving candidate, not a theorem. A total-action half weight would ordinarily introduce a $\sqrt{N}$ factor after Gibbs averaging, so it would not by itself give the required volume-uniform covariance estimate.

The same certificate also obstructs the immediate moment shortcut. For the exact rational center

$$\psi=(-1,-1,0,0,2,0)\log(101/100),$$

an alternating rational upper bound on $\log(101/100)$ proves

$$\frac{\psi\cdot\nabla A(\psi)}{A(\psi)}<2.$$

Thus the pointwise virial inequality with coefficient two cannot be used to deduce $\mathbb{E}[S_\lambda]/N<1/2$ by Gibbs integration by parts. The required action-density control nevertheless follows from an affine virial theorem. Put

$$s_x=\sum_{y\sim x} e^{\psi_y-\psi_x},\qquad r_x=s_x-q,\qquad t_x=\sum_{y\sim x} e^{\psi_y-\psi_x}(\psi_y-\psi_x).$$

Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_AFFINE_VIRIAL_ACTION_DENSITY_V1` uses convexity and superadditivity of $u\log u$ to prove, on every finite q-regular graph,

$$D=\sum_x r_x t_x\geq 2A-Nq^2\left(1+\frac{1}{4}\log q\right).$$

For $q=8$, the rational Taylor certificate $e^{7/10}>482921/240000>2$ gives

$$D\geq 2A-\frac{488}{5}N.$$

Radial integration by parts for the actual measure $e^{-A/\lambda^2}d\psi$ on the $(N-1)$-dimensional mean-zero hyperplane then gives, at $\lambda=2/5$,

$$\frac{1}{N}\mathbb{E}[A]\leq\frac{1222}{25},\qquad \mathbb{E}\sqrt{1+\frac{A}{N}}\leq\frac{\sqrt{1247}}{5}.$$

The constant is crude but independent of volume. The same theorem and the certified bilaplacian envelope give genuine actual-measure bounds $\mathbb{E}[B_\psi^2]/N^2\leq 1251328/25$ and $\mathbb{E}[B_\phi]/N\leq 40\sqrt{1222}$. These bounds do not control how bilaplacian energy is distributed among the lowest modes. Consequently the annealed half-action factor is established, but this is not yet an H^{-1} theorem.

The remaining global curvature gate is now exactly obstructed. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_ORTHOGONAL_HESSIAN_BLOCK_OBSTRUCTION_V1` uses, on every L^4 torus with $4\mid L$, the period-four background and direction

$$\psi_x=\log(4/3)\sum_{\mu=1}^{4}(-1)^{x_\mu},\qquad v_x=\prod_{\mu=1}^{4}(1,0,-1,0)_{x_\mu}.$$

The direction is orthogonal to the full eight-dimensional lowest axial Fourier eigenspace. Exact rational differentiation on one 4^4 cell gives

$$A = \frac{80458}{81}, \qquad \operatorname{Hess} A[v,v] = 72 - \frac{19712}{81} = -\frac{13880}{81} < 0.$$

Period-four replication multiplies both quantities by $(L/4)^4$. Thus the orthogonal Hessian block is indefinite on an unbounded volume sequence, so a globally positive inverse block and the proposed pointwise half-action Schur curvature do not exist as formulated. This closes a proof route, not the probabilistic question. The direct normalized low-frequency marginal, the interacting H^{-1} moment, tightness, continuum measure, positive-Hilbert reconstruction, Krein bridge, Born rule, and every Lorentzian statement remain open.

There is now an exact coordinate system for that normalized marginal. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_RESIDUAL_SPECTRAHEDRAL_PUSHFORWARD_V1` sets $\Omega = e^{\psi} > 0$, $r = (\Delta\Omega)/\Omega$, and

$$K(r) = -\Delta + \operatorname{diag}(r), \qquad \mathcal{C}_G = \{r : K(r) \succeq 0\}.$$

On every finite connected graph, the residual map is an analytic diffeomorphism from positive fields modulo scale onto the smooth boundary of the convex spectrahedron $\mathcal{C}_G$. The inverse extracts the unique positive ground state of $K(r)$. The exact ground-state transform is

$$v^T K(r) v = \sum_{\{x,y\}\in E} \Omega_x \Omega_y \left(\frac{v_x}{\Omega_x} - \frac{v_y}{\Omega_y} \right)^2 .$$

If $J_\psi = Dr(\psi)$, the directed matrix-tree theorem gives the restricted coarea Jacobian

$$\mathcal{J}_H(\psi) = \sqrt{N}\, \|\Omega^2\|_2\, \tau_\psi, \qquad \tau_\psi = \sum_T \frac{\prod_{\{x,y\}\in T} \Omega_x \Omega_y}{\prod_x \Omega_x^2}.$$

Thus the actual normalized measure pushes forward exactly to

$$Z_L^{-1} \frac{e^{-\|r\|_2^2/(2\lambda^2)}}{\sqrt{N}\, \|\Omega(r)^2\|_2 \tau(r)}\, d\mathcal{H}^{N-1}(r), \qquad r \in \partial\mathcal{C}_G.$$

The logarithm of the displayed Jacobian is convex under every multiplicative field tilt, because both factors are log-sum-exp functions. On every vertex-transitive graph, AM–GM over spanning trees and $\sum_x \psi_x = 0$ sharpen this to the global bound

$$\mathcal{J}_H(\psi) \geq N\kappa(G),$$

where $\kappa(G)$ is the number of spanning trees; equality holds at the constant field. Thus the reciprocal Jacobian cannot favor a nonconstant field relative to its vacuum value. This replaces an implicit orthogonal-field partition factor by an explicit Gaussian surface, positive-ground-state, and spanning-tree problem. It is a normalized reformulation, not the missing estimate: an all-volume bound for the lowest Fourier coefficient of $\lambda^{-1} \log \Omega(r)$, followed by a dyadic shell sum, remains open. Foundationally, the graph identities and cofactors are finite exact algebra, Perron–Frobenius and coarea are finite-dimensional analysis, and the uniform limit is a third unsupplied layer. This is only `USED-BY-DISPLAYED-PROOF`; no weakest-base reversal is claimed.

The most immediate convex-boundary shortcut is also decided. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_RESIDUAL_BOUNDARY_CURVATURE_OBSTRUCTION_V1` derives the outward second fundamental form

$$\mathrm{II}_r(h,h) = \frac{2}{\|\Omega^2\|_2} \left\langle \operatorname{diag}(h)\Omega, K(r)^+ \operatorname{diag}(h)\Omega \right\rangle .$$

It is strictly positive at each finite principal-boundary point. However, on the four-cycle the exact family $\Omega(q) = (q, 1, 1, q^{-1})$ and tangent $h(q) = (0, 0, 1, -q^2)$ give

$$\kappa_{\mathrm{trial}}(q) = \frac{2q^3(2q+1)}{(q+1)^2(q^4+1)^2}, \qquad \lim_{q\to\infty} q^6 \kappa_{\mathrm{trial}}(q) = 4.$$

Thus no positive field-uniform principal-curvature lower bound exists even on C_4. For the Gaussian surface reference density at $\lambda = 2/5$, its weighted mean curvature at $q = 2$ is exactly

$$H_{2/5} = -\frac{398039}{88434} < 0.$$

Consequently the standard boundary spectral-gap route requiring uniform positive second fundamental form and positive weighted mean curvature is obstructed as formulated. This is not an obstruction to every intrinsic boundary inequality and not a divergence theorem for the actual inverse-tree-Jacobian BT measure. The normalized low-mode moment remains open.

The natural multiplicative-tilt calculation also removes a possible misreading of that inverse Jacobian. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_RESIDUAL_TILT_JACOBIAN_CANCELLATION_V1` lets $R : H \to \partial\mathcal{C}_G$ denote the residual chart, with surface Jacobian $\mathcal{J}_H(\psi)$, and induces from $\psi \mapsto \psi + th$ the boundary map $T_{t,h} = R \circ (\psi \mapsto \psi + th) \circ R^{-1}$. The chain rule gives

$$\mathrm{Jac}_{\partial\mathcal{C}_G} T_{t,h}(R(\psi)) = \frac{\mathcal{J}_H(\psi + th)}{\mathcal{J}_H(\psi)}.$$

Since the BT surface density is proportional to $e^{-S(\psi)}/\mathcal{J}_H(\psi)$, its full pullback ratio is exactly

$$\frac{\rho(T_{t,h}R(\psi))\,\mathrm{Jac}_{\partial\mathcal{C}_G} T_{t,h}(R(\psi))}{\rho(R(\psi))} = e^{-S(\psi+th)+S(\psi)}.$$

Thus the coarea factors cancel rather than supplying an extra confining potential. On the exact four-cycle fixture the surface and reciprocal-density ratios are $9/10$ and $10/9$, so $\frac{9}{10}\frac{10}{9} = 1$; at $\lambda = 2/5$ only the action produces the remaining exponent gap $5325/128$. Equivalently, after $\psi = \eta + sh$, the one-mode marginal is the flat conditional-fiber integral $Z^{-1}\int e^{-S(\eta+sh)}d\eta$, with no tree factor. The action-weighted fiber ratio, normalized one-mode moment, and interacting H^{-1} estimate remain open. Foundationally, the chart cancellation is finite exact algebra plus finite-dimensional change of variables; the all-volume estimate is still an unsupplied analytic layer, and no weakest-base reversal is claimed.

The centered pointwise fiber comparison is now also decided negatively. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_CENTERED_FIBER_DOMINATION_OBSTRUCTION_V1` works on the fixed 6^4 lattice with the lowest time-circle mode $h = (2, 1, -1, -2, -1, 1)$. For

$$\eta_n = n \log 2\,(-1, -1, 1, -3, 3, 1), \qquad t_n = -n \log 2,$$

one has $\eta_n \perp h$ and the exact all-n estimate

$$\frac{A(\eta_n + t_n h)}{A(\eta_n)} \le \frac{9}{4\,4^n} \longrightarrow 0.$$

Thus no fixed $c > 0$ and finite C can make $A(\eta + th) \ge cA(\eta) - C$ on every centered orthogonal fiber, and the corresponding centered Boltzmann ratio is unbounded. This is not marginal divergence: translation by three time sites preserves the action and the orthogonal fiber while sending h to $-h$, so the fully integrated marginal satisfies

$$m_h(t) = m_h(-t).$$

The result separates two claims that a pointwise proof would conflate. Individual fibers can contain arbitrarily distant points whose action is smaller than at their centered point by an exponentially vanishing ratio, while the annealed marginal remains centered by symmetry. The live gate is an annealed or background-recentered fiber estimate; marginal evenness alone does not bound its second moment.

The successor obstruction shows that recentering is not optional. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_CONDITIONAL_MASS_ESCAPE_OBSTRUCTION_V1` uses the subsequence

$$\eta_m = 4m \log 2\,(-1, -1, 1, -3, 3, 1), \qquad m \geq 2,$$

and writes the fiber coordinate as $t = u \log 2$. If q_m is the normalized conditional Gibbs density on this fiber, then exact action barriers and a certified distant well give

$$q_m\{u \geq -m\} \leq 2^{-m}, \qquad \mathbb{E}_{q_m}[u^2] \geq m^2(1 - 2^{-m}).$$

Moreover every global fiber minimizer satisfies $u_m^* < -m$. Thus no bound on the raw conditional second moment can be uniform in the orthogonal background, even at this fixed volume. This still does not prove divergence of the integrated marginal: the exceptional backgrounds can have vanishing marginal weight. The remaining problem has two distinct terms, a recentered conditional width and the annealed second moment of the moving center.

The same runaway family has a sharply different behavior after recentering. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_RUNAWAY_FIBER_WIDTH_BOUND_V1` writes $R = 2^{4m} \geq 256$, $z = 2^u$, and differentiates the complete six-site fiber action exactly. Its curvature is

$$A_m''(u) = (\log 2)^2 K_m(z), \qquad K_m(z) \geq \frac{115}{4} \quad (m \geq 2, z > 0).$$

The spatial factor and $\lambda = 2/5$ give conditional inverse temperature $216/\lambda^2 = 1350$. The one-dimensional Brascamp–Lieb inequality therefore yields

$$\mathrm{Var}_{q_m}(u) \leq \frac{2}{77625(\log 2)^2} \leq \frac{8}{77625},$$

where the last step uses $\log 2 = \int_1^2 dx/x \geq 1/2$. If the conditional mean were at least $-m/2$, the predecessor bound $q_m\{u < -m\} \geq 1 - 2^{-m}$ would force the variance to be at least $(1 - 2^{-m})m^2/4 \geq 3/4$, a contradiction. Hence

$$\mathbb{E}_{q_m}[u] < -\frac{m}{2}.$$

Thus the certified raw-moment growth on this family is caused by a moving conditional center, not a widening conditional packet. This is uniform only in m on the declared exact family. It neither proves an all-background recentered-width estimate nor controls the Gibbs-weighted distribution of centers.

A complementary theorem now reaches the correct volume scaling on a larger, but still nongeneric, carrier. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_SEPARABLE_LOWEST_MODE_CURVATURE_V1` considers every separable background $\psi(t, x) = a_t + b_x$ on the periodic L^4 lattice, $L \geq 4$. For any phase of the real lowest temporal mode, $h_t = \cos(2\pi t/L + \alpha)$, and $\omega_L = 4\sin^2(\pi/L)$, it proves

$$\operatorname{Hess} A_\psi[h,h] \ge \frac{N\omega_L^2}{3}, \qquad \operatorname{Var}(t \mid a+b) \le \frac{3}{N\omega_L^2}.$$

The free curvature is $N\omega_L^2/2$, so at least two thirds survives. The proof reduces each temporal line exactly, uses $2\cosh(2u) - 2\cosh u \ge 3u^2$, and pairs every spatial edge as $e^\delta + e^{-\delta} - 2 \ge 0$. This is the first interacting conditional bound in this line with the desired $N^{-1}\omega_L^{-2}$ scaling across all volumes. It is not yet a Gibbs estimate: separable backgrounds form only a special sector. Indeed the same certificate gives an exact centered, mode-orthogonal, correlated $L = 4$ fixture whose spatial correlation remainder is $-456623975/262144$. Thus the next missing lemma must absorb the signed correlation term rather than assume it is positive. All-background width, the annealed center moment, the normalized marginal, and interacting H^{-1} tightness remain open.

The signed correlation term can in fact be absorbed. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_ALL_BACKGROUND_LOWEST_MODE_CURVATURE_V1` groups each temporal edge with each of the six incident spatial edges. Its exact plaquette inequality retains one fifth of the two endpoint functions $f(z) = z^2 + z^{-2} - z - z^{-1}$. Summing the six edge groups and completing the time-cycle squares gives, for every background and every real phase of every lowest axial mode,

$$\operatorname{Hess} A_\psi[h,h] \ge \frac{2}{9} N\omega_L^2, \qquad \operatorname{Var}(t \mid \eta) \le \frac{9}{2N\omega_L^2}.$$

The first inequality retains four ninths of the free curvature and is uniform in the orthogonal background, axis, phase, coupling, and integer $L \ge 4$. This proves the all-background recentered-width half of the split. It does not bound the Gibbs-weighted motion of the unique fiber centers. Consequently the integrated lowest-mode moment and the interacting H^{-1} estimate still remain open, but their missing ledger has narrowed to the annealed center term followed by Fourier-shell summation.

Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_ANNEALED_CENTER_SCORE_REDUCTION_V1` makes that center term explicit. If $V_\eta(t) = S_\lambda(\eta + th)$, $\kappa_L = (2/9)N\omega_L^2$, and $m(\eta)$ is the unique zero of V_η', then strong monotonicity gives

$$m(\eta)^2 \le \frac{V_\eta'(0)^2}{\kappa_L^2}.$$

Integration by parts also puts the conditional mean within squared distance κ_L^{-1} of m. Hence the single sufficient estimate

$$\mathbb{E}_\nu[V_\eta'(0)^2] \le C_s N\omega_L^2$$

would imply

$$\mathbb{E}_\mu[t^2] \le \frac{27 + 81C_s}{2N\omega_L^2}.$$

An exact shifted-Gaussian family proves that curvature, inversion symmetry, and extensive virial/action control alone cannot imply this score estimate. Deterministic $L = 4, 6$ Metropolis diagnostics give the scaled center moments $0.0359, 0.0400$ and scaled score moments $0.0101, 0.0122$, but these are only finite-volume observations.

The first perturbative test of the sufficient estimate produces a sharper method boundary. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_CUBIC_SCORE_LOG_OBSTRUCTION_V1` derives the exact lattice cubic vertex

$$V_3(a,b,c) = a^2 + b^2 + c^2 - 2(ab + ac + bc) = -16\mathcal{A}^2,$$

where a, b, c are the three lattice dispersions and $\mathcal{A}$ is the Heron triangle area of their square roots. Thus every leg carries the expected soft factor. Nevertheless the free orthogonal-background coefficient C_L in

$$\mathrm{Var}_0[V_\eta'(0)] = \lambda^2 N \omega_L^2 C_L + O(\lambda^3)$$

obeys the exact lower bound

$$C_L \geq \frac{J_L}{4665600}, \qquad J_L = \#\{j \geq 0 : 16\, 2^j \leq L\}.$$

It is therefore logarithmically unbounded. This obstructs a fixed-bare- coupling proof that bounds perturbative coefficients separately. It does not prove divergence of the resummed Gibbs score or the actual moment: running coupling, field-strength renormalization, or a nonperturbative multiscale cancellation may still supply the missing estimate. The live gate has accordingly moved from a bare score bound to an explicit scale setting, its finite-volume Ward/RG normalization, and then a nonperturbative annealed estimate. At fixed physical torus size the logarithm is read along the ultraviolet refinement; at fixed lattice spacing it accompanies the soft lowest-momentum/large-volume limit.

Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_SCORE_RG_MATCHING_V1` now computes that normalization. The lattice annulus has exact logarithmic residue

$$C_L = \frac{5}{16\pi^2} \log L + O(1).$$

The continuum source defines its MS coupling with an explicit 4π factor. For the unrescaled perfect-square coefficient g used by the lattice,

$$\beta_g = -\frac{5}{16\pi^2} g^3 - \frac{15}{128\pi^4} g^5 + O(g^7).$$

On a fixed-physical-volume refinement, where the cutoff scale grows proportionally to L, asymptotic freedom therefore gives

$$g_L^2 \log L \longrightarrow \frac{8\pi^2}{5}, \qquad g_L^2 C_L \longrightarrow \frac{1}{2}.$$

The leading logarithmic nonuniformity is thus restored to a finite nonzero coefficient on the tuned trajectory. The Ward relation $\gamma_\sigma = \beta_g/g$ also makes $g\sigma$, the exponential lattice field, RG invariant. Neither fact is an all-order lattice score bound or a nonperturbative Gibbs theorem. Fixed lattice spacing and growing physical volume is a distinct limit: there the cutoff coupling is not tuned with L, so this cancellation cannot be imported. The next gate is the finite-lattice Ward identity for the conditional score composite. The ordinary equation-of-motion identity $\mathbb{E}[(D_h S)^2] = \mathbb{E}[D_h^2 S]$ is exact, but evaluates the score at the sampled field rather than at zero fiber coordinate. The same shifted- Gaussian fixture has full-score variance (2) and zero-fiber-score variance (100), so no general transfer exists. A BT-specific projected identity, followed by all-order and nonperturbative control on the declared scale-setting branch, remains necessary.

The constrained version of that proposed identity has now been classified. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_ZERO_FIBER_WARD_WEIGHT_OBSTRUCTION_V1` uses the exact disintegration $\nu(d\eta) = Z_\eta d\eta/Z$ and $q_\eta(t) = e^{-V_\eta(t)}/Z_\eta$. If ρ is the integrated mode density, $s_\eta = V_\eta'(0)$, and $H_\eta = V_\eta''(0)$, then

$$\rho''(0) = \mathbb{E}_\nu[q_\eta(0)(s_\eta^2 - H_\eta)].$$

The background measure constrained to $t = 0$ satisfies

$$\frac{d\mu_0}{d\nu} = \frac{q_\eta(0)}{\rho(0)}, \qquad \mathbb{E}_\nu[s_\eta^2] = \rho(0)\mathbb{E}_{\mu_0}\left[\frac{s_\eta^2}{q_\eta(0)}\right].$$

Thus every local constrained Ward insertion carries the wrong background weight for the annealed-center target. This weight cannot be removed by a pointwise BT bound. On the already certified fixed-6^4 runaway family, strict convexity, a unique mode below $-m$, and $q_m^{(u)}\{u \geq -m\} \leq 2^{-m}$ imply

$$q_m^{(u)}(0) \leq \frac{2^{-m}}{m}.$$

Here $q_m^{(u)}$ is the density with respect to du; since $t = u \log 2$, $q_{\eta_m}^{(t)}(0) = q_m^{(u)}(0)/\log 2$. Hence no positive background-uniform lower bound on $q_\eta(0)$ exists even inside the BT lattice. This obstructs a purely local constrained-Ward transfer as formulated; it does not obstruct an annealed theorem, because the same runaway backgrounds may be exponentially rare under ν. The next viable estimate must retain their Gibbs weight and jointly control center displacement, score size, and inverse conditional density.

The next fixed-order calculation exposes a separate power-counting barrier. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_QUARTIC_SCORE_POWER_OBSTRUCTION_V1` extracts the quartic zero-fiber-score polynomial directly from the positive exponential lattice action. If

$$d_\delta(k) = e^{ik\cdot\delta} - 1, \qquad B_j(k_1, \ldots, k_j) = \sum_\delta \prod_{a=1}^{j} d_\delta(k_a),$$

then its fully symmetric four-leg kernel is

$$K_4 = \frac{1}{24}\left[\sum_i B_1(k_i)B_3(k_{\widehat{i}}) + B_2(k_1, k_2)B_2(k_3, k_4) + \text{two pairings}\right].$$

At $p = (\varepsilon, 0, 0, 0)$, $q = (\pi/2, 0, 0, 0)$, $r = (0, \pi/2, 0, 0)$, and $s = -p-q-r$, exact Gaussian-rational dual-number arithmetic gives

$$K_4(0, q, r, -q-r) = 0, \qquad \partial_\varepsilon K_4|_{\varepsilon=0} = -\frac{1}{3}.$$

The external soft degree is therefore genuinely linear. Fixed disjoint ultraviolet boxes around this fixture and positivity of the corresponding third-Wiener-chaos block imply, for all sufficiently large L,

$$\frac{\mathbb{E}_0[Q_L^2]}{N\omega_L^2} \geq \frac{c}{\omega_L} \geq c'L^2.$$

Thus the isolated quartic-score square cannot be bounded uniformly term by term. On the tuned refinement trajectory g_L^4 is only of order $(\log L)^{-2}$, so running of the coupling does not by itself suppress this power. This is not a divergence theorem for the interacting score. The complete order-g^4 coefficient also contains signed measure, normalization, projection, and higher Taylor terms, and those terms may cancel the displayed L^2 sector. The immediate gate is to put every order-g^4

contribution in one Wiener-chaos basis and decide that exact cancellation before attempting a whole-composite or nonperturbative annealed bound.

That complete fixed-order bookkeeping is now available. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_UV_NONCANCELLATION_V1` first integrates the free lowest-mode fiber. With $A = D_h S_1$, $B = D_h S_2$, $C = D_h S_3$ and effective background-action coefficients

$$W_1 = \mathbb{E}_T[S_1], \qquad W_2 = \mathbb{E}_T[S_2] - \frac{1}{2}\operatorname{Var}_T(S_1),$$

normalization of the background measure gives the exact coefficient

$$M_4 = \mathbb{E}_0\left[B^2 + 2AC - 2ABW_1 + A^2\left(\frac{1}{2}W_1^2 - W_2 - \mathbb{E}_0\left[\frac{1}{2}W_1^2 - W_2\right]\right)\right].$$

Equivalently, this is obtained by multiplying the score by $\sqrt{d\nu_g/d\nu_0}$ and expanding its free squared norm. This second form checks that the Gibbs-density and partition-function terms have not been omitted.

On fixed inversion-symmetric ultraviolet carriers, the formula decides the cancellation question. The exact cubic score is $O(p^2)$, the quartic score is $O(p)$ with the nonzero derivative above, and the quintic score is at least $O(p)$ because each field leg occurs in a directed-edge factor. The effective density introduces no inverse external power. Hence B^2 has a positive p^2 third-chaos coefficient, while the apparent p^3 cross terms vanish by real-cosine parity and begin at p^4. No local, fixed-UV, or diagram-by-diagram cancellation of the power sector is possible.

The unrestricted lattice coefficient nevertheless remains open. The fixed- carrier Taylor expansion is not uniform in momentum regions that shrink toward zero with p. Any cancellation must now come from that infrared complement and occur only after summation across regions. The next gate is a polylogarithmic bound on the signed complement, or an exact computation of its $N\omega_p$ coefficient. The UV theorem alone is not an interacting divergence or continuum result.

The signed complement has now been reduced to one kernel. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_CHAOS_GATE_V1` uses the square-root-density normal form to write

$$M_4 = \|D\|_0^2 + 2\langle A, E\rangle_0.$$

For $L \geq 4$, the actual real-cosine-conditioned covariance cannot close the external $\pm p$ momentum of the quadratic score A, so A is centered and belongs entirely to second homogeneous Wiener chaos. Polynomial degree and parity put D in chaoses $1, 3, 5$ and E in chaoses $0, 2, 4, 6, 8$. Orthogonality thus gives the exact identity

$$M_4 = \|D\|_0^2 + 2\langle A, \Pi_2 E\rangle_0.$$

All possible negative cancellation is concentrated in the effective three-leg kernel $\Pi_2 E$; every other contribution is part of the nonnegative norm. Since $\|A\|_0^2 = O(N\omega_p^2 \log L)$, a single estimate

$$\|\Pi_2 E\|_0^2 \leq C_E N\omega_p(1 + \log L)^b$$

for any fixed finite b would make the cross term $o(N\omega_p)$ by Cauchy–Schwarz. Combined with the certified positive UV block of $\|D\|_0^2$, this would prove survival of the full order-g^4 power. The estimate has not yet been proved. Its derivation must Wick-contract the W_1, W_2, and normalization terms together before splitting the resulting kernel into hard, one-soft, and all-soft regions. Ordinary continuum off-shell IR finiteness is not itself this projected lattice-composite bound.

The remaining projection can nevertheless be compressed one step further. Certificate REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_EFFECTIVE_HESSIAN_V1 uses Gaussian integration by parts to represent it by the expected Hessian

$$K_E = \mathbb{E}_0[D^2 E], \qquad \|\Pi_2 E\|_0^2 = \frac{1}{2}\operatorname{Tr}(CK_E CK_E).$$

This expression retains the W_1, W_2, and normalization terms in one product-rule kernel, so they need not be expanded into a separate Wick-diagram inventory. The real-cosine-conditioned covariance is $C = C_0 - R_h$: a translation-invariant bulk minus one rank-one covariance. The bulk cross trace selects transfer $\pm p$; a single rank-one insertion can also sample $\pm 3p$; and the double-rank cross term vanishes for $L \geq 4$. Thus the open calculation is an explicit momentum-diagonal bulk plus a finite-mode correction. The Fourier sums and their weighted norm bound have not been evaluated, so the whole-lattice order-g^4 decision remains open.

There is, however, a normalization subtlety that changes the correct object to estimate. Certificate REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_CONNECTED_NORMALIZATION_V1 proves

$$M_4 = \mathbb{E}_0[B^2 + 2AC - 2ABW_1] + \operatorname{Cov}_0\left(A^2, \frac{1}{2}W_1^2 - W_2\right).$$

In the square-root representation the coefficient multiplying A has mean $-\mathbb{E}_0[W_1^2]/8$, and $\mathbb{E}_0[W_1^2] \geq cN$ on fixed ultraviolet boxes. Its disconnected cross contribution cancels the matching term in $\|D\|_0^2$ only after the two pieces are recombined. Consequently a separate or triangle estimate of that aligned sector is obstructed, although an internal cancellation in the fully combined $\Pi_2 E$ remains possible. The original labeled Wick table concerns the translation-invariant C_0 bulk. Rank-one insertions from $C = C_0 - R_h$ can revive entries forbidden in that bulk table, but a separate exact signed-source audit over every bulk/rank choice proves that the fully conditioned connected formula still has at most two free loop momenta for every $L \geq 4$. Certificate REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_L4_DECISION_V1 then evaluates every zero-, one-, and two-loop contribution exactly on the 4^4 lattice:

$$M_4(4) = -\frac{338835474713437}{204838502400000} < 0.$$

An independent C++ enumeration agrees modulo four 61-bit primes whose product exceeds twice a rigorous bound on the cleared integer difference. Thus this is an exact rational decision, not a numerical sign test, and it refutes the proposed all-volume identity $M_4(L) = 0$. The dominant bulk two-loop sector $(+21.490\ldots)$ nearly cancels the dominant single-rank two-loop sector $(-23.376\ldots)$, leaving the finite negative remainder. Certificate REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_GENERAL_L_TWO_LOOP_V1 now performs the required recombination for every integer $L \geq 5$. Of 96 source-conserving oriented topology flows, 48 contain an identically zero propagator; the rest reduce to 21 common integrands, five of which cancel exactly. If X_L is the cubic one-loop bubble and Y_L the quartic tadpole, all Y_L^2 and $X_L Y_L$ contributions cancel between fixed-p bulk lines and rank-one covariance terms. The conditioning-scale remainder is

$$R_L = \frac{162 X_L^2}{N\omega_p^2} = O\big((1 + \log L)^2\big).$$

Consequently $g_L^4 R_L$ is bounded on the already certified tuned asymptotically free refinement branch. This closes only the factorized conditioning sector. Certificate REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_SEVEN_KERNEL_REDUCTION_V1 pairs the remaining fourteen entries into seven exact inversion pairs. It also proves

$$\frac{\omega_k\omega_r}{6} \le K_4(k,-k,r,-r) \le \frac{19}{6}\omega_k\omega_r$$

and factorizes one pair as a strictly negative nested tadpole carrier T_L with

$$T_L \le -\frac{N-1}{4N\omega_p} \le -\frac{624}{625}\frac{L^2}{16\pi^2}, \qquad L \ge 5.$$

Thus the tuned running coupling cannot control the order-g^4 terms separately. The successor certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_SUBPOWER_PAIR_BOUNDS_V1` proves explicit $O(\log^2 L)$ bounds for inversion pairs 1, 2, and 5. They are $o(N\omega_p)$, become uniformly bounded after multiplication by the tuned g_L^4, and cannot contribute to a nonzero power coefficient. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_LINEAR_PAIR_BOUNDS_V1` then proves pair 3 is $O(L)$ and pair 6 is $O(L\log L)$. They are also $o(N\omega_p)$, without needing a parity cancellation. Cancellation of the common $N\omega_p$ coefficient is therefore confined to negative pair 4 and positive pair 7. Their combination and the lower-loop completion are decided below; uniform tuned-coupling control, the nonperturbative score, and the interacting H^{-1} moment remain open. The next certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_TWO_PAIR_COEFFICIENT_NORMAL_FORM_V1` proves that both surviving normalized limits exist. Pair 4 becomes an absolutely convergent integer sum times the four-dimensional lattice Green constant and obeys $c_4 < -0.01613$. Pair 7 becomes one positive finite Brillouin-zone integral whose six-term soft derivative collapses to three terms. Certificate `REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_TWO_PAIR_NONCANCELLATION_V1` then proves the sharp lattice inequality

$$\|\omega(q)\sin q + \omega(r)\sin r + \omega(s)\sin s\|^2 \le 9\omega(q)\omega(r)\omega(s), \qquad q+r+s=0.$$

Axis symmetry reduces c_7 to at most three times a three-Green convolution. A 96-cell-per-axis rational box cover, with a 16^4 refinement of only the singular origin box and exact outward rounding, gives $c_7 < 0.016103194 < 0.01613$. Therefore $c_4 + c_7 < 0$: the leading two-loop power coefficient does not cancel. Certificate

`REVERSE_PHYSICS_BT_EUCLIDEAN_COMPLETE_G4_LOWER_LOOP_BOUNDS_V1`

then closes the omitted conditioned sectors for every $L \ge 7$. Exact affine enumeration leaves ten zero-loop rows and twenty-seven one-loop rows. The zero-loop rows combine to $M_{4,0}(L) = L^{-4}Z(\omega_p)$, with

$$\lim_{L\to\infty} M_{4,0}(L) = \frac{111}{32\pi^4} > 0.$$

Selectable cubic bounds, all-leg quartic and quintic bounds, and a common five-centre lattice-shell estimate give

$$|M_{4,1}(L)| \le \frac{900315}{4}\pi^2\big(1+\log\lfloor L/2\rfloor\big).$$

The four rows that appear power-capable under the generic bounds close by the exact extra-soft identity

$$K_3(-2p,p,p) = -\frac{2}{3}\cos^2(p_1/2)\omega_p^3.$$

Thus both lower-loop sectors are $o(N\omega_p)$, and the complete perturbative coefficient M_4 inherits the strictly negative leading $N\omega_p$ coefficient of its two-loop sector. This is a fixed-order coefficient theorem. Uniformity of the perturbation series at tuned coupling, the nonperturbative score, and the actual interacting H^{-1} moment remain open.

Certificate

`REVERSE_PHYSICS_BT_EUCLIDEAN_TUNED_REMAINDER_COMPENSATION_V1`

turns that open uniformity issue into an exact obstruction. Let $F_L(g) = \mathbb{E}_{\nu_{L,g}}[s_{L,g}^2] \geq 0$ be the full annealed zero-fiber-score moment. Background inversion makes the cubic norm coefficient vanish exactly. On the tuned branch the quadratic contribution is $O(1)$, whereas

$$g_L^4 M_4 \sim c_* g_L^4 N\omega_p \longrightarrow -\infty, \qquad c_* < -\frac{13403}{500000000}.$$

Consequently the exact complement

$$Q_L = F_L(g_L) - g_L^2 M_2(L) - g_L^4 M_4(L)$$

must satisfy

$$\liminf_{L\to\infty} \frac{Q_L}{g_L^4 N\omega_p} \geq -c_* > \frac{13403}{500000000}.$$

Thus a uniformly bounded fixed-order remainder, or one that is little-o of $g_L^4 N\omega_p$, is impossible. All-order or nonperturbative compensation is mathematically required. This does not determine whether the full positive moment F_L is bounded or divergent; it says that a proof must estimate the whole resummed score square rather than its orders separately.

A natural nonperturbative shortcut would be a homogeneous radial virial bound. Certificate

`REVERSE_PHYSICS_BT_EUCLIDEAN_RADIAL_CONVEXITY_OBSTRUCTION_V1`

rules out its coefficient-one and convexity forms. On the exact period-six four-torus it constructs a layered field with rational edge weights for which

$$0 < D(\psi) < A(\psi), \qquad \frac{d^2}{d\rho^2} A(\rho\psi)\bigg|_{\rho=1} < 0.$$

The signs follow from direct enumeration of all 6^4 sites and the rational alternating bound $\log(101/100) < 29851/3000000$. Hence radial convexity cannot imply $D \geq A$, and $D \geq A$ itself is false. This strengthens the earlier coefficient-two counterexample without refuting the certified affine virial theorem. At this stage a weaker pointwise inequality $D \geq cA$ with $0 < c < 1$ remained possible.

Certificate

`REVERSE_PHYSICS_BT_EUCLIDEAN_LOG_BUBBLE_VIRIAL_NO_GO_V1`

closes that remaining homogeneous route. It defines one fixed smooth radial profile on a logarithmic annulus in the flat four-torus. With $X = \Delta\psi$ and $Y = |\nabla\psi|^2$, rational polynomial integration gives, after removing the positive factor $2\pi^2$,

$$\frac{1}{2}\int (X+Y)^2 = \frac{19349691}{5173168} > 0, \qquad \int (X+Y)(X+2Y) = -\frac{2896611}{1293292} < 0.$$

Uniform Taylor–Peano expansion transfers these limits to samples on the finite periodic L^4 grids. The sampled fields therefore have $D_L < 0$ for every sufficiently large L. No inequality $D_L \geq cA_L$ with $c \geq 0$ can hold on all finite lattices. The result obstructs a pointwise proof method; it does not decide a Gibbs-weighted block estimate or the actual interacting H^{-1} moment.

Certificate

`REVERSE_PHYSICS_BT_EUCLIDEAN_LOG_BUBBLE_ENTROPY_SOFT_SCORE_BALANCE_V1` separates bubble probability from the target observable. An optimized reflected wall has

$$A_{\mathrm{red}} = \frac{1965963925}{733296564} < \frac{16}{5}, \qquad D_{\mathrm{red}} = -\frac{2157475}{16665831} < 0.$$

On the tuned branch its bare activity exponent is $5A_{\mathrm{red}}/4 < 4$. Four-dimensional position entropy therefore defeats an action-only Peierls estimate. Radial reflection, however, makes the lowest-mode insertion quadratic in the bubble radius. Its square cancels the position count, and the dilute score-weighted activity tends to zero. The surviving route is an observable-weighted block estimate that preserves this soft factor. Fluctuation neighborhoods, interacting multibubbles, and the actual Gibbs score remain open.

The next exact reduction uses the full lowest axial phase plane rather than a single cosine. For

$$E_p = \mathrm{span}\{\cos(2\pi x_1/L), \sin(2\pi x_1/L)\},$$

lattice translations rotate E_p and preserve its orthogonal complement. Consequently the exact background marginal is translation invariant. The certificate
`REVERSE_PHYSICS_BT_EUCLIDEAN_FULL_PHASE_WEIGHTED_CURRENT_GATE_V2`
combines the all-phase curvature theorem with the canonical oriented current

$$J_{x,i} = r_x e^{\psi_{x+e_i} - \psi_x} - r_{x+e_i} e^{\psi_x - \psi_{x+e_i}}.$$

The two real zero-fiber scores are exactly its Fourier divergence. For the lowest axial momentum p,

$$s_{\cos}^2 + s_{\sin}^2 = \frac{\omega_p}{g^2} |\widehat{J}_1(p)|^2.$$

Thus the center theorem is reduced to the single translation-invariant susceptibility estimate

$$\mathbb{E}_{\nu_p} |\widehat{J}_1(p)|^2 \le C_J g^2 N \omega_p.$$

The V2 certificate corrects the scope of the first rational fixture: that row was not in $E_p^{\perp}$. A replacement two-axis rational 4^4 field is mean zero and exactly orthogonal to both lowest phases, yet its current has

$$\sum_x J_{x,1} = -24.$$

Thus the canonical current is not an ordinary periodic gradient even on the required slice. It does, however, have the exact weighted normal form

$$J_{xy} = \Omega_x \Omega_y (u_x - u_y), \qquad u_x = \frac{r_x}{\Omega_x^2}, \qquad \sum_x \Omega_x^3 u_x = 0.$$

The second soft factor is therefore a stationary nonlinear random-conductance flux-corrector problem. Multiplication by the fluctuating conductance mixes Fourier momenta, so the weighted-gradient identity alone is not the susceptibility estimate. In mean-log gauge the exact split

$$J_{x,i} = (u_x - u_{x+e_i}) + K_{x,i}, \qquad K_{x,i} = (\Omega_x \Omega_{x+e_i} - 1)(u_x - u_{x+e_i})$$

reduces it to two explicit subgates:

$$\mathbb{E}_{\nu_p} |\widehat{u}(p)|^2 \le C_u g^2 N, \qquad \mathbb{E}_{\nu_p} |\widehat{K}_1(p)|^2 \le C_K g^2 N \omega_p.$$

The slice-valid fixture's entire nonzero current zero mode lies in K, so the corrector gate cannot be omitted. Both estimates, the interacting H^{-1} bound, and the continuum limit remain open.

The next certificate

`REVERSE_PHYSICS_BT_EUCLIDEAN_FLUX_CORRECTOR_POINTWISE_ENERGY_NO_GO_V1`

closes a deterministic shortcut to the corrector estimate. An exact $E_p^{\perp}$ localized slab exists on every L^4 torus with $4 \mid L$. Its action and weighted Dirichlet energy are

$$A_L = \frac{837}{128}L^3, \qquad E_{\mathrm{dir,L}} = \frac{290423}{1024}L^3,$$

while elementary exact Fourier bounds give

$$|\widehat{K}_1(p_L)| \geq \frac{3}{16}L^3 \qquad (L \geq 300,\ 4 \mid L).$$

Consequently, even the generously volume-losing comparison satisfies

$$\frac{|\widehat{K}_1(p_L)|^2}{N\omega_p A_L} \geq \frac{49}{360096}L.$$

The corresponding ratios with $E_{\mathrm{dir,L}}$ or $A_L + E_{\mathrm{dir,L}}$ also diverge linearly. Thus no pointwise action/flux-energy argument can manufacture the missing external momentum. This is not a Gibbs counterexample: Gibbs rarity and correlation of the slab environments may still prove the averaged hyperuniformity estimate. The susceptibility, interacting H^{-1} bound, and continuum limit remain open.

5.6 A finite BV contraction does not freeze the quantum theory

The energy-two free BV block provides another finite exact case. The certificate `FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1` verifies integer matrices for a strong deformation retract and checks the required identities in PRA. Because the contraction and projection are supplied explicitly, no Hahn–Banach extension or abstract complement selection is needed for this fixture.

The result is intentionally small. It does not verify all energies, the field-derived infinite complex, the Fock lift, a Green operator, a state, or a renormalization construction. It does not pass the classical import freeze gate for the quantum programme. The dependency tag is exactly `LOCAL-ALGEBRAIC`. A bounded exact certificate can remove foundational ambiguity from one algebraic step while leaving every analytic and causal obligation open.

This is the intended interaction with the Weyl BV–BFV programme. The classical complex is authoritative, but its import into quantum work must be checked object by object. Local counterterm and anomaly classification must precede coefficient calculations; restoration of the local quantum master equation must precede residual transfer. No reduced-mode or Euclidean result may be used as evidence for a Lorentzian causal claim. The reverse- foundational atlas makes these dependency cuts visible before a calculation is promoted.

6 What the case studies teach

The examples support several general lessons. They are methodological conclusions from the displayed certificates, not universal theorems about all physical theories.

6.1 The first cost of infinity is often completion, not Choice

In explicit separable carriers, countable infinity can enter through a named index set and canonical completion without any selection from an arbitrary family. The finite-to-infinite boundary still adds real mathematical content: summability, completion, operator domains, and limits. It simply need not add Countable Choice in the same step. "Infinite" and "choice-dependent" are not synonyms.

6.2 Physical input often acts as data, not as a foundational axiom

A mode grading, detector projection, boundary condition, or convergence modulus can remove an existential search. The physical postulate does not thereby become a set-theoretic axiom. It supplies a witness inside the chosen foundation. This is one of the most promising ways physics can lower the mathematical strength of a formulation: replace arbitrary existence with canonical physical data.

6.3 Representation sensitivity is scientific content

The RCA_0 wave theorem works because fast Cauchy rates and rational polygonal dense data are part of the representation. That is not a defect to hide. It tells us precisely which information makes the evolution effective and weakly formalizable. A later theorem may show that another natural representation is equivalent over the same base, or it may expose a genuine difference. Until then, representation invariance is an open obligation.

6.4 Positive structure migrates rather than disappears

Krein geometry removes positive definiteness from the distinguished pairing but restores a positive companion topology through a fundamental symmetry. Algebra-first formulations remove state vectors from the primitive vocabulary but recover Hilbert representations through GNS. Operational reconstructions remove Hilbert space from the primitive postulates but recover it from continuity, composition, and informational principles. In each case the assumption load moves. An honest analysis follows it.

6.5 The difficult cells are interfaces

Finite algebra, explicit coordinate carriers, and named states are relatively tractable. The sparse region begins where several interfaces must be crossed at once: weak or constructive analysis plus variable-coefficient Lorentzian PDE; internal topos structure plus external experimental probability; gauge cohomology plus microlocal renormalization; or an operational reconstruction plus infinite-dimensional continuum limits. These gaps are not random. They mark where entire mathematical toolchains must be retyped.

7 A research protocol

The matrix becomes scientifically useful only if filling a cell means more than attaching a citation. We propose the following protocol.

7.1 Declare the target before auditing the proof

Every project begins with one obligation, one carrier, one representation, and one dependency boundary. "Wave existence" should say whether the solution is a Cauchy name, a Sobolev function, a distribution, or a smooth field. "State" should say algebraic state, normal density state, KMS state, Hadamard state, or physical selector. "Causality" should say graph-step support, finite propagation, or advanced/retarded Green support.

7.2 Build a proof ledger

The displayed proof is factored into individual dependencies: finite algebra, completion, compactness, extension, subsequence selection, spectral calculus, operator-domain closure, Green existence, support, duality, and so on. Each dependency gets one of the seven relation types. A proof use is never promoted to necessity.

7.3 Search for avoidance before proving a lower bound

Before attempting a reversal, ask whether the physical representation supplies a witness or canonical enumeration. An explicit diagonal operator may avoid a general spectral theorem. A named mode basis may avoid a basis-selection principle. Chiral coordinates may turn a PDE evolution into translation. This often produces a useful upper bound quickly and exposes the precise residue for a later lower bound.

7.4 Separate producer and checker

Exact finite claims should carry machine-readable witnesses and an independent checker that does not merely rerun the producer. Analytic claims should pin the theorem statement, hypotheses, representation, and source. Every result records what would falsify it, what was not checked, and which stronger claim it does not establish. Exact and numerical evidence remain distinct types.

7.5 Promote only through typed gates

For quantum Weyl work, the dependency tags

`LOCAL-ALGEBRAIC`, `EUCLIDEAN-SPECTRAL`, `REDUCED-MODE`, `LORENTZIAN-CAUSAL`

are noninterchangeable. Likewise the lifecycle states

`CLASSIFIED`, `COEFFICIENT_COMPUTED`, `QME_RESTORED`, `RESIDUAL_TRANSFERRED`,
`LORENTZIAN_CERTIFIED`

must not be collapsed. The atlas should show missing typed edges, not draw an unlabelled arrow around them.

7.6 Calibrate maturity without turning absence into failure

The assembly layer therefore reports model-specific maturity rails independently. Obligation coverage, cross-cell composition, prediction derivation, observable identification, numerical reproducibility, empirical comparison, and robustness are related, but they are not one pass/fail bit. In particular, the classical-standard mixed-carrier reference has direct records for all sixteen obligations in the current atlas, while only two of seven required cross-cell joins are certified. The remaining joins are unassessed, not refuted. Downstream prediction and comparison records are likewise missing or premature, not failed empirical tests.

For exposition, those envelopes are now presented as nine recognizable research-programme lenses rather than anonymous carrier maxima: mainstream GR/QFT, algebraic QFT, finite/discrete exact models, Bateman–Turok, reverse mathematics, Mannheim conformal gravity, the present pure-Weyl BV–BFV programme, constructive/computable physics, and topos/internal quantum foundations. Each lens states its central question, intellectual lineage, signature ideas, the narrower region actually sampled by the atlas, and a scope warning. This organization is illustrative rather than sociological: it neither exhausts a research community nor implies that a named researcher endorses every cell selected by the deterministic coverage rule.

The added numerical rail matters for the BT lattice prototype: independent algorithms can reproduce a finite-volume calculation coarsely while the empirical-comparison and robustness rails remain empty. Likewise, the certified Euclidean/Krein carrier non-identity is an explicit scoped incompatibility, whereas an unconstructed continuum or analytic bridge remains missing rather than refuted.

The first end-to-end control now fixes one model identity and one observational sector: four-dimensional standard general relativity in the static, spherically symmetric, asymptotically flat solar vacuum exterior. Within that scope, an exact rational certificate connects the vacuum Einstein equations to the Schwarzschild exterior, translates the solution to isotropic weak-field form, derives the PPN value $\gamma = 1$, and obtains the first-order null-delay coefficient $1 + \gamma = 2$. A separately typed literature bridge imports the Cassini operational parameter and the published estimate $\gamma = 1 + (2.1 \pm 2.3) \times 10^{-5}$ [3]. Thus the exact prediction $\gamma - 1 = 0$ lies inside the displayed reported band. This is a bounded prediction assembly, not a raw-data or likelihood reanalysis: its applicability mask requires three of the atlas's sixteen obligations, touches two more without requiring them, and places the remaining eleven outside this calculation.

A second bounded assembly gives a more informative mixed outcome. It holds fixed the Mannheim–Kazanas NGC 3198 thin-disk model used by Mannheim and O'Brien, joins the Weyl action to the locally certified static vacuum family and weak-field circular-orbit law, imports the published disk formula and parameter row, and evaluates the displayed prediction independently. At the published outer radius, the predicted speed is $153.9\,\mathrm{km\,s^{-1}}$, compared with $149.6\,\mathrm{km\,s^{-1}}$ reconstructed from the paper's endpoint acceleration: a 2.893% residual, inside our declared 5% coarse audit gate. With no parameter refitted, a cross-check against 39 later SPARC points inside the same radius gives an unweighted RMS residual of $4.538\,\mathrm{km\,s^{-1}}$, inside a declared $5\,\mathrm{km\,s^{-1}}$ coarse shape gate, but reduced $\chi^2 = 5.592$ using the tabulated random errors alone, outside the declared $\chi^2_\nu \leq 2$ gate. We therefore retain the result as a partial assembly with a failed uncertainty-sensitive comparison, not as empirical support. SPARC is a later, nonidentical photometric reduction, and the massive-tracer matter-coupling assumption remains an explicit boundary; Appendix A.6 records the full typed chain.

That no-refit audit is now supplemented by a common-protocol control. We fit the same 39 velocities and the same analytic stellar/gas geometry with three families: Newtonian baryons alone, standard weak-field GR with an NFW dark halo, and Mannheim conformal gravity. Every family fits one stellar-mass scale; NFW additionally fits halo speed and concentration. The resulting reduced χ^2 values are 128.72, 0.965, and 3.202, respectively. Only GR+NFW passes the declared $\chi^2_\nu \leq 2$ random-error gate, and it remains first after AICc penalizes its two extra parameters. Mannheim has the smaller *unweighted* RMS (4.694 versus $5.148\,\mathrm{km\,s^{-1}}$), which illustrates why visual or unweighted agreement must not replace an uncertainty-sensitive comparison. This is one galaxy with random errors only, a shared analytic disk rather than the full SPARC numerical baryonic model, and no halo concentration–mass prior. It selects neither a complete theory nor a population-level winner; the generated appendix records the protocol, fitted parameters, metrics, and boundary.

The broader calibration ledger retains four primary-source records across solar-system propagation, compact binaries, and gravitational waves: Cassini, the Double Pulsar timing test, the GWTC–3 general-relativity analyses, and the GW170817 multimessenger propagation bound [3, 13, 14, 1].

These controls are not cube-selected Weyl-gravity assemblies, do not cover the galactic or cosmological benchmark families in the present ledger, do not establish that general relativity is complete, and transfer no observational support to a Weyl-gravity prototype. Appendix A.6 gives the model-scoped chain, prototype envelopes, and calibration table.

8 Near-term research programme

The most valuable next problems are not simply the emptiest cells. They are cells for which one can isolate a small theorem that connects two mature literatures.

8.1 Compare the represented union with the full LF topology

The finite localized coefficient identity, its named H^2 completion, the fixed-support translator, and the support-indexed name equivalence are now established. The next target is the genuinely topological question: compare the quotient topology induced by the declared representation with the full locally convex strict inductive-limit topology, identifying which universal properties and bounded-set statements survive over a weak base. This must remain separate from H2 surjectivity, arbitrary-distribution uniqueness, and causal propagation.

8.2 The scalar biwave and the Weyl–BV transfer boundary

The flat scalar benchmark now extends from $P = \Box$ to $B = P^2$. On the declared rational code carrier, the canonical maps

$$H_{\mathrm{ret}} = G_{\mathrm{ret}} \circ G_{\mathrm{ret}}, \qquad H_{\mathrm{adv}} = G_{\mathrm{adv}} \circ G_{\mathrm{adv}}$$

satisfy both Green identities, retain the original causal cone by causal transitivity, and obey advanced–retarded adjoint duality. Factoring through $w = P\phi$ exposes two zero-data wave solves and therefore selects the four past-zero Cauchy data of the fourth-order equation. The represented extension uses a supplied source width and finite observation horizon; it does not claim globally bounded energy for the persistent retarded solution. The exact authority is `FOUNDATIONAL_SCALAR_MINKOWSKI_BIWAVE_GREEN_V1`.

This closes the scalar calculation but not the theory transfer. The generated dependency delta names sixteen gates between this formula and a full Lorentzian Weyl BV propagator. In particular, tensor and graded carriers, gauge fixing, constraint propagation, curved lower-order estimates, degreewise inverse identities, BRST compatibility, microlocal control, and a passed classical import freeze remain independent requirements. The corpus now contains a portable Bach-flat strict minimal unary complex and an exact arity-two master-identity receiver: all 18 typed $q_1 q_2$ channels and 51 composable paths vanish, with every unary and ordered bilinear component used and representative sign mutations detected (`STRICT_PORTABLE_LOCAL_Q1_AST_V1`, `STRICT_LOCAL_Q1_Q2_IDENTITY_V1`, `STRICT_MINIMAL_BV_CYCLIC_SIGN_RECONCILIATION_V1`, `CLASSICAL_IMPORT_GATE_V5_RECONCILIATION`). Adding the canonical odd pairing revealed 540 exact non-Bach cyclicity defects in the source receiver convention. The involution $(c^*, \omega^*) \mapsto (-c^*, -\omega^*)$ changes exactly two unary and four ordered bilinear rows, preserves $q_1^2 = 0$ and the full $q_1 q_2$ identity by conjugation, and reduces the defect to zero on all 932 expanded non-Bach coefficients. The Bach unary and cubic sectors are respectively the second and third variations of the pinned local Weyl action, so the rank-thirty minimal pairing is cyclic modulo compact-support boundary terms. This repairs a local-algebraic layer only; the common local D action, the nonminimal/auxiliary/residual pairing extension, full support-local contraction and accepted common snapshot hashes remain open. The corpus also answers a sharper causal question: the sign repair does not invalidate the existing 386-row strict pure-Weyl Green-homotopy architecture (`STRICT_386_CAUSAL_SIGN_TRANSPORT_V1`, `FOUNDATIONAL_LORENTZIAN_WEYL_BV_COMPLETION_ATLAS_V4`). The minimal generators have dimensions $5 + 10 + 10 + 5$, exactly the causal endpoint profile. In the certified $356 + 30$ algebraic/endpoint splitting, extend the repair as

$$T_{386} = I_{356} \oplus \mathrm{diag}(I_5, I_{10}, I_{10}, -I_5).$$

Then $Q' = T_{386}QT_{386}$ and $\Lambda'_\pm = T_{386}\Lambda_\pm T_{386}$ retain nilpotency, both Green-homotopy identities, causal support and advanced/retarded orientation; the adjoint relation is retained with the transported pairing. This finite sign wrapper is formalizable in PRA and introduces neither Choice nor an infinite selection, but that calibration does not apply automatically to the imported analytic Green theorem. More importantly, the bridge is exact only at the level of types, dimensions and complex roles.

That last boundary has now been resolved for the minimal unary operator, but not for the full paired carrier (`STRICT_386_ENDPOINT_Q1_CONTENT_BRIDGE_V1`, `FOUNDATIONAL_LORENTZIAN_WEYL_BV_COMPLETION_ATLAS_V5`). An exact rational change of ghost, metric, equation and identity coordinates identifies all five gauge-arrow tables, all 700 independent Bach four-jet input columns and all five Noether-arrow tables. Thus all 80 unary multiindex tables agree and the common Gate-coordinate q_1 has 619 nonzero coefficients. The Bach comparison uses a 700-column coordinate-to-covariant witness satisfying 490,000 exact triangular equations; it is not a sampled component check. The apparent endpoint mismatch is instead the exact suspension character of the pairing convention (`STRICT_386_SUSPENDED_ADJOINT_BRIDGE_V1`, `FOUNDATIONAL_LORENTZIAN_WEYL_BV_COMPLETION_ATLAS_V6`). Reconstructing the endpoint DeWitt/ghost pairing from its 54 nonzero ordered entries before Gate pullback gives $T^{\sharp_G} = \mathrm{diag}(-I_5, I_{10}, I_{10}, I_5)$ and hence

$$R = T^{\sharp_G}T = \mathrm{diag}(-I_5, I_{10}, I_{10}, -I_5).$$

With the suspended adjoint $A^\ddagger = RA^{\sharp_G}R$, conjugation by T transports the source relation without a residual sign: $(\Lambda'_+)^\ddagger = \Lambda'_-$. Extending $R_{386} = I_{356} \oplus R$ over the certified orthogonal projector splitting yields 376 positive and 10 negative signs and replays the full projector-level Green adjoint together with the two homotopy identities and causal support. This resolves the sign as a convention theorem, not as a new causal obstruction.

The component serialization now closes the next, narrower carrier object (`STRICT_386_COMPONENT_PAIRING_SERIALIZATION_V1`, `FOUNDATIONAL_LORENTZIAN_WEYL_BV_COMPLETION_ATLAS_V7`). Its hybrid basis contains 386 named rows: 30 Gate-canonical endpoint rows, 36 generalized- auxiliary rows and 320 mapping-cylinder cone/cotangent rows. The full odd pairing has 410 ordered rational nonzero entries and exact rank 386. After pullback, the endpoint part has 30 entries; the earlier count 54 referred to the pre-pullback coordinates and was a coordinate-label imprecision, not an algebraic discrepancy. On these same row indices the complete T, $T^{\sharp_G}$ and R diagonals are serialized, and $T^T\Omega = \Omega T^{\sharp_G}$ replays componentwise. This still does not provide portable coefficient tables for the full q_1, H_{alg}, endpoint inclusion/projection or advanced/retarded Green operators. Hence an independent replay of every component operator adjoint and homotopy identity, local D, and same-carrier q_2 compatibility remain open, so Gate A remains fail closed. The corpus does contain a positive four-row metric Bach BV Green homotopy on unit Nariai and an exact but analytically incomplete Berger route. Conversely, the 24-field scalar-symbol and fixed-temporal 46-parameter first-order families are scoped no-go theorems only. The authoritative classical import gate still fails closed, so none of these ingredients is promoted to a full-complex propagator, Hadamard state, renormalized product, causal pAQFT construction, or Lorentzian quantum master equation. This boundary is machine-readable in `FOUNDATIONAL_SCALAR_BIWAVE_TO_WEYL_BV_DEPENDENCY_DELTA_V1`.

8.3 From the scalar Green benchmark to variable coefficients

The flat scalar $1 + 1$ formula now supplies the control case. The next target is one explicitly presented variable-coefficient normally hyperbolic scalar operator. Factor the classical proof into Sobolev completion, energy estimate, local solution, exhaustion, uniqueness, duality, and support, and compare every dependency with the canonical flat construction. That audit should then be compared gate by gate with the generated scalar-biwave-to-Weyl- BV delta, rather than treating "fourth order" as an operator identification.

8.4 Proof strength of an operational reconstruction

Select one finite-dimensional reconstruction, preferably the continuity branch in Hardy or the purification branch of Chiribella–D'Ariano–Perinotti. Formalize the exact real-algebraic and compactness principles used. Then separate the finite theorem from any proposed countable or field-theoretic extension. This is a promising place to obtain an actual equivalence or avoidance theorem because the physical postulates are already explicit.

8.5 Internal BV and the external probability bridge

The topos literature supplies internal commutative algebra, locale spectra, state representations, and versions of dynamics. A useful Weyl target is smaller than "construct QFT in a topos": internalize a finite local BRST complex and verify nilpotency and a finite contraction, then state precisely what additional structure would be required for integration modulo boundary terms, anomaly classification, and external probabilities. Even an obstruction ledger would convert a vague gap into typed missing morphisms.

8.6 Physical selectors on explicit state spaces

The state-existence problem is already easy in several explicit ZF carriers. The next scientific question is which physical assumptions select a state. Candidate selectors include a Hamiltonian ground condition, KMS condition, incoming detector preparation, residual symmetry, or a local Hadamard condition. Each should be tested independently for existence, uniqueness, foundational strength, and compatibility with gauge structure. A no-selection theorem is valuable only when its symmetry and state class are physically justified.

8.7 The BT reconstruction fork

The next BT question is no longer whether a positive finite measure exists. Ordinary OS reconstruction is obstructed both near the free endpoint and, through an exact density-kernel witness, at $\lambda = 0.4$ on the 6^4 lattice. The active analytic gate is now an L-uniform interacting negative-Sobolev estimate. Only after that estimate should one attempt tightness and Schwinger-function convergence. A specifically indefinite-metric reconstruction is a separate branch requiring its own axioms and carrier-matching theorem; no Euclidean result here supplies a Krein/Lorentzian observable map. Global bilaplacian strong convexity cannot supply the estimate: an exact fixed-volume family makes its comparison ratio tend to zero. A direct or annealed covariance estimate is therefore the next analytic gate. The direct pointwise Schwinger–Dyson sign route is now also obstructed, although the exact residual-sum identity supplies uniform quartic action confinement of each lowest normalized Fourier mode. The remaining gate is specifically annealed or normalized-marginal control. A sharper bilaplacian envelope already gives the required uniform H^{-1} bound for its normalized radial reference measure, but an exact negative Hessian rules out transferring that bound by standard convex-perturbation comparison. A mode-specific repair also fails: ultraviolet relaxation drives the exact lowest-mode Hessian Schur complement to zero. The exact affine virial theorem now supplies the previously missing volume-uniform expectation of the half-action-density weight under the actual Gibbs measure. But an exact period-four family makes the entire Hessian block orthogonal to the lowest axial eigenspace indefinite on every volume L^4 with $4 \mid L$. Hence the global inverse-Hessian/Schur route, including its pointwise half-action curvature refinement, is obstructed as formulated. The live gate is now a direct normalized low-frequency marginal estimate, first for one axial mode and then across Fourier shells. Neither the action-density nor bilaplacian-density theorem alone is an H^{-1} estimate because energy may concentrate in the infrared modes. The residual-spectrahedral pushforward now makes the normalization exact: the field is the logarithm of the positive ground state of $-\Delta + \mathrm{diag}(r)$, while the measure is a Gaussian surface weight divided by its spanning-tree coarea Jacobian. The next theorem must control this log-ground-state marginal uniformly in volume; the coordinate change itself supplies no moment bound. The boundary is pointwise strictly convex, but an exact four-cycle family makes its least-curvature upper bound tend to zero and makes the Gaussian weighted mean curvature negative at $\lambda = 0.4$. Hence a global positive-curvature boundary spectral gap cannot supply the estimate. The tree-Jacobian factor also cancels exactly under the induced multiplicative tilt, so its log convexity supplies no extra confinement. The live gate is not a comparison of each orthogonal background with its mode-zero value: an exact fixed-volume family makes the centered action ratio tend to zero under unbounded lowest-mode shifts. Half-period translation keeps the integrated marginal even. On a subsequence of those backgrounds the global fiber minima and all but at most 2^{-m} of the conditional mass escape below $u = -m$, so a uniform raw conditional moment is also impossible. The remaining object was therefore an all-background recentered width together with an annealed bound on the moving center in the flat log-field chart. The first term is now proved: a plaquette absorption inequality controls the signed spatial remainder and gives conditional variance at most $9/(2N\omega_L^2)$ for every background and every $L \geq 4$. The live gate is the Gibbs-weighted second moment of the unique conditional centers, and it reduces exactly to a zero-fiber-score second moment. The leading cubic score does retain a soft factor on every leg, but its free orthogonal-background coefficient has an exact logarithmic lower bound in the lattice-size limit. Therefore a fixed-bare-coupling, coefficient-by-coefficient estimate cannot close the gate. The physical scale setting now gives a precise fork. On a fixed-physical-volume refinement the logarithmic residue is $5/(16\pi^2)$, asymptotic freedom gives $g_L^2 \log L \to 8\pi^2/5$, and the leading score coefficient tends to $1/2$. This restores leading-log uniformity on the tuned branch. At fixed lattice spacing, increasing L instead sends the lowest external momentum to zero without the same coupling tuning, so that conclusion does not transfer. The ordinary finite-lattice equation-of-motion Ward identity controls the sampled full score, not the zero-fiber score, and an exact shifted-Gaussian fixture obstructs any general transfer between them. Exact disintegration now sharpens the barrier: a constrained zero-fiber identity weights every background by its conditional density $q_\eta(0)$, whereas the target requires the inverse of that weight under the constrained measure. On the actual BT runaway family $q_m^{(u)}(0) \leq 2^{-m}/m$ (and the t density differs only by the fixed factor $1/\log 2$), so no pointwise uniform density lower bound can repair the transfer. The next viable route is therefore a genuinely annealed BT multiscale estimate coupling center displacement to the Gibbs rarity of its background, with the RG normalization retained on the declared refinement branch. The next quartic calculation also proves that the isolated free square of the quartic zero-fiber score grows at least as L^2: its exact four-leg kernel is only linearly soft, with derivative $-1/3$ at a quarter-period fixture. Running g_L^4 is only logarithmically small and cannot suppress that power by itself. The complete order-g^4

formula is now assembled, and its signed corrections cannot cancel the positive p^2 term on any fixed ultraviolet carrier: they begin at p^4 after real-cosine parity. The unrestricted coefficient still remains open because a nonuniform contribution from the shrinking infrared complement could cancel the UV sector only after summation. Bounding or computing that complement is now the precise fixed-order gate. Wiener-chaos orthogonality reduces this signed complement to one effective second-chaos three-leg kernel: a linear-soft weighted norm bound for that kernel would make the only negative cross term negligible. Its expected-Hessian formula is exact, and real-cosine conditioning has been isolated as a rank-one correction. The extensive normalization-aligned sector cancels only in the connected covariance form of the complete coefficient. A bulk Wick table plus an exact all-volume rank-insertion audit reduces that form to at most two loop sums. Exact rational evaluation gives $M_4(4) = -338835474713437/204838502400000$, independently certified by modular enumeration. This refutes an exact all-volume zero identity, but does not decide the large-volume sign or scaling. For every $L \geq 5$, an exact affine-flow atlas now combines the bulk and rank-one two-loop sectors into 21 common integrands. Five cancel before absolute values, including every power-sized quartic-tadpole square and cubic-bubble–quartic-tadpole cross. The remaining conditioning factor is the positive $162X_L^2/(N\omega_p^2) = O((1 + \log L)^2)$, which is controlled by g_L^4 on the tuned refinement branch. The fourteen other entries reduce to seven inversion pairs. The paired quartic vertex is nonnegative and comparable to the product of its two dispersions, and one negative nested pair has magnitude at least cL^2. Hence termwise tuned-g_L^4 bounds are obstructed: the common $N\omega_p$ coefficient must be computed before absolute values. Exact soft-factor and shifted-convolution estimates prove that pairs 1, 2, and 5 are only $O(\log^2 L)$. An all-leg quintic estimate and a new three-weight torus convolution bound further prove that pair 3 is $O(L)$ and pair 6 is $O(L \log L)$. All five are $o(N\omega_p)$. Thus only the common power coefficient of negative pair 4 and positive pair 7 decides the leading two-loop scale. Both normalized limits exist: the negative coefficient has an exact integer-sum formula and is less than -0.01613. A sharp lattice vector-dispersion inequality and exact outward Green-function cubature give $0 < c_7 < 0.016103194$, so their sum is strictly negative. The missing lower-loop atlas is now exact for $L \geq 7$: its ten zero-loop rows have the positive finite limit $111/(32\pi^4)$, while a five-centre shell bound and the exact collinear cubic soft factor prove that all twenty-seven one-loop rows sum to $O(\log L)$. They are therefore $o(N\omega_p)$, and complete perturbative M_4 has a strictly negative leading $N\omega_p$ coefficient. On the tuned branch, parity kills the cubic norm coefficient and the quadratic contribution remains bounded, while the quartic truncation tends to minus infinity. Positivity of the exact score square therefore forces the omitted sector to compensate on the same $g_L^4 N\omega_p$ scale. A volume-uniform fixed-order remainder is now obstructed, but the all-order score and interacting H^{-1} estimate remain open. The logarithmic coefficient is not evidence that the resummed interacting moment diverges. Only after the normalized marginal closes should the programme sum Fourier shells. Another fixed-origin, pointwise, global-curvature, bare perturbative, or coarea comparison cannot close that gate.

9 Limits of the present paper

This paper introduces a programme and reports bounded examples. It does not establish a global metatheorem relating physical postulates to set-theoretic strength. It does not prove that the three atlas axes form a literal Cartesian product of coherent theories. It does not prove that the six regimes, six carriers, or sixteen obligations are canonical or exhaustive.

The literature survey is a structured seed corpus, not a systematic review of all mathematical physics. "No reviewed source" means no source in the declared corpus, not that no result exists.

The 576-cell count and coverage statistics describe a versioned research instrument. They are not novelty counts, evidence weights, or probabilities that a theorem is true.

The local foundational results are upper bounds unless a reversal is stated. The RCA_0 coded-wave theorem does not show RCA_0 is weakest. ZF sufficiency for the explicit Krein and separable C^* carriers is not a necessity theorem. Exact finite certificates do not settle infinite completions. Computable or constructive results are not automatically reverse-mathematical classifications.

Finally, none of the case studies constructs a complete Lorentzian off-shell BV propagator, a BRST-compatible Hadamard state for the full metric complex, renormalized Lorentzian time-ordered products, a causal perturbative AQFT, or a Lorentzian quantum-master-equation theorem. The certificates carrying `REDUCED-MODE` or `LOCAL-ALGEBRAIC` tags provide no evidence for those `LORENTZIAN-CAUSAL` obligations.

10 Reproducibility

The paper has a machine-readable claim map generated from current certificate inputs. It pins the source paths and hashes, extracts the atlas dimensions and status counts, checks the exact dependency tags of every local case study, and records the paper's nonclaims. The reproduction commands are:

```
python3 paper/generate_21_reverse_foundations_appendices.py --check
python3 paper/generate_21_reverse_foundations_claim_map.py --check
python3 paper/verify_21_reverse_foundations_claim_map.py
cd paper
pdflatex -interaction=nonstopmode -halt-on-error \
  21-reverse-foundations-of-physics.tex
```

The principal machine-readable authorities are `FOUNDATIONAL_INTERSECTION_CUBE_V14`, `FOUNDATIONAL_FULL_SURFACE_GAP_AUDIT_V1`, `FOUNDATIONAL_MATRIX_EXPLORER_SITE_V2`, `FOUNDATIONAL_THEORY_ASSEMBLY_ATLAS_V1`, `FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1`, `FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1`, `FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1`, `FOUNDATIONAL_CODED_WAVE_OBSERVABLE_RECONSTRUCTION_V1`, `FOUNDATIONAL_CODED_LOCAL_WEAK_WAVE_TEST_CLASS_V1`, `FOUNDATIONAL_CODED_WEAK_WAVE_H2_TEST_COMPLETION_V1`, `FOUNDATIONAL_FIXED_SUPPORT_SMOOTH_TO_H2_TRANSLATOR_V1`, `FOUNDATIONAL_SUPPORT_INDEXED_TEST_SPACE_COMPARISON_V1`, `FOUNDATIONAL_SCALAR_MINKOWSKI_GREEN_CHOICE_AUDIT_V1`, `STRICT_386_COMPONENT_PAIRING_SERIALIZATION_V1`, `FOUNDATIONAL_LORENTZIAN_WEYL_BV_COMPLETION_ATLAS_V7`, `REVERSE_PHYSICS_BT_EUCLIDEAN_FREE_RECONSTRUCTION_OBSTRUCTION_V1`, `REVERSE_PHYSICS_BT_EUCLIDEAN_OS_WITNESS_PREFLIGHT_V1`, `REVERSE_PHYSICS_BT_EUCLIDEAN_RESIDUAL_TILT_JACOBIAN_CANCELLATION_V1`, `REVERSE_PHYSICS_BT_EUCLIDEAN_CENTERED_FIBER_DOMINATION_OBSTRUCTION_V1`, `REVERSE_PHYSICS_BT_EUCLIDEAN_CONDITIONAL_MASS_ESCAPE_OBSTRUCTION_V1`, `FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1`, `FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1`, and `FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1`. The static atlas appendices are generated from `foundations/site/data.json`.

11 Conclusion

The mathematical assumptions of a physical theory are neither inert notation nor a single hidden axiom. They are a structured dependency layer. Changing the carrier can move positivity, completion, and representation obligations. Changing the foundation can alter what counts as existence. Physical postulates can supply canonical data that avoid otherwise arbitrary choices. And the same physical slogan can have different strength under different encodings.

The practical response is to replace undifferentiated questions—"does physics need Choice?", "can Krein replace Hilbert space?", "can finite mathematics recover the continuum?"—with typed judgements of the form

$$L + S + M + \mathrm{Enc}(P) \;\vdash_R\; O,$$

where every symbol is explicit and every arrow has a relation type. The first case studies already change the picture. Explicit infinity need not introduce Choice at its first step. State existence does not select a physical state. Evolution does not establish causal support. Exact finite causality does not establish a continuum light cone. And a finite algebraic BV certificate does not promote an analytic quantum theory.

The resulting atlas is useful precisely because it distinguishes completed results from reviewed holes. A reviewed gap is not an absence theorem; it is a request for a well-typed obligation, an encoding, a proof ledger, and an independently checkable result. Resolving those gaps one theorem at a time offers a realistic route toward understanding not only which physics follows from which physical assumptions, but which mathematics the physics truly needs.

A Static companion to the evidence atlas

The interactive atlas exposes seven views of the same normalized evidence and derived assessments: matrix, dimensions guide, theory profiles, assemblies, typed implications, strength ladder, and evidence catalogue. This appendix preserves their key research content in a citable static form. It is a snapshot, not a replacement for cell inspection, filtering, neighboring-cell comparison, or the complete downloadable dataset.

Website view	Static counterpart
Matrix	Overall counts, a 6×6 three-way coverage roll-up, and status counts for every obligation.
Dimensions guide	All 28 axis options with their non-specialist descriptions.
Theory profiles	Coverage-envelope and Pareto navigation; these are not composed theories.
Assemblies	Nine cube-selected prototypes, two model-scoped chains with independent maturity rails, typed joins, and an external standard-GR calibration control.
Implications	The complete typed relation ledger: ten directed edges with their exact assertion and evidence.
Strength ladder	All six cylinder-wave gates, including what each adds, establishes, and leaves open.
Evidence	The complete literature register, local-certificate register, and usage crosswalk for all 83 records.

A.1 Dimensions guide

Every coordinate combines one foundational regime, one carrier, and one theorem-level obligation. The tables below reproduce the website's plain-language guide rather than assuming the reader already knows the programme vocabulary.

A.1.1 Under which rules may the mathematics reason and assert that objects exist?

Option	Plain-language purpose
Classical standard	Mainstream mathematics: classical logic, completed infinite structures, and ordinary analysis, with Choice available unless a proof explicitly avoids it.
Weak formal base	Use a deliberately small formal system and ask exactly how much arithmetic or set existence the proof needs.
ZF with weakened Choice	Keep classical set theory but remove or isolate principles that choose objects from infinitely many sets at once.
Constructive/computable	An existence claim must provide a witness, construction, or algorithm—not only show that nonexistence would be contradictory.
Topos/internal	Do the mathematics inside an alternative logical universe, where truth may be local and classical either/or reasoning may fail.
Finite/discrete restriction	Replace an infinite or continuous system by finite exact data or finitely many modes. This is not automatically the same as rejecting infinity as a foundation.

A.1.2 What mathematical object carries states, observables, fields, and evolution?

Option	Plain-language purpose
Finite exact algebra	Finite matrices, rational arrays, or other finite algebraic data that can be checked exactly.
Hilbert/operator	The positive-norm vector spaces and operators used in standard quantum mechanics and spectral theory.
Krein/indefinite	A vector space whose inner product can be positive, negative, or zero, as often occurs before unphysical gauge directions are removed.
Algebraic C*-system	Start from an algebra of observable quantities; a state is a rule assigning expectation values rather than primarily a wavefunction.
Smooth/PDE/distributional	Continuum fields on space or spacetime, including derivatives, PDEs, Sobolev spaces, generalized functions, and Green operators.
Localic/synthetic/internal	Describe spaces through regions, logical relations, or internal geometry instead of beginning with a set of individual points.

A.1.3 Which precise physical or theorem-level job is established?

Option	Plain-language purpose
Kinematics/observables	Say what the possible configurations and measurable quantities are before specifying how they evolve.
State existence	Show that at least one mathematically valid state actually exists.
State representation	Explain how an abstract state is encoded—for example by a vector, density matrix, measure, valuation, or GNS construction.
Probability rule	Turn states and events into normalized probabilities, such as a Born-type prediction rule.
Physical state selection	Explain why a particular vacuum, thermal, Hadamard, or other state should count as physically distinguished.
Generator/spectral dynamics	Construct what generates time evolution and, where relevant, identify its allowed frequencies or energy spectrum.

Option	Plain-language purpose
Evolution/well-posedness	Show that admissible initial data produce a solution that exists, is unique, and changes stably or computably with the data.
Causal propagation/Green	Show that disturbances propagate within the permitted causal region and construct retarded or advanced response maps.
Gauge/BV/cohomology	Handle redundant gauge descriptions consistently and identify the quantities or states that remain physically meaningful.
Interaction construction	Build a genuine coupling or nonlinear theory rather than only a collection of free, noninteracting fields.
Counterterm classification	List every local correction that quantum calculations are allowed to require before attempting to calculate its coefficient.
Anomaly classification	List the possible ways a classical symmetry or consistency condition could fail after quantization.
Renormalized products	Define products and correlation functions that would otherwise be singular when fields meet at the same spacetime point.
QME restoration	Repair the quantum master equation, the BV consistency condition that encodes quantum gauge symmetry.
Residual quantum transfer	After quantum consistency is restored, transfer the correction to the smaller complex that represents the surviving physical content.
Reconstruction/limits	Connect the formulation back to operational predictions, a continuum or standard theory, or a demonstrated notion of empirical equivalence.

A.2 Matrix overview

The full atlas has 576 coordinates. The website exposes sixteen 6×6 slices, one for each obligation. For print, Table 4 aggregates those slices without implying that the statuses are truth values. Each entry is *direct / seeded / reviewed / unmapped*: direct combines local and reviewed literature results; seeded combines pieces-only and priority-gap cells; reviewed means a formulated open question with a typed missing certificate but no direct result; unmapped means that no coverage classification is made.

Regime ↓ / carrier →	Finite exact	Hilbert	Krein	Algebraic	PDE	Localic
Classical standard	8/7/1/0	12/4/0/0	9/3/4/0	12/4/0/0	9/6/1/0	5/4/7/0
Weak formal base	9/7/0/0	4/1/11/0	3/1/12/0	0/3/13/0	5/8/3/0	0/0/16/0
ZF with weakened Choice	7/8/1/0	10/6/0/0	7/4/5/0	10/6/0/0	2/8/6/0	2/2/12/0
Constructive/computable	7/8/1/0	10/6/0/0	3/3/10/0	5/3/8/0	1/9/6/0	5/4/7/0
Topos/internal	7/7/2/0	2/4/10/0	4/3/9/0	5/4/7/0	2/7/7/0	3/13/0/0
Finite/discrete restriction	10/6/0/0	12/4/0/0	7/8/1/0	7/8/1/0	8/7/1/0	5/4/7/0

Table 4: Aggregate coverage matrix. Each cell reports direct results / open cells with a starting point / reviewed open gaps / not mapped, across all sixteen obligations.

Table 5: Coverage status by physical obligation. Every row sums to 36.

Obligation	Local	Literature	Pieces	Priority gap	Reviewed gap	Not mapped	Total
Kinematics/observables	11	16	0	0	9	0	36
State existence	12	10	6	0	8	0	36
State representation	12	15	2	0	7	0	36
Probability rule	10	6	11	0	9	0	36
Physical state selection	3	1	23	0	9	0	36
Generator/spectral dynamics	15	11	7	0	3	0	36
Evolution/well-posedness	16	6	11	0	3	0	36
Causal propagation/Green	4	2	20	0	10	0	36

Obligation	Local	Literature	Pieces	Priority gap	Reviewed gap	Not mapped	Total
Gauge/BV/cohomology	4	2	15	5	10	0	36
Interaction construction	15	2	3	3	13	0	36
Counterterm classification	5	2	10	4	15	0	36
Anomaly classification	5	2	10	4	15	0	36
Renormalized products	0	2	15	4	15	0	36
QME restoration	6	0	13	4	13	0	36
Residual quantum transfer	4	0	13	4	15	0	36
Reconstruction/limits	5	13	1	2	15	0	36
All obligations	127	90	160	30	169	0	576

A local or literature result remains bounded by its attached claim boundary. Pieces-only means relevant ingredients do not yet compose the target. Priority gap marks a selected current-programme gap. Reviewed gap marks an explicitly formulated open question with a typed missing certificate, not a result or literature-absence claim. The atlas has 169 reviewed gaps and 0 not-mapped coordinates; all 576 coordinates are emitted by the authoritative cube.

A.3 Typed implication ledger

The implication view is directional. Table 6 is the authoritative print reading of its arrows: the relation column says how the edge may be used, and the assertion column says exactly what has and has not crossed that edge.

Table 6: Complete typed relation ledger from the implications tab.

From	Relation	To	Exact assertion	Evidence
Finite resolution	Enough for this step	Finite Laurent wave	Resolving finitely many Fourier modes is enough to construct the exact finite Laurent-wave fixture.	`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1`
Supplied error control	Depends on the coding	Energy-tail modulus	A usable cutoff rule supplies a tail modulus only because that quantitative rule is part of the chosen representation.	`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1`
Energy-tail modulus	Enough for this step	Coded Hilbert evolution	A prescribed fast Cauchy rate and finite-code time moduli support the explicit diagonal extension in RCA_0.	`FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1`
Finite total energy	Unproved bridge	Energy-tail modulus	Finite total energy alone does not supply a rate of tail convergence; deriving one uniformly remains an open bridge.	Open bridge—no direct certificate
Coded Hilbert evolution	Unproved bridge	Finite-test wave identity	The completed evolution name has been constructed, but verifying the wave identity against the declared finite test class remains open.	`FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1`
Finite-test wave identity	Unproved bridge	Localized spacetime distribution	An identity on finite Fourier tests has not yet been extended to a localized spacetime test-function class.	Open bridge—no direct certificate
Finite Laurent wave	This method fails	Causal Green maps	Finite spectral projection is globally nonzero, so this route cannot certify causal support.	`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1`
Localized spacetime distribution	Not enough by itself	Causal Green maps	A spacetime distribution alone is not enough: energy uniqueness and support propagation are additional requirements.	`FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1`

From	Relation	To	Exact assertion	Evidence
Finite propagation speed	Unproved bridge	Causal Green maps	Finite propagation is the physical requirement, but an explicit theorem must still construct advanced and retarded maps with controlled support.	`FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1`
Computable wave propagation	Literature contrast	Noncomputable solution value	Computability changes with topology, regularity, and representation; the two literature results are not contradictory.	`weihrauch-zhong-2002`; `pour-el-richards-1981`

A.4 Cylinder-wave strength ladder

The ladder prevents a certified lower level from being read as a theorem at a stronger level. "Adds" names the new mathematical data at the gate; "establishes" records the result available there; and "still open or excluded" blocks silent promotion.

Table 7: All six gates in the cylinder-wave strength ladder.

Gate, object, and sufficient base	Adds	Establishes	Still open or excluded
`L0_FINITE_LAURENT / CERTIFIED`. A labelled finite family of rational-complex chiral modes. Base: PRA for every fixed fixture.	finite sums; integer differentiation weights; decidable rational equality	exact wave-equation residual zero; Galerkin nesting; exact positive finite energy	completion; spatial localization; causal support
`L1_NAMED_TAIL_MODULUS / CERTIFIED`. The explicit datum c_n=1/n^2 together with N(k)=2^k. Base: Primitive-recursive rational inequalities plus induction for the displayed telescoping bound.	a tolerance-to-cutoff function supplied as data	a constructive Cauchy name for this named energy datum	a uniform modulus extractor for every classically convergent energy series
`L2_CODED_ENERGY_CARRIER / CERTIFIED`. Fast-Cauchy completion of mean-zero rational polygonal chiral pairs. Base: RCA_0 for the declared representation.	coded Cauchy completion with prescribed rate; exact rational translation group; finite-code time moduli; primitive-recursive diagonal for named real times	completed energy state; energy-conserving real-time solution name; Cauchy existence and uniqueness in the declared carrier	necessity or reversal; representation invariance; localized weak equation; causal support
`L3_COEFFICIENT_WEAK_SOLUTION / FORMALIZATION_TARGET`. An energy solution tested against finite Fourier sequences. Base: Not classified. This coefficient-weak notion avoids claiming a spacetime distribution or localized support.	finite-support test-sequence pairing; termwise weak equation	—	coded localized test class; coefficient-to-distribution comparison; weakest base; L2 supplies a completed evolution name, but the present certificate does not formalize test-function integration.
`L4_SPACETIME_DISTRIBUTION / OPEN`. A distribution on R x S^1 tested against localized smooth functions. Base: Not classified.	coded test-function space; integration or Fourier/test-function comparison; continuity in the test-function topology	—	Neither finite Laurent algebra nor a Hilbert coefficient sequence supplies this automatically.

Gate, object, and sufficient base	Adds	Establishes	Still open or excluded
L5_CAUSAL_GREEN_OPERATOR / CONDITIONAL_IMPORT_ONLY. Advanced and retarded solution maps with finite propagation and uniqueness. Base: Not classified.	localized energy estimates; Cauchy uniqueness; support propagation; slab globalization; distributional duality	—	The imported biwave result is conditional and its weakest base and Choice strength remain open.

A.5 Complete evidence and literature registers

The normalized catalogue contains 83 evidence records: 32 local result records and 51 literature records. Tables 8–10 print the complete catalogue rather than only the records highlighted by the graph and ladder. Literature entries preserve their artifact status: metadata-only records are bibliographic leads, not content-pinned evidence. A catalogue entry supports only its stated role and never silently crosses its recorded boundary.

Catalogue lifecycle or artifact status	Records
CONTENT_PINNED	39
METADATA_ONLY	12
CERTIFIED	9
SUFFICIENCY_PROVED	8
SEPARATED	6
CLASSIFIED	2
L2_UPPER_BOUND_CERTIFIED	2
LITERATURE_SCOPED	2
CERTIFIED_DEPENDENCY_DELTA	1
RESIDUAL_TRANSFERRED	1
VERIFIED_BOUNDED_MODEL	1

A.5.1 Complete literature reference list

The stable link is the catalogue's preferred public locator. CONTENT_PINNED means that the reviewed artifact is content-addressed in the evidence ledger; METADATA_ONLY means that the bibliographic identity is resolved but the source content is not pinned. The latter is a research lead, not equivalent evidential weight.

Table 8: Complete literature reference and evidence register (51 records).

Reference and provenance	Recorded evidential role and boundary
abramsky-coecke-2004. Samson Abramsky and Bob Coecke, A categorical semantics of quantum protocols, 2004. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/quant-ph/0402130	*Supports:* Compact closed categories with biproducts express finite quantum protocols compositionally, with scalars and a Born-rule-like probabilistic interpretation emerging from the categorical structure. *Boundary:* Categorical semantics is not itself a selected topos, a continuum field carrier, a constructive metatheory, or a derivation of physical Hilbert space from Weyl gravity.
baer-2015. Christian Bär, Green-hyperbolic operators on globally hyperbolic spacetimes, Communications in Mathematical Physics 333 (2015), 1585-1615. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1310.0738	*Supports:* Normally hyperbolic wave operators are Green-hyperbolic on globally hyperbolic spacetimes; advanced and retarded maps have the declared support and extend continuously to several support classes.; For symmetric hyperbolic systems the paper proves Cauchy uniqueness, existence, finite propagation and continuous dependence. *Boundary:* This is a classical smooth/distributional theorem. It does not code the proof in second-order arithmetic, eliminate Choice, give a Bishop-constructive proof, or construct the Weyl metric BV propagator.

Reference and provenance	Recorded evidential role and boundary
`bahr-dittrich-2009`. Benjamin Bahr and Bianca Dittrich, Breaking and restoring of diffeomorphism symmetry in discrete gravity, 2009. `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://arxiv.org/abs/0909.5688	*Supports:* Discretization can break diffeomorphism symmetry and coarse-graining or perfect-action methods can be used to study its restoration. *Boundary:* The analysis does not give a Weyl-BV cohomology certificate, establish quantum anomaly cancellation, or prove a continuum limit for this repository.
`barnich-brandt-henneaux-2000`. Glenn Barnich, Friedemann Brandt, and Marc Henneaux, Local BRST cohomology in gauge theories, Physics Reports 338 (2000), 439-569. `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://arxiv.org/abs/hep-th/0002245	*Supports:* Local BRST cohomology classifies the anomaly consistency condition, counterterms, and classical deformations for broad classes of gauge theories. *Boundary:* This is a classical-standard local cohomology framework. It does not compute Weyl-gravity coefficients, restore this repository's QME, or calibrate weak, constructive, or choice-free foundations.
`bateman-turok-2026`. Sam Bateman and Neil Turok, Escape from Ostrogradsky via Hidden Ghost Parity (2026). `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://arxiv.org/abs/2607.00096	*Supports:* The Bateman-Turok proposal changes the state-space metric and adjoint/probability structure by using a Krein carrier and a fundamental ghost parity. *Boundary:* The paper's tree-level positivity and deferred operator claims retain the boundaries audited in the repository; no foundational-strength result follows from this citation.
`bender-boettcher-1998`. Carl M. Bender and Stefan Boettcher, Real spectra in non-Hermitian Hamiltonians having PT symmetry, Physical Review Letters 80 (1998), 5243-5246. `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://arxiv.org/abs/physics/9712001	*Supports:* Families of non-Hermitian PT-symmetric Hamiltonians can have real positive spectra in an unbroken-symmetry regime. *Boundary:* PT symmetry is not identical to Krein self-adjointness, and real spectrum alone does not supply a positive physical inner product, unitary dynamics, or a gauge-QFT state space.
`blackadar-farah-2026`. Bruce Blackadar and Ilijas Farah, Separable C*-algebras Without the Countable Axiom of Choice (2026). `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://arxiv.org/abs/2602.15812	*Supports:* A substantial theory of separable C*-algebras, including representation, polynomial spectral mapping, and continuous functional calculus for commuting normal elements, can be developed in ZF.; Without Choice, nonseparable examples can break familiar state-space and commutative representation conclusions. *Boundary:* Does not automatically remove Choice from AQFT, interacting QFT, state selection, or nonseparable operator-algebra arguments.
`blackadar-farah-karagila-2026`. Bruce Blackadar, Ilijas Farah, and Asaf Karagila, Hilbert Spaces Without The Countable Axiom of Choice, revised 2026. `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://arxiv.org/abs/2304.09602	*Supports:* Hilbert spaces and their operator theory can be studied in ZF without countable Choice, and pathologies expose distinctions hidden by ZFC.; The assertion that every Hilbert space has the familiar basis behaviour is not a definition-level truth independent of set theory. *Boundary:* Foundational Hilbert-space analysis, not evidence that a particular physical Hilbert space exhibits the pathological examples.
`brattka-2008`. Vasco Brattka, How incomputable is the separable Hahn-Banach theorem? (2008). `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://arxiv.org/abs/0808.1663	*Supports:* The separable Hahn-Banach theorem also admits a uniform computability analysis, connecting its reverse-mathematical WKL calibration to computable analysis. *Boundary:* A Weihrauch or computability classification is not automatically the same relation as implication over RCA_0 or derivability in constructive set theory.
`brenna-flori-2012`. W. Brenna and Cecilia Flori, Complex Numbers, One-Parameter of Unitary Transformations and Stone's Theorem in Topos Quantum Theory, 2012. `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://arxiv.org/abs/1206.0809	*Supports:* The topos approach admits an internal treatment of one-parameter groups and a counterpart of Stone's theorem. *Boundary:* An internal Stone theorem is not spacetime propagation, Green-hyperbolicity, renormalization, or an empirical equivalence theorem.
`bridges-svozil-2000`. Douglas Bridges and Karl Svozil, Constructive Mathematics and Quantum Physics, International Journal of Theoretical Physics 39 (2000), 503-515. `PRIMARY_RESEARCH` / `CONTENT_PINNED`. https://doi.org/10.1023/A:1003613131948	*Supports:* Bishop-style constructive mathematics can isolate constructive properties of Hilbert subspaces and projections and formulate a constructive quantum-logical axiom system. *Boundary:* The paper is an attempt at constructive quantum foundations, not a complete constructive dynamics, continuum QFT, or gravity construction.
`bridges-wang-1998-dirichlet`. Douglas Bridges and Luminiţa Vîţă Wang, Constructive aspects of the Dirichlet problem, Bulletin of the London Mathematical Society 30 (1998), 579-588. `PRIMARY_RESEARCH` / `METADATA_ONLY`. https://doi.org/10.1112/S0024610798006243	*Supports:* The paper constructively treats weak solutions of the Dirichlet problem and dependence on boundary or domain data in its elliptic setting. *Boundary:* This is genuine constructive PDE evidence, but it is elliptic rather than hyperbolic and supplies no causal Green maps.

Reference and provenance	Recorded evidential role and boundary
`brown-simpson-1986`. Douglas K. Brown and Stephen G. Simpson, Which set existence axioms are needed to prove the separable Hahn-Banach theorem?, Annals of Pure and Applied Logic 31 (1986), 123-144. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1016/0168-0072(86)90066-7	*Supports:* Over RCA_0, the separable Hahn-Banach theorem studied in the paper is equivalent to WKL_0. *Boundary:* This is a reverse-mathematical result for a coded separable theorem, not the ZF choice strength of unrestricted Hahn-Banach and not proof that a concrete physics calculation invokes it.
`brunetti-fredenhagen-rejzner-2013`. Romeo Brunetti, Klaus Fredenhagen, and Katarzyna Rejzner, Quantum gravity from the point of view of locally covariant quantum field theory, Communications in Mathematical Physics 345 (2016), 741-779. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1306.1058	*Supports:* Perturbative quantum gravity can be formulated as an effective locally covariant theory using renormalized BV methods and relational observables. *Boundary:* This does not establish perturbative renormalizability, the Weyl-gravity QME, a nonperturbative theory, or a weak/constructive foundational calibration.
`brunetti-fredenhagen-verch-2001`. Romeo Brunetti, Klaus Fredenhagen, and Rainer Verch, The generally covariant locality principle: A new paradigm for local quantum field theory, Communications in Mathematical Physics 237 (2003), 31-68. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/math-ph/0112041	*Supports:* Locally covariant QFT is formulated as a functor from globally hyperbolic spacetimes to star-algebras, with admissible state spaces and relative Cauchy evolution. *Boundary:* The framework supplies architecture, not a Weyl-gravity model, a preferred state, a weak-foundation audit, or the missing full metric-BV Hadamard construction.
`chiribella-dariano-perinotti-2011`. Giulio Chiribella, Giacomo Mauro D'Ariano, and Paolo Perinotti, Informational derivation of quantum theory, Physical Review A 84, 012311 (2011). `PRIMARY_RESEARCH / CONTENT_PINNED`. https://doi.org/10.1103/PhysRevA.84.012311	*Supports:* Within an operational-probabilistic framework, five informational axioms define a broad class and purification selects quantum theory without taking Hilbert space as the starting postulate. *Boundary:* A reconstruction result in its declared operational framework; its proof has not yet been audited for logical or choice strength here.
`constantin-doring-2020`. Carmen Maria Constantin and Andreas Doring, A topos theoretic notion of entropy, 2020. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/2006.03139	*Supports:* For finite-dimensional systems of dimension at least three, contextual entropy determines the density matrix and the paper gives an explicit reconstruction algorithm. *Boundary:* This is finite-dimensional mathematical state reconstruction, not physical state selection, field dynamics, or an internal renormalized gauge theory.
`coquand-spitters-2009`. Thierry Coquand and Bas Spitters, Constructive Gelfand duality for C*-algebras, Math. Proc. Cambridge Philos. Soc. 147 (2009), 339-344. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://doi.org/10.1017/S0305004109002515	*Supports:* Commutative Gelfand duality admits a constructive formulation in which spectra are handled point-free. *Boundary:* Constructive commutative duality is not a complete constructive formulation of interacting quantum field theory.
`dittrich-2012`. Bianca Dittrich, From the discrete to the continuous: Towards a cylindrically consistent dynamics, New Journal of Physics 14 (2012), 123004. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1205.6127	*Supports:* Cylindrical consistency, embedding maps, and coarse graining provide explicit obligations for connecting discrete dynamics to a continuum theory. *Boundary:* This is a continuum-construction framework, not a completed continuum theorem for Weyl gravity or a foundational rejection of actual infinity.
`doring-2008`. Andreas Doring, Quantum states and measures on the spectral presheaf, 2008. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/0809.4847	*Supports:* Normal states on a von Neumann algebra are related to measures on the spectral presheaf in the topos approach to quantum theory. *Boundary:* A state-measure representation is not a physical state-selection rule, a Born-rule derivation, or an internal interacting field theory.
`esmeral-ferrer-wagner-2015`. K. Esmeral, O. Ferrer, and E. Wagner, Frames in Krein spaces arising from a non-regular W-metric, Banach Journal of Mathematical Analysis 9 (2015), 1-16. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://doi.org/10.15352/bjma/09-1-1	*Supports:* A Krein space has an indefinite product together with a fundamental symmetry whose associated positive product supplies a Hilbert-space topology. *Boundary:* A mathematical structural fact; it neither validates a generalized Born rule nor removes set-theoretic assumptions from infinite-dimensional analysis.

Reference and provenance	Recorded evidential role and boundary
fewster-verch-2011. Christopher J. Fewster and Rainer Verch, Dynamical locality of the free scalar field, Annales Henri Poincare 13 (2012), 1675-1709. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1109.6732	*Supports:* Dynamical locality compares kinematic and dynamically defined local content and is established for classical and quantized scalar models, with explicit exceptional cases. *Boundary:* The scalar examples are not a theorem for gauge theories or Weyl gravity; the paper itself identifies gauge-theoretic complications.
fredenhagen-rejzner-2011. Klaus Fredenhagen and Katarzyna Rejzner, Batalin-Vilkovisky formalism in perturbative algebraic quantum field theory, Communications in Mathematical Physics 317 (2013), 697-725. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1110.5232	*Supports:* The BV formalism can be combined with perturbative AQFT, including renormalized time-ordered products and the anomalous master Ward identity. *Boundary:* This does not instantiate the full Weyl metric complex or certify its Lorentzian QME, and it contains no reverse-mathematical or constructive strength analysis.
gibbons-hoffman-wootters-2004. Kathleen S. Gibbons, Matthew J. Hoffman, and William K. Wootters, Discrete phase space based on finite fields, Physical Review A 70, 062101 (2004). PRIMARY_RESEARCH / CONTENT_PINNED. https://doi.org/10.1103/PhysRevA.70.062101	*Supports:* For finite-dimensional quantum systems of prime-power dimension, finite-field phase space supports discrete Wigner functions and mutually unbiased line-associated bases. *Boundary:* A finite kinematics for selected dimensions is not a finite replacement for continuum dynamics, Lorentzian QFT, or a convergence theorem.
gottschalk-2004. Hanno Gottschalk, Complex velocity transformations and the Bisognano-Wichmann theorem for quantum fields acting on Krein spaces, 2004. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/math-ph/0408048	*Supports:* Relativistic local fields on Krein space admit analytic-vector and modular-theoretic results under stated axioms, extending a Bisognano-Wichmann-type characterization. *Boundary:* This is an indefinite-metric QFT result under strong axioms, not a construction of the full Weyl metric BV theory, positive Born probabilities, or choice-free analysis.
grinkevich-1996. Yuri B. Grinkevich, Synthetic Differential Geometry: A Way to Intuitionistic Models of General Relativity in Toposes (1996). PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/gr-qc/9608013	*Supports:* Synthetic differential geometry in a suitable topos supports intuitionistic models of Riemannian geometry and Einstein equations on formal manifolds. *Boundary:* This is a classical-gravity reformulation route; it does not supply a constructive quantum Weyl theory or an external equivalence theorem for every classical spacetime.
haag-kastler-1964. Rudolf Haag and Daniel Kastler, An Algebraic Approach to Quantum Field Theory, Journal of Mathematical Physics 5 (1964), 848-861. PRIMARY_RESEARCH / METADATA_ONLY. https://doi.org/10.1063/1.1704187	*Supports:* Local quantum field theory can be axiomatized algebra-first in terms of observables and locality rather than beginning with a preferred global Hilbert-space realization. *Boundary:* Algebra-first formulation moves but does not erase representation, state-existence, topology, infinity, or set-existence questions.
harding-heunen-2019. John Harding and Chris Heunen, Topos quantum theory with short posets, 2019. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1903.01897	*Supports:* A smaller context poset yields a different topos while retaining core results on Kochen-Specker obstruction, spectral representation, state measures, and dynamics. *Boundary:* Changing the context poset changes internal logic; preservation of these core results is not a general invariance theorem for field theory, BV, or empirical predictions.
hardy-2001. Lucien Hardy, Quantum Theory From Five Reasonable Axioms (2001). PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/quant-ph/0101012	*Supports:* Finite-dimensional quantum formalism can be reconstructed from operational axioms, with continuity doing explicit work in separating the quantum and classical cases in this framework. *Boundary:* Finite-dimensional reconstruction under its stated framework; it does not establish the foundations of infinite-dimensional QFT or the set-theoretic strength of the reconstruction proof.
henry-2014. Simon Henry, Constructive Gelfand duality for non-unital commutative C*-algebras (2014). PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1412.2009	*Supports:* Constructive, localic Gelfand duality extends to non-unital commutative C*-algebras and locally compact locales. *Boundary:* Commutative localic duality does not by itself construct noncommutative interacting QFT or its states and dynamics.
heunen-landsman-spitters-2009. Chris Heunen, Nicolaas P. Landsman, and Bas Spitters, A topos for algebraic quantum theory, Communications in Mathematical Physics 291 (2009), 63-110. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/0709.4364	*Supports:* An algebraic quantum system can be represented internally in a topos with an intuitionistic Heyting logic and a localic spectrum. *Boundary:* A reformulation of algebraic quantum theory; it does not establish empirical superiority, eliminate all external classical reasoning, or construct Weyl QFT.

Reference and provenance	Recorded evidential role and boundary
humphreys-simpson-1996. A. James Humphreys and Stephen G. Simpson, Separable Banach space theory needs strong set existence axioms, Transactions of the American Mathematical Society 348 (1996), 4231-4255. PRIMARY_RESEARCH / CONTENT_PINNED. https://doi.org/10.1090/S0002-9947-96-01742-6	*Supports:* Some weak-star closure existence statements for separable Banach-space duals require Pi^1_1 comprehension, so separability alone does not guarantee uniformly weak foundational strength. *Boundary:* This does not make every theorem about a separable Banach space strong; the coded target and its representation determine the reversal.
humphreys-simpson-1999. A. James Humphreys and Stephen G. Simpson, Separation and Weak Konig's Lemma, Journal of Symbolic Logic 64 (1999), 268-278. PRIMARY_RESEARCH / CONTENT_PINNED. https://doi.org/10.2307/2586763	*Supports:* Over RCA_0, separation for open convex sets is equivalent to WKL_0, while separation for separably closed convex sets is equivalent to ACA_0.; The representation of a closed convex set changes the logical strength of an apparently similar separation claim. *Boundary:* The result calibrates two precise separable Banach-space statements; it does not assign one strength to every use of geometric separation.
kogut-susskind-1975. John Kogut and Leonard Susskind, Hamiltonian formulation of Wilson's lattice gauge theories, Physical Review D 11 (1975), 395-408. PRIMARY_RESEARCH / METADATA_ONLY. https://doi.org/10.1103/PhysRevD.11.395	*Supports:* Wilson lattice gauge theory admits a Hamiltonian formulation with gauge constraints and a continuum-limit programme. *Boundary:* The historical formulation is not finite in every mathematical sense and does not supply a controlled Weyl-gravity continuum or BV-QME certificate.
kostrykin-potthoff-schrader-2011. Vadim Kostrykin, Jürgen Potthoff, and Robert Schrader, Finite propagation speed for solutions of the wave equation on metric graphs, 2011. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1106.0817	*Supports:* A class of self-adjoint Laplace operators on metric graphs has existence and uniqueness for the wave equation and strict finite propagation, proved by localized energy methods. *Boundary:* Metric graphs retain continuous edges and Hilbert/Sobolev analysis. They are not finite exact algebra, a continuum-limit theorem, or a choice-free construction.
mostafazadeh-2001. Ali Mostafazadeh, Pseudo-Hermiticity versus PT symmetry: The necessary condition for the reality of the spectrum of a non-Hermitian Hamiltonian, Journal of Mathematical Physics 43 (2002), 205-214. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/math-ph/0107001	*Supports:* Pseudo-Hermiticity is introduced and shown to be a necessary structural condition for Hamiltonians with real spectrum in the setting studied. *Boundary:* The result assumes analytic and spectral hypotheses and does not by itself define physical probabilities, interacting QFT, or a weakest foundational base.
muehlhoff-2010. Rainer Mühlhoff, Cauchy Problem and Green's Functions for First Order Differential Operators and Algebraic Quantization, Journal of Mathematical Physics 52 (2011), 022303. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1001.4091	*Supports:* Prenormally hyperbolic first-order operators have unique advanced and retarded Green functions and a globally well-posed Cauchy problem under the stated globally hyperbolic hypotheses. *Boundary:* The reduction imports the normally-hyperbolic second-order theorem and remains classical; it is not a foundational-strength or Weyl-BV result.
nachtergaele-raz-schlein-sims-2007. Bruno Nachtergaele, Hillel Raz, Benjamin Schlein, and Robert Sims, Lieb-Robinson Bounds for Harmonic and Anharmonic Lattice Systems, Communications in Mathematical Physics 286 (2009), 1073-1098. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/0712.3820	*Supports:* Harmonic and specified anharmonic lattice systems satisfy Lieb-Robinson bounds, including exponentially small commutators outside an effective cone for Weyl observables. *Boundary:* An exponentially small tail is not strict support and is not an advanced/retarded Green operator. The result must not be promoted to continuum Lorentzian causality.
neumann-pape-streicher-2018. Eike Neumann, Martin Pape, and Thomas Streicher, Computability in Basic Quantum Mechanics, Logical Methods in Computer Science 14(2:14) (2018), 1-20. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1610.09209	*Supports:* For separable infinite-dimensional Hilbert spaces, states and observables can be represented by measures and valuations so that an effective spectral-theorem analysis becomes possible. *Boundary:* Computable representations are not equivalent to a Bishop-constructive proof or an RCA_0 reversal, and the result does not supply interacting QFT.
pischke-2025-semigroups. Nicholas Pischke, A proof-theoretic metatheorem for nonlinear semigroups generated by an accretive operator and applications, Selecta Mathematica 31 (2025), 32. PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/2304.01723	*Supports:* Logical metatheorems cover nonlinear semigroups generated by accretive operators and support extraction of quantitative convergence information in the stated systems and case studies. *Boundary:* Proof mining in higher-type systems is not an RCA_0/WKL_0/ACA_0 reversal, and the operator and semigroup structures are encoded assumptions rather than a causal PDE construction.

Reference and provenance	Recorded evidential role and boundary
`pour-el-richards-1981`. Marian B. Pour-El and J. Ian Richards, The wave equation with computable initial data such that its unique solution is not computable, Advances in Mathematics 39 (1981), 215-239. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1016/0001-8708(81)90001-3	*Supports:* Under the paper's representations, computable initial data can evolve to a unique wave solution with noncomputable values. *Boundary:* The result is representation-sensitive and is not a no-go theorem for computable Sobolev wave propagation or for the explicit coefficient carrier used here.
`richman-bridges-1999`. Fred Richman and Douglas Bridges, A Constructive Proof of Gleason's Theorem, Journal of Functional Analysis 162 (1999), 287-312. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1006/jfan.1998.3372	*Supports:* Gleason's theorem has a constructive proof after its hypotheses and conclusion are given an appropriate constructive formulation. *Boundary:* This is a reformulation and proof of a specific probability-representation theorem; it is not a constructive derivation of all quantum mechanics.
`selivanova-selivanov-2013`. Svetlana Selivanova and Victor Selivanov, Computing Solution Operators of Boundary-value Problems for Some Linear Hyperbolic Systems of PDEs, Logical Methods in Computer Science 13(4:13) (2017). `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1305.2494	*Supports:* For symmetric hyperbolic systems on a cube with computable coefficients, the Cauchy solution operator is computable in the stated TTE representations; dissipative boundary-value problems are also treated under additional hypotheses.; The proof uses rational finite-difference approximants with effective error estimates rather than an explicit solution formula. *Boundary:* A TTE computability theorem is not a Bishop-constructive derivation, an RCA_0 upper bound or reversal, or a theorem for globally hyperbolic manifolds and advanced/retarded Green support.
`selivanova-selivanov-2018`. Svetlana Selivanova and Victor Selivanov, Bit Complexity of Computing Solutions for Symmetric Hyperbolic Systems of PDEs with Guaranteed Precision, 2020. `PRIMARY_RESEARCH / CONTENT_PINNED`. https://arxiv.org/abs/1807.03140	*Supports:* The symmetric-hyperbolic computability programme admits explicit bit-complexity upper bounds under the paper's representations and coefficient hypotheses. *Boundary:* Complexity of represented solution operators neither supplies strict causal Green support nor calibrates a subsystem of second-order arithmetic.
`simpson-1984-ode`. Stephen G. Simpson, Which set existence axioms are needed to prove the Cauchy/Peano theorem for ordinary differential equations?, Journal of Symbolic Logic 49 (1984), 783-802. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.2307/2274131	*Supports:* Over RCA_0, the Cauchy-Peano existence theorem for ordinary differential equations is equivalent to WKL_0; stronger related ODE solution principles reach ACA_0. *Boundary:* This calibrates ODE existence, not a hyperbolic PDE energy theorem, finite propagation, or a Green map.
`weihrauch-zhong-2002`. Klaus Weihrauch and Ning Zhong, Is wave propagation computable or can wave computers beat the Turing machine?, Proceedings of the London Mathematical Society 85 (2002), 312-332. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1112/S0024611502013643	*Supports:* The paper proves computability of the wave propagator for specified continuously differentiable and Sobolev representations, with a loss of one derivative in the former setting and no loss in the stated Sobolev setting. *Boundary:* This is TTE computable analysis, not a reverse-mathematical RCA_0/WKL_0/ACA_0 classification and not specifically the cylinder fixture.
`weihrauch-zhong-2006-fundamental`. Klaus Weihrauch and Ning Zhong, An Algorithm for Computing Fundamental Solutions, SIAM Journal on Computing 35 (2006), 1283-1294. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1137/S0097539704446360	*Supports:* The paper computes a distributional fundamental solution for every constant-coefficient differential operator in its represented setting. *Boundary:* A fundamental solution is not automatically a retarded or advanced Green operator and the record does not establish causal support.
`weihrauch-zhong-2007-cauchy`. Klaus Weihrauch and Ning Zhong, Computable analysis of the abstract Cauchy problem in a Banach space and its applications I, Mathematical Logic Quarterly 53 (2007), 511-531. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1002/malq.200710015	*Supports:* The paper gives necessary and sufficient conditions for computability of the solution operator of an abstract linear Cauchy problem with a possibly unbounded Banach-space operator in the representation approach. *Boundary:* This is TTE computability, not Bishop constructivity or reverse mathematics, and it does not prove finite propagation.
`zhong-1999-sobolev`. Ning Zhong, Computability structure of the Sobolev spaces and its applications, Theoretical Computer Science 219 (1999), 487-510. `PRIMARY_RESEARCH / METADATA_ONLY`. https://doi.org/10.1016/S0304-3975(98)00302-8	*Supports:* The paper defines computability structures on Sobolev spaces and applies them to classes of second-order hyperbolic and parabolic PDEs. *Boundary:* A computable-analysis result is not a Bishop proof, a subsystem reversal, or a causal-support theorem.

Reference and provenance	Recorded evidential role and boundary
zhong-weihrauch-2003-distributions. Ning Zhong and Klaus Weihrauch, Computability theory of generalized functions, Journal of the ACM 50 (2003), 469-505. PRIMARY_RESEARCH / METADATA_ONLY. https://doi.org/10.1145/792538.792542	*Supports:* The paper defines TTE computability for test functions and distributions and proves computability of the solution operator for the distributional inhomogeneous three-dimensional wave equation. *Boundary:* The reviewed abstract does not certify retarded or advanced selection or a strict cone-support theorem; this is therefore not direct causal-Green evidence.
zohar-burrello-2014. Erez Zohar and Michele Burrello, Formulation of lattice gauge theories for quantum simulations, Physical Review D 91, 054506 (2015). PRIMARY_RESEARCH / CONTENT_PINNED. https://arxiv.org/abs/1409.3085	*Supports:* Hamiltonian lattice gauge models can encode local gauge invariance and Gauss-law constraints for finite, compact, and truncated gauge groups. *Boundary:* A simulator-oriented lattice Hamiltonian is not a finite Weyl-gravity BV complex, a regulator-independent continuum limit, or a QME certificate.

A.5.2 Complete local result and certificate list

Local records are repository results, not external publications. Their positive flags state what the registered certificate establishes; the complete negative list prevents a bounded result from being promoted to a stronger mathematical or physical claim.

Table 9: Complete local result and certificate register (32 records).

Record and repository locator	Certified scope and complete boundary
FOUNDATIONAL_BT_CORNER_BORN_INTERFACE_V1. CERTIFIED_CROSS_CELL_INTERFACE / SUFFICIENCY_PROVED. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: foundations/results/FOUNDATIONAL_BT_CORNER_BORN_INTERFACE_V1.json. Report: foundations/reports/bt-corner-born-interface.md.	*Establishes:* Certified positive flags: cross cell interface certified; same corner state used on both sides; event effect map constructed; conditional probabilities nonnegative; conditional probabilities normalized; interface independent rederivation passed; predecessor note only provenance drift recorded. *Does not establish:* a probability rule for arbitrary Krein operators without the five displayed hypotheses; identity of the full algebraic C* and Krein carriers; a canonical choice of the finite incoming projection; a thermodynamic normal state or finite trace of the identity; the nonlinear Bateman-Turok Eq. (19), a complete NLO probability, or an all-order process; a gravitational, BV-BRST, QME, residual-transfer, or LORENTZIAN-CAUSAL result; empirical agreement or a complete physical theory; repair or successful replay of the predecessor certificate's stale narrative-note provenance pin.
FOUNDATIONAL_BT_EUCLIDEAN_LATTICE_IMPORT_V1. FOUNDATIONAL_EVIDENCE_IMPORT_AND_SCOPED_CARRIER_INTERFACE / CLASSIFIED. Dependency tags: LOCAL-ALGEBRAIC, EUCLIDEAN-SPECTRAL. Result: foundations/results/FOUNDATIONAL_BT_EUCLIDEAN_LATTICE_IMPORT_V1.json. Report: foundations/reports/bt-euclidean-lattice-foundational-import.md.	*Establishes:* Certified positive flags: five finite euclidean capabilities imported; finite partition function supports normalized gibbs state; independent sampler coarse reproduction recorded. *Does not establish:* a finite exact carrier or finite probability sample space; a physical-state-selection theorem from zero-mode fixing; reflection positivity or Osterwalder-Schrader reconstruction; a continuum or infinite-volume limit; analytic continuation to the BT Krein theory; a Born rule, scattering probability, or laboratory event rate; empirical validation or out-of-sample robustness; a graviton or full Weyl-gravity lattice theory; a new physical dimension; anything LORENTZIAN-CAUSAL.
FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1. FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SEPARATED. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: foundations/results/FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1.json. Report: foundations/reports/bt-separable-cstar-state-chain.md.	*Establishes:* Certified positive flags: separable detector algebra constructed; explicit zf states constructed; explicit corner gns constructed. *Does not establish:* that B(l2(Z)) is separable; a finite trace of the identity; a canonical choice of incoming detector projection; a thermodynamic normal state; that the coherent CCR state is the BT physical state; full nonlinear Moller dynamics or Eq. (19); a gravitational or BRST lift; anything LORENTZIAN-CAUSAL.

Record and repository locator	Certified scope and complete boundary
`FOUNDATIONAL_CODED_LOCAL_WEAK_WAVE_TEST_CLASS_V1`. `FINITE_LOCALIZED_SEPARATING_TEST_CLASS_AND_COEFFICIENT_WEAK_EQUATION / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_CODED_LOCAL_WEAK_WAVE_TEST_CLASS_V1.json`. Report: `foundations/reports/coded-local-weak-wave-test-class-v1.md`.	*Establishes:* Certified positive flags: finite localized test class constructed; labelled finite carrier separated; coefficient transport identities proved; coefficient scalar weak wave identity proved; pra finite certificate; rca0 completion transfer. *Does not establish:* separation after forgetting the declared right/left chiral labels; a separating algebra for arbitrary completed L2 states or gauge equivalence classes; the weak wave equation against every smooth compactly supported test; a representation-independent distributional solution theorem; pointwise differentiability of the step components; strict finite propagation or causal support; an advanced or retarded Green operator; a variable-coefficient, curved-spacetime, Weyl, or metric-BV equation; a probability rule or empirically calibrated detector; a new LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1`. `CODED_SECOND_ORDER_ARITHMETIC_UPPER_BOUND / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1.json`. Report: `foundations/reports/coded-polygonal-wave-rca0.md`.	*Establishes:* Certified positive flags: rca0 upper bound for declared representation; completed energy state constructed; real time solution name constructed; energy conservation proved; cauchy uniqueness in declared carrier. *Does not establish:* that RCA_0 is necessary or the weakest base; a WKL_0, ACA_0, or Choice lower bound; the same upper bound for bare finite-energy existence without a fast Cauchy name; representation invariance; a localized spacetime-distribution theorem; finite propagation or an advanced/retarded Green map; a variable-coefficient or curved-spacetime Cauchy theorem; the biwave or metric-BV propagator; a new LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_CODED_WAVE_FRONTIER_V2`. `CODED_UPPER_BOUND_AND_LITERATURE_FRONTIER / L2_UPPER_BOUND_CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_CODED_WAVE_FRONTIER_V2.json`. Report: `foundations/reports/coded-wave-frontier-v2.md`.	*Establishes:* Certified positive flags: declared representation rca0 upper bound; literature screen completed. *Does not establish:* literature completeness; that RCA_0 is the weakest base; a reverse-mathematical equivalence for wave evolution; a Bishop-constructive hyperbolic Green theorem; a ZF-without-Choice PDE theorem; that a distributional fundamental solution is retarded or advanced; strict causal support; a new Lorentzian Weyl result.
`FOUNDATIONAL_CODED_WAVE_OBSERVABLE_RECONSTRUCTION_V1`. `EXPLICIT_UNIFORM_OBSERVABLE_RECONSTRUCTION / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_CODED_WAVE_OBSERVABLE_RECONSTRUCTION_V1.json`. Report: `foundations/reports/coded-wave-observable-reconstruction-v1.md`.	*Establishes:* Certified positive flags: declared rational initial data; declared bounded linear observable; finite rational approximants constructed; uniform bounded time convergence proved; explicit cutoff function proved; rca0 upper bound proved; fixed fixture arithmetic primitive recursive. *Does not establish:* that RCA_0 is necessary or the weakest base; uniform reconstruction for unnamed convergent data without a supplied rate or finite rational code; reconstruction of the full wave state from this one smeared observable; a point-local field observable or probability rule; a localized spacetime-distributional weak equation; finite propagation, causal support, or an advanced/retarded Green operator; a variable-coefficient, curved-spacetime, biwave, or metric-BV theorem; empirical calibration of the detector profile; a new LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_CODED_WEAK_WAVE_H2_TEST_COMPLETION_V1`. `REPRESENTATION_AWARE_SOBOLEV_TEST_COMPLETION_AND_WEAK_SOLUTION_EXTENSION / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_CODED_WEAK_WAVE_H2_TEST_COMPLETION_V1.json`. Report: `foundations/reports/coded-weak-wave-h2-test-completion-v1.md`.	*Establishes:* Certified positive flags: rational h2 test code carrier constructed; named h2 test completion constructed; explicit residual modulus proved; weak solution extended to every named h2 test; represented smooth tests covered; continuous distributional state map constructed; energy image evolution wellposed. *Does not establish:* a uniform algorithm assigning an H2 name to every bare extensional smooth test; the nonmetrizable LF topology of the unrestricted classical test-function space; a representation-independent distribution theory or weakest-base reversal; uniqueness among arbitrary distributional weak solutions outside the coded energy image; pointwise differentiability of the step chiral derivatives; strict finite propagation or causal support; an advanced or retarded Green operator; a variable-coefficient, curved-spacetime, Weyl, or metric-BV equation; a probability rule or empirical calibration; a new LORENTZIAN-CAUSAL result.

Record and repository locator	Certified scope and complete boundary
`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1`. `FOUNDATIONAL_STRENGTH_LADDER / SEPARATED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1.json`. Report: `foundations/reports/cylinder-wave-strength-ladder.md`.	*Establishes:* Certified positive flags: finite cylinder wave exact; explicit tail modulus exact; finite spectral locality obstruction exact; typed physics to mathematics graph constructed. *Does not establish:* that PRA proves an infinite completed energy-space theorem; RCA_0 sufficiency for the selected representation; a WKL_0, ACA_0, or stronger reversal; a uniform constructive modulus extractor from bare finite-energy existence; equivalence between coefficient-weak and spacetime-distributional solutions; finite propagation or causal support from Fourier truncations; a normally-hyperbolic Green theorem; the full biwave or metric-BV propagator; renormalized Lorentzian products or a QME theorem; a new LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V2`. `FOUNDATIONAL_STRENGTH_LADDER / L2_UPPER_BOUND_CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V2.json`. Report: `foundations/reports/cylinder-wave-strength-ladder-v2.md`.	*Establishes:* Certified positive flags: finite cylinder wave exact; explicit tail modulus exact; finite spectral locality obstruction exact; typed physics to mathematics graph constructed; arbitrary energy completion formalized in rca0; declared representation rca0 upper bound. *Does not establish:* that RCA_0 is necessary or the weakest base; a WKL_0, ACA_0, or Choice reversal; an upper bound for representations lacking prescribed Cauchy rates; representation invariance; a coefficient-weak or localized spacetime-distribution theorem; finite propagation or causal support from Fourier or polygonal evolution; a normally-hyperbolic Green theorem; the full biwave or metric-BV propagator; renormalized Lorentzian products or a QME theorem; a new LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_EXPLICIT_ENERGY_SPECTRAL_FRAGMENT_ZF_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_EXPLICIT_ENERGY_SPECTRAL_FRAGMENT_ZF_V1.json`. Report: `foundations/reports/explicit-energy-spectral-fragment-audit.md`.	*Establishes:* Certified positive flags: explicit energy self adjointness route classified; fock fixed energy finiteness route classified. *Does not establish:* ZF is the weakest base; a reversal or independence theorem; foundations of arbitrary self-adjoint operators; Euclidean Hessian spectral measures; determinants or traces; an interacting Hamiltonian; a LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_EXPLICIT_MODE_DYNAMICS_ZF_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_EXPLICIT_MODE_DYNAMICS_ZF_V1.json`. Report: `foundations/reports/explicit-mode-dynamics-zf.md`.	*Establishes:* Certified positive flags: explicit strongly continuous unitary group constructed; explicit j unitary group constructed; explicit point norm cstar dynamics constructed; explicit fock unitary group constructed. *Does not establish:* that ZF is the weakest sufficient base; exponentiation of the other unbounded conformal generators; a representation of the full conformal group; nonlinear Bateman-Turok Moller or full-orbit dynamics; an interacting Weyl Hamiltonian or automorphism group; causal Green propagation or a Lorentzian off-shell BV propagator; state selection, a KMS state, or a generalized Born rule; thermodynamic-limit implementability in an inequivalent representation; anything LORENTZIAN-CAUSAL.
`FOUNDATIONAL_FINITE_BRST_TWENTY_CELL_CLOSURE_V1`. `EXACT_FINITE_BRST_MULTI_CELL_CERTIFICATE / RESIDUAL_TRANSFERRED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FINITE_BRST_TWENTY_CELL_CLOSURE_V1.json`. Report: `foundations/reports/finite-brst-twenty-cell-closure.md`.	*Establishes:* Certified positive flags: exactly twenty previously unmapped cells classified; seventeen local results; three pieces only results; counterterms classified before qme; anomalies classified before qme; finite one loop qme restored; residual transfer after restoration. *Does not establish:* that the six-generator toy complex is the classical Weyl BV complex; Weyl-gravity counterterm or anomaly classification; a coefficient calculation for a physical quantum field theory; a continuum renormalized product, regulator-independent limit, or time-ordered product; a nonlinear BV Laplacian/antibracket realization beyond the declared linearized one-loop cochain equation; an all-loop, strict pure-Weyl, or Lorentzian QME; transfer to the certified Weyl residual complex; positivity, unitarity, a physical state, scattering, or empirical agreement; equivalence of general finite, Hilbert, and Krein carriers; a weakest mathematical base or reverse-mathematics lower bound; a complete physical theory or LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_FINITE_FIELD_FINITE_MODE_NON_EQUIVALENCE_V1`. `FOUNDATIONAL_NON_EQUIVALENCE_LEDGER / SEPARATED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FINITE_FIELD_FINITE_MODE_NON_EQUIVALENCE_V1.json`. Report: `foundations/reports/finite-field-versus-finite-mode.md`.	*Establishes:* Certified positive flags: four objects typed; pairwise non equivalence witnessed. *Does not establish:* a no-go theorem for finite-field physics; a continuum limit of finite-field Wigner systems; equivalence of varying prime-power dimensions; convergence of Weyl mode cutoffs; a finitist construction of real or complex analysis; empirical evidence for or against actual infinity; a finite formulation of interacting Weyl gravity; anything LORENTZIAN-CAUSAL.

Record and repository locator	Certified scope and complete boundary
`FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1`. `EXACT_FINITE_DISCRETE_CAUSAL_GREEN_KERNEL / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1.json`. Report: `foundations/reports/finite-graph-wave-causality.md`.	*Establishes:* Certified positive flags: finite exact retarded kernel constructed; finite exact advanced kernel constructed; graph step support certified. *Does not establish:* continuum finite propagation; a Lorentzian advanced or retarded Green operator; CFL stability or convergence under refinement; a regulator-independent continuum limit; a Weyl metric BV propagator; a reverse-mathematical classification of continuum PDE.
`FOUNDATIONAL_FINITE_OPERATOR_TEN_CELL_CLOSURE_V1`. `SCOPED_MULTI_CELL_LOCAL_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FINITE_OPERATOR_TEN_CELL_CLOSURE_V1.json`. Report: `foundations/reports/finite-operator-ten-cell-closure.md`.	*Establishes:* Certified positive flags: exactly ten previously unmapped cells classified; nine local results; one pieces only result; finite hilbert interaction constructed; finite krein interaction constructed; constructive krein state probability constructed; fixed model counterterm space classified; finite regulated products constructed. *Does not establish:* equivalence of finite exact, Hilbert, and Krein carrier classes beyond the named finite realization; an infinite-dimensional or continuum interacting theory; a Weyl-gravity or Bateman–Turok interaction vertex; Weyl counterterm or anomaly classification; a continuum renormalized product or regulator-independent limit; QME restoration or residual quantum transfer; a general constructive probability rule for arbitrary Krein processes; a weakest mathematical base or reverse-mathematics lower bound; causal propagation, empirical agreement, or a LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_FINITE_QUBIT_INTERACTION_CORE_V1`. `EXACT_FINITE_INTERACTION_WITNESS / VERIFIED_BOUNDED_MODEL`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FINITE_QUBIT_INTERACTION_CORE_V1.json`. Report: `foundations/reports/finite-qubit-interaction-core.md`.	*Establishes:* Certified positive flags: exact finite interaction model; exact state and probability witness; exact entanglement generation witness; exact finite krein companion. *Does not establish:* a weakest logical or set-theoretic base; that PRA, ZF without Choice, Bishop constructivism, and finitism are equivalent regimes; an infinite-dimensional Hilbert, Krein, or C*-completion; a continuum field theory or controlled continuum limit; a Weyl-gravity interaction vertex; counterterm or anomaly classification; renormalization or restoration of the quantum master equation; a physical selection principle for the displayed states; a Lorentzian propagator, Hadamard state, or causal quantum theory.
`FOUNDATIONAL_FIXED_SUPPORT_SMOOTH_TO_H2_TRANSLATOR_V1`. `REPRESENTATION_TRANSLATOR_WITH_EXPLICIT_SUPPORT_AND_MODULUS / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_FIXED_SUPPORT_SMOOTH_TO_H2_TRANSLATOR_V1.json`. Report: `foundations/reports/fixed-support-smooth-to-h2-translator-v1.md`.	*Establishes:* Certified positive flags: fixed support smooth name translated; rational h2 fast name constructed; explicit cutoff and modulus constructed. *Does not establish:* a name for a bare extensional smooth function without support and rate advice; a uniform selection of compact support bounds; the unrestricted LF topology of compactly supported smooth tests; a weakest-base reversal or equivalence; strict causal support or an advanced or retarded Green operator; a variable-coefficient, curved-spacetime, Weyl, or metric-BV theorem.
`FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC. Result: `foundations/results/FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1.json`. Report: `foundations/reports/free-bv-energy2-pra-sdr.md`.	*Establishes:* Certified positive flags: fixed energy2 integer sdr verified; pra sufficiency for fixed checker; hahn banach avoided for displayed certificate. *Does not establish:* That PRA is the weakest possible base theory.; A reversal or necessity theorem for any comprehension or choice principle.; That the source producer, all energy levels, the Fock lift, or the complete field-derived BV complex is independently verified.; That arbitrary finite-dimensional linear algebra avoids Choice without an explicit presentation and witness.; A choice-free countable or infinite-dimensional Hilbert, Krein, spectral, Green-operator, state, or renormalization construction.; A complete classical import freeze or any publishable quantum promotion.; Any EUCLIDEAN-SPECTRAL, REDUCED-MODE, or LORENTZIAN-CAUSAL claim..
`FOUNDATIONAL_FULL_SURFACE_GAP_AUDIT_V1`. `TYPED_FULL_CARTESIAN_GAP_ASSESSMENT / CLASSIFIED`. Dependency tags: LOCAL-ALGEBRAIC, EUCLIDEAN-SPECTRAL, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_FULL_SURFACE_GAP_AUDIT_V1.json`. Report: `foundations/reports/full-surface-gap-audit.md`.	*Establishes:* Certified positive flags: all 175 remaining coordinates reviewed; all 576 coordinates formulated; new reviewed gap status defined. *Does not establish:* a direct mathematical or physical result for any of the 175 coordinates; that every formulated coordinate is realizable in one common physical theory; that adjacent evidence transfers across a foundation, carrier, or obligation axis; that any newly reviewed gap is a programme priority; literature completeness or absence of relevant literature; impossibility, independence, inconsistency, or a no-go theorem; a weakest foundation, carrier equivalence, or continuum limit; a complete Weyl theory, quantum completion, empirical agreement, or LORENTZIAN-CAUSAL result.

Record and repository locator	Certified scope and complete boundary
`FOUNDATIONAL_HARDY_CONTINUITY_KN_AUDIT_V1`. `FOUNDATIONAL_PROOF_DEPENDENCY_AUDIT / SEPARATED`. Dependency tags: LOCAL-ALGEBRAIC. Result: `foundations/results/FOUNDATIONAL_HARDY_CONTINUITY_KN_AUDIT_V1.json`. Report: `foundations/reports/hardy-continuity-kn-foundational-audit.md`.	*Establishes:* Certified positive flags: hardy kn step audited; rca0 sufficient for explicit modulus route. *Does not establish:* a weakest-base theorem; a reversal to RCA_0 or WKL_0; the strength of extracting a modulus from every pointwise-continuous code; the full K=N^r derivation; the K=N^2 quantum reconstruction; that Axiom 5 is empirically necessary; infinite-dimensional quantum theory; anything LORENTZIAN-CAUSAL.
`FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1.json`. Report: `foundations/reports/krein-explicit-j-zf-audit.md`.	*Establishes:* Certified positive flags: finite integral j verified; explicit mode index countable in zf; zf one particle completion sufficient; zf explicit bosonic fock sufficient. *Does not establish:* That ZF is the weakest foundation for the countable completion or Fock construction.; A reverse-mathematical equivalence or necessity theorem for Infinity, comprehension, Countable Choice, or Choice.; That an arbitrary indefinite inner-product space admits a fundamental decomposition without Choice.; That an arbitrary Hilbert space has an orthonormal basis in ZF.; Foundational classifications of the all-real-order Sobolev scale, closed generator domains, exponentiation, or the conformal group representation.; A trace-class density operator, normal state, generalized Born rule, particle-unitarity theorem, or interacting implementer.; A positive graviton Hilbert space, complete covariant BV propagator, Hadamard state, QME theorem, or any LORENTZIAN-CAUSAL claim..
`FOUNDATIONAL_KREIN_FOCK_GROUND_STATE_DYNAMICS_INTERFACE_V1`. `CERTIFIED_CROSS_CELL_INTERFACE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_KREIN_FOCK_GROUND_STATE_DYNAMICS_INTERFACE_V1.json`. Report: `foundations/reports/krein-fock-ground-state-dynamics-interface.md`.	*Establishes:* Certified positive flags: cross cell interface certified; free ground state selected; unique vector ground state proved; unique normal zero energy density state proved; vacuum dynamics invariance proved; shared fock objects identified; interface independent rederivation passed. *Does not establish:* that stationarity alone selects a unique state; excited energy eigenstates and mixtures can also be stationary; an interacting Weyl or Bateman–Turok ground state; a KMS, Hadamard, incoming, outgoing, detector-conditioned, or BRST-compatible state; selection among non-normal states without a density operator; a thermodynamic limit or implementability in an inequivalent representation; causal propagation, a Green operator, or a Lorentzian off-shell BV propagator; a generalized Born rule, prediction chain, or empirical agreement; a weakest-base reverse-mathematics theorem; a gravitational, QME, residual-transfer, or LORENTZIAN-CAUSAL result.
`FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1`. `FOUNDATIONAL_DEPENDENCY_CERTIFICATE / SUFFICIENCY_PROVED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1.json`. Report: `foundations/reports/krein-state-selection-zf.md`.	*Establishes:* Certified positive flags: explicit zf krein vector states constructed; explicit rank one density witnesses constructed; explicit fock coordinate states constructed. *Does not establish:* a unique or canonical state from J; nonexistence of singular permutation-invariant states; that sign-preserving coordinate permutations are a required physical symmetry; a KMS, Hadamard, BRST-compatible, incoming, outgoing, or interacting state; a generalized Born rule or physical probability interpretation; trace-norm thermodynamic convergence of the Bateman-Turok construction; positivity with respect to the Krein adjoint where Hilbert-adjoint positivity was used; a weakest-base reverse-mathematics theorem; anything LORENTZIAN-CAUSAL.
`FOUNDATIONAL_LOW_HANGING_CELL_CLOSURE_AUDIT_V1`. `FOUNDATIONAL_SCOPE_AUDIT / SEPARATED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_LOW_HANGING_CELL_CLOSURE_AUDIT_V1.json`. Report: `foundations/reports/low-hanging-cell-closure-audit.md`.	*Establishes:* Certified positive flags: three existing result closures identified; local bv cohomology imported with scope; local counterterm anomaly classes imported with scope; finite cutoff dynamics imported with scope; remaining assessed open cells exhaustively triaged. *Does not establish:* a complete global smooth or distributional off-shell BV complex; the classical import freeze gate; a complete classification beyond the regular Bach locus and declared derivative bounds; counterterm or anomaly coefficients; regulated Slavnov breaking, renormalized products, or QME restoration; residual transfer or a one-particle interpretation of residual classes; a finite-to-continuum comparison or regulator-independent limit; causal propagation or any LORENTZIAN-CAUSAL theorem; that any remaining open cell is impossible or unimportant; that any of the 157 NOT_MAPPED cells has been assessed.

Record and repository locator	Certified scope and complete boundary
`FOUNDATIONAL_NORMAL_HYPERBOLIC_FACTOR_ATLAS_V1`. `LITERATURE_AND_FOUNDATIONAL_DEPENDENCY_ATLAS / LITERATURE_SCOPED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_NORMAL_HYPERBOLIC_FACTOR_ATLAS_V1.json`. Report: `foundations/reports/normal-hyperbolic-factor-foundations.md`.	*Establishes:* Certified positive flags: classical factor theorem identified; computable upper bound identified; finite exact support constructed. *Does not establish:* literature completeness; a weakest subsystem; an RCA_0, WKL_0 or ACA_0 equivalence; a Bishop-constructive globally hyperbolic Green theorem; Choice avoidance for Sobolev/distribution theory; a continuum limit from finite graphs; a full off-shell Weyl metric BV propagator; a BRST-compatible Hadamard state; renormalized Lorentzian products or a Lorentzian QME.
`FOUNDATIONAL_SCALAR_BIWAVE_TO_WEYL_BV_DEPENDENCY_DELTA_V1`. `FAIL_CLOSED_CROSS_THEORY_DEPENDENCY_DELTA / CERTIFIED_DEPENDENCY_DELTA`. Dependency tags: LOCAL-ALGEBRAIC, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_SCALAR_BIWAVE_TO_WEYL_BV_DEPENDENCY_DELTA_V1.json`. Report: `foundations/reports/scalar-biwave-to-weyl-bv-dependency-delta-v1.md`.	*Establishes:* Certified positive flags: scalar biwave green imported; transfer requirements classified; scoped nariai weyl result recorded; scoped architectural no gos recorded. *Does not establish:* a full off-shell Lorentzian Weyl metric BV propagator; a BRST-compatible Hadamard state; renormalized Lorentzian time-ordered products; a causal perturbative AQFT construction; a Lorentzian quantum-master-equation theorem; a passed classical import freeze gate; that the two scoped no-go theorems rule out other architectures; that the centered Weyl-square deformation classes are particles; a weakest-base reversal for curved tensor PDE; empirical adequacy or a complete physical theory.
`FOUNDATIONAL_SCALAR_MINKOWSKI_BIWAVE_GREEN_V1`. `SCALAR_FOURTH_ORDER_LORENTZIAN_GREEN_CONSTRUCTION / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_SCALAR_MINKOWSKI_BIWAVE_GREEN_V1.json`. Report: `foundations/reports/scalar-minkowski-biwave-green-v1.md`.	*Establishes:* Certified positive flags: scalar biwave retarded green constructed; scalar biwave advanced green constructed; two sided test code identities proved; strict causal support proved; adjoint duality proved; four zero data selection proved; named finite horizon extension proved; canonical construction avoids choice. *Does not establish:* a global bounded-energy estimate for persistent retarded biwave solutions; uniqueness among arbitrary distributional solutions; support, horizon, or convergence data selected from a bare extensional source; a curved or variable-coefficient tensor Green operator; a gauge-fixed Green-hyperbolic Weyl BV complex; BRST-compatible causal homotopies; a Hadamard state or wavefront-set theorem; renormalized Lorentzian time-ordered products, causal pAQFT, or a Lorentzian QME; a weakest-base reversal; empirical adequacy or a complete physical theory.
`FOUNDATIONAL_SCALAR_MINKOWSKI_GREEN_CHOICE_AUDIT_V1`. `SCALAR_LORENTZIAN_GREEN_OPERATOR_AND_CHOICE_DEPENDENCY_AUDIT / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, LORENTZIAN-CAUSAL. Result: `foundations/results/FOUNDATIONAL_SCALAR_MINKOWSKI_GREEN_CHOICE_AUDIT_V1.json`. Report: `foundations/reports/scalar-minkowski-green-choice-audit-v1.md`.	*Establishes:* Certified positive flags: scalar retarded green constructed; scalar advanced green constructed; two sided test code identities proved; strict causal support proved; adjoint duality proved; named energy extension proved; canonical construction avoids choice; lorentzian causal scalar claim. *Does not establish:* uniqueness among all arbitrary distributional solutions; support or convergence data selected from a bare extensional source; a variable-coefficient or curved-spacetime Green operator; a Green operator for conformal Weyl gravity or the metric BV complex; a BRST-compatible Hadamard state or renormalized time-ordered products; a causal perturbative AQFT construction or Lorentzian quantum master equation; a weakest-base reversal; empirical adequacy or a complete physical theory.
`FOUNDATIONAL_SUPPORT_INDEXED_TEST_SPACE_COMPARISON_V1`. `SUPPORT_INDEXED_REPRESENTED_UNION_COMPARISON_WITH_LF_BOUNDARY / CERTIFIED`. Dependency tags: LOCAL-ALGEBRAIC, REDUCED-MODE. Result: `foundations/results/FOUNDATIONAL_SUPPORT_INDEXED_TEST_SPACE_COMPARISON_V1.json`. Report: `foundations/reports/support-indexed-test-space-comparison-v1.md`.	*Establishes:* Certified positive flags: support indexed represented union constructed; conventional and tagged names equivalent; stage inclusion compatibility proved; every represented smooth test embedded in h2; weak identity assembled over named tests. *Does not establish:* a support bound selected uniformly from a bare extensional function; equality between the represented quotient topology and the full classical locally convex LF topology; surjectivity of the smooth-test embedding onto the H2 completion; a single metrization or metric completion of the classical test-function LF space; a weakest-base reversal; causal Green propagation, a Weyl equation, or a metric-BV theorem.
`FOUNDATIONAL_TOPOS_WEYL_BV_OBSTRUCTION_LEDGER_V0`. `FOUNDATIONAL_GLOSSARY_AND_OBSTRUCTION_LEDGER / LITERATURE_SCOPED`. Dependency tags: LOCAL-ALGEBRAIC. Result: `foundations/results/FOUNDATIONAL_TOPOS_WEYL_BV_OBSTRUCTION_LEDGER_V0.json`. Report: `foundations/reports/topos-weyl-bv-obstruction-ledger.md`.	*Establishes:* Certified positive flags: glossary complete for first artifact; obstruction dag closed. *Does not establish:* a formalized topos-internal Weyl-gravity BV complex; that every ordinary Weyl-gravity construction internalizes in an arbitrary topos; a constructive theory of distributions or wavefront sets; internal retarded or advanced Green operators; a topos-internal Krein or Hilbert completion; a physical state-selection or Born-rule theorem; renormalized time-ordered products; restoration of the quantum master equation; equivalence with the repository's external classical or quantum theory; anything LORENTZIAN-CAUSAL.

Record and repository locator	Certified scope and complete boundary
FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1. FOUNDATIONAL_PROOF_DEPENDENCY_AUDIT / SEPARATED. Dependency tags: LOCAL-ALGEBRAIC, LORENTZIAN-CAUSAL. Result: foundations/results/FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1.json. Report: foundations/reports/typed-biwave-green-foundational-dependencies.md.	*Establishes:* Certified positive flags: dependency cut complete for source theorem; finite resolvent algebra replayed; conditional lorentzian theorem imported. *Does not establish:* the weakest subsystem for normally-hyperbolic PDE; Choice avoidance for Sobolev or distribution theory; the energy hypotheses for an arbitrary Weyl operator; a full off-shell metric BV propagator; a BRST-compatible Hadamard state; renormalized Lorentzian time-ordered products; a Lorentzian QME theorem; quantum theory.

A.5.3 Evidence usage crosswalk

Every evidence record is used by at least one matrix coordinate. The matrix column reports the number of coordinates carrying the record and their coverage-status composition. Graph entries use edge numbers from Table 6; ladder entries use the gate identifiers from Table 7. Clicking a record identifier returns to its full literature or local-certificate entry above.

Table 10: Complete crosswalk from the atlas views to all 83 evidence records.

Evidence record	Matrix use	Graph and ladder use
FOUNDATIONAL_BT_CORNER_BORN_INTERFACE_V1 (local)	2 coordinates (Local 2).	No graph or ladder use.
FOUNDATIONAL_BT_EUCLIDEAN_LATTICE_IMPORT_V1 (local)	6 coordinates (Local 5, Priority gap 1).	No graph or ladder use.
FOUNDATIONAL_BT_SEPARABLE_STATE_CHAIN_ZF_V1 (local)	11 coordinates (Local 6, Literature 1, Pieces 4).	No graph or ladder use.
FOUNDATIONAL_CODED_LOCAL_WEAK_WAVE_TEST_CLASS_V1 (local)	3 coordinates (Local 3).	No graph or ladder use.
FOUNDATIONAL_CODED_POLYGONAL_WAVE_RCA0_V1 (local)	3 coordinates (Local 3).	graph edges 3, 5; ladder L2_CODED_ENERGY_CARRIER.
FOUNDATIONAL_CODED_WAVE_FRONTIER_V2 (local)	2 coordinates (Local 2).	No graph or ladder use.
FOUNDATIONAL_CODED_WAVE_OBSERVABLE_RECONSTRUCTION_V1 (local)	2 coordinates (Local 2).	No graph or ladder use.
FOUNDATIONAL_CODED_WEAK_WAVE_H2_TEST_COMPLETION_V1 (local)	4 coordinates (Local 4).	No graph or ladder use.
FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V1 (local)	6 coordinates (Local 4, Literature 1, Pieces 1).	graph edges 1, 2, 7.
FOUNDATIONAL_CYLINDER_WAVE_STRENGTH_LADDER_V2 (local)	2 coordinates (Local 2).	No graph or ladder use.
FOUNDATIONAL_EXPLICIT_ENERGY_SPECTRAL_FRAGMENT_ZF_V1 (local)	8 coordinates (Local 6, Pieces 2).	No graph or ladder use.
FOUNDATIONAL_EXPLICIT_MODE_DYNAMICS_ZF_V1 (local)	15 coordinates (Local 11, Pieces 4).	No graph or ladder use.
FOUNDATIONAL_FINITE_BRST_TWENTY_CELL_CLOSURE_V1 (local)	20 coordinates (Local 17, Pieces 3).	No graph or ladder use.
FOUNDATIONAL_FINITE_FIELD_FINITE_MODE_NON_EQUIVALENCE_V1 (local)	2 coordinates (Local 1, Priority gap 1).	No graph or ladder use.
FOUNDATIONAL_FINITE_GRAPH_WAVE_CAUSALITY_V1 (local)	1 coordinates (Local 1).	No graph or ladder use.
FOUNDATIONAL_FINITE_OPERATOR_TEN_CELL_CLOSURE_V1 (local)	10 coordinates (Local 9, Pieces 1).	No graph or ladder use.
FOUNDATIONAL_FINITE_QUBIT_INTERACTION_CORE_V1 (local)	99 coordinates (Local 45, Literature 2, Pieces 52).	No graph or ladder use.
FOUNDATIONAL_FIXED_SUPPORT_SMOOTH_TO_H2_TRANSLATOR_V1 (local)	2 coordinates (Local 2).	No graph or ladder use.
FOUNDATIONAL_FREE_BV_ENERGY2_PRA_SDR_V1 (local)	15 coordinates (Local 5, Pieces 10).	No graph or ladder use.
FOUNDATIONAL_FULL_SURFACE_GAP_AUDIT_V1 (local)	175 coordinates (Local 6, Reviewed gap 169).	No graph or ladder use.
FOUNDATIONAL_HARDY_CONTINUITY_KN_AUDIT_V1 (local)	2 coordinates (Local 2).	No graph or ladder use.

Evidence record	Matrix use	Graph and ladder use
FOUNDATIONAL_KREIN_EXPLICIT_J_ZF_V1 (local)	18 coordinates (Local 12, Pieces 6).	No graph or ladder use.
FOUNDATIONAL_KREIN_FOCK_GROUND_STATE_DYNAMICS_INTERFACE_V1 (local)	2 coordinates (Local 2).	No graph or ladder use.
FOUNDATIONAL_KREIN_STATE_SELECTION_ZF_V1 (local)	12 coordinates (Local 8, Pieces 4).	No graph or ladder use.
FOUNDATIONAL_LOW_HANGING_CELL_CLOSURE_AUDIT_V1 (local)	7 coordinates (Local 3, Pieces 4).	No graph or ladder use.
FOUNDATIONAL_NORMAL_HYPERBOLIC_FACTOR_ATLAS_V1 (local)	4 coordinates (Local 2, Pieces 2).	No graph or ladder use.
FOUNDATIONAL_SCALAR_BIWAVE_TO_WEYL_BV_DEPENDENCY_DELTA_V1 (local)	2 coordinates (Local 1, Priority gap 1).	No graph or ladder use.
FOUNDATIONAL_SCALAR_MINKOWSKI_BIWAVE_GREEN_V1 (local)	2 coordinates (Local 2).	No graph or ladder use.
FOUNDATIONAL_SCALAR_MINKOWSKI_GREEN_CHOICE_AUDIT_V1 (local)	3 coordinates (Local 3).	No graph or ladder use.
FOUNDATIONAL_SUPPORT_INDEXED_TEST_SPACE_COMPARISON_V1 (local)	4 coordinates (Local 4).	No graph or ladder use.
FOUNDATIONAL_TOPOS_WEYL_BV_OBSTRUCTION_LEDGER_V0 (local)	22 coordinates (Literature 2, Pieces 17, Priority gap 3).	No graph or ladder use.
FOUNDATIONAL_TYPED_BIWAVE_GREEN_DEPENDENCY_AUDIT_V1 (local)	8 coordinates (Local 5, Pieces 3).	graph edges 8, 9; ladder L5_CAUSAL_GREEN_OPERATOR.
abramsky-coecke-2004 (literature)	22 coordinates (Local 4, Literature 10, Pieces 8).	No graph or ladder use.
baer-2015 (literature)	2 coordinates (Local 2).	No graph or ladder use.
bahr-dittrich-2009 (literature)	7 coordinates (Local 2, Literature 1, Pieces 4).	No graph or ladder use.
barnich-brandt-henneaux-2000 (literature)	4 coordinates (Literature 4).	No graph or ladder use.
bateman-turok-2026 (literature)	5 coordinates (Local 3, Literature 1, Pieces 1).	No graph or ladder use.
bender-boettcher-1998 (literature)	6 coordinates (Literature 2, Pieces 4).	No graph or ladder use.
blackadar-farah-2026 (literature)	9 coordinates (Local 1, Literature 4, Pieces 4).	No graph or ladder use.
blackadar-farah-karagila-2026 (literature)	6 coordinates (Literature 4, Pieces 2).	No graph or ladder use.
brattka-2008 (literature)	2 coordinates (Local 1, Pieces 1).	No graph or ladder use.
brenna-flori-2012 (literature)	17 coordinates (Literature 10, Pieces 7).	No graph or ladder use.
bridges-svozil-2000 (literature)	2 coordinates (Literature 1, Pieces 1).	No graph or ladder use.
bridges-wang-1998-dirichlet (literature)	1 coordinates (Literature 1).	No graph or ladder use.
brown-simpson-1986 (literature)	2 coordinates (Local 2).	No graph or ladder use.
brunetti-fredenhagen-rejzner-2013 (literature)	2 coordinates (Literature 2).	No graph or ladder use.
brunetti-fredenhagen-verch-2001 (literature)	11 coordinates (Literature 7, Pieces 4).	No graph or ladder use.
chiribella-dariano-perinotti-2011 (literature)	7 coordinates (Literature 5, Pieces 2).	No graph or ladder use.
constantin-doring-2020 (literature)	14 coordinates (Local 2, Literature 7, Pieces 5).	No graph or ladder use.
coquand-spitters-2009 (literature)	14 coordinates (Literature 10, Pieces 4).	No graph or ladder use.
dittrich-2012 (literature)	7 coordinates (Local 2, Literature 3, Pieces 2).	No graph or ladder use.
doring-2008 (literature)	18 coordinates (Literature 9, Pieces 9).	No graph or ladder use.
esmeral-ferrer-wagner-2015 (literature)	1 coordinates (Literature 1).	No graph or ladder use.
fewster-verch-2011 (literature)	2 coordinates (Literature 2).	No graph or ladder use.
fredenhagen-rejzner-2011 (literature)	18 coordinates (Literature 13, Pieces 5).	No graph or ladder use.
gibbons-hoffman-wootters-2004 (literature)	10 coordinates (Literature 7, Pieces 3).	No graph or ladder use.
gottschalk-2004 (literature)	13 coordinates (Literature 7, Pieces 6).	No graph or ladder use.
grinkevich-1996 (literature)	2 coordinates (Literature 2).	No graph or ladder use.
haag-kastler-1964 (literature)	5 coordinates (Local 3, Literature 1, Pieces 1).	No graph or ladder use.
harding-heunen-2019 (literature)	39 coordinates (Literature 21, Pieces 18).	No graph or ladder use.

Evidence record	Matrix use	Graph and ladder use
hardy-2001 (literature)	7 coordinates (Literature 5, Pieces 2).	No graph or ladder use.
henry-2014 (literature)	2 coordinates (Literature 2).	No graph or ladder use.
heunen-landsman-spitters-2009 (literature)	33 coordinates (Literature 20, Pieces 13).	No graph or ladder use.
humphreys-simpson-1996 (literature)	1 coordinates (Local 1).	No graph or ladder use.
humphreys-simpson-1999 (literature)	2 coordinates (Local 2).	No graph or ladder use.
kogut-susskind-1975 (literature)	11 coordinates (Local 2, Literature 3, Pieces 6).	No graph or ladder use.
kostrykin-potthoff-schrader-2011 (literature)	3 coordinates (Literature 3).	No graph or ladder use.
mostafazadeh-2001 (literature)	9 coordinates (Literature 4, Pieces 5).	No graph or ladder use.
muehlhoff-2010 (literature)	1 coordinates (Local 1).	No graph or ladder use.
nachtergaele-raz-schlein-sims-2007 (literature)	1 coordinates (Pieces 1).	No graph or ladder use.
neumann-pape-streicher-2018 (literature)	14 coordinates (Literature 8, Pieces 6).	No graph or ladder use.
pischke-2025-semigroups (literature)	1 coordinates (Local 1).	No graph or ladder use.
pour-el-richards-1981 (literature)	4 coordinates (Literature 1, Pieces 3).	graph edges 10.
richman-bridges-1999 (literature)	9 coordinates (Literature 4, Pieces 5).	No graph or ladder use.
selivanova-selivanov-2013 (literature)	3 coordinates (Literature 1, Pieces 2).	No graph or ladder use.
selivanova-selivanov-2018 (literature)	2 coordinates (Literature 1, Pieces 1).	No graph or ladder use.
simpson-1984-ode (literature)	1 coordinates (Local 1).	No graph or ladder use.
weihrauch-zhong-2002 (literature)	3 coordinates (Local 1, Literature 1, Pieces 1).	graph edges 10.
weihrauch-zhong-2006-fundamental (literature)	1 coordinates (Pieces 1).	No graph or ladder use.
weihrauch-zhong-2007-cauchy (literature)	1 coordinates (Literature 1).	No graph or ladder use.
zhong-1999-sobolev (literature)	1 coordinates (Literature 1).	No graph or ladder use.
zhong-weihrauch-2003-distributions (literature)	1 coordinates (Pieces 1).	No graph or ladder use.
zohar-burrello-2014 (literature)	22 coordinates (Local 5, Literature 4, Pieces 13).	No graph or ladder use.

A.6 Model-scoped assembly, prototype envelopes, and empirical calibration

The assembly view no longer treats missing downstream work as a failed test. Its model-specific maturity rails are reported independently: direct obligation coverage may be complete while composition is partial and numerical, prediction, or empirical records remain separately typed. Red is reserved for an explicit incompatibility, obstruction, or failed comparison.

First bounded end-to-end assembly. Standard GR solar-system prediction assembly: field equations to Cassini. This is one model identity and one declared observational sector, not a maximum assembled from unrelated cells. Its applicability mask requires 3 of the atlas's sixteen obligations, touches 2 others without requiring them, and explicitly places 11 outside this prediction.

The exact rail derives $g_{tt} = -1 + 2U - 2\beta U^2 + O(U^3)$ and $g_{ij} = (1 + 2\gamma U + O(U^2))\delta_{ij}$, obtains $\beta = \gamma = 1$, and therefore fixes the first-order null-delay coefficient $1 + \gamma = 2$. The separately typed empirical rail imports the publisher's displayed Cassini estimate $\gamma = 1 + (2.1 \pm 2.3) \times 10^{-5}$. Exact rational comparison puts the prediction $\gamma - 1 = 0$ inside that displayed band, at absolute standardized distance 21/23 from its centre.

All three required obligations are satisfied and all five stage interfaces are registered (three exact and two literature-scoped). The resulting disposition is bounded prediction assembly complete. It does not reproduce the Cassini raw-data reduction or likelihood, assess an independent held-out test, establish a complete theory, or transfer empirical support to Weyl gravity.

Second bounded assembly: mixed result. Mannheim conformal-gravity NGC 3198 assembly: field equation to rotation curve. This chain retains one declared Mannheim–Kazanas phenomenolog-

Stage	Status	What is established
Vacuum Einstein equations	certified exact	The declared model uses $G_{\mu\nu} = 0$ in the exterior sector.
Static spherical exterior	certified exact	The field equations integrate to $f(r) = 1 - 2m/r$ under the declared boundary normalization.
Isotropic weak-field reduction	certified exact	Exact coordinate translation and formal series give $\beta = \gamma = 1$.
Null-delay observable	certified exact	The first-order delay coefficient is $1 + \gamma = 2$.
Cassini fitted parameter	literature scoped	The publisher abstract identifies bending/delay and the measured frequency shift with $\gamma + 1$.
Published Cassini comparison	supported reported band	The exact prediction $\gamma - 1 = 0$ lies inside the displayed reported plus-minus band.

Table 11: The six typed stages of the standard-GR solar-exterior assembly. The first four are exact local derivations; the last two are explicitly literature-scoped bridges.

ical model from the Weyl action through certified local static-vacuum and orbit-law predecessors, the published thin-disk formula and NGC 3198 parameter row, an independently evaluated endpoint, and a no-refit comparison with a later SPARC rotation curve.

Stage	Status	What is established
Weyl action and Bach equation	declared model input	The classical model is four-dimensional pure metric conformal gravity with Bach equation B_mu_nu=0 in the exterior.
Mannheim–Kazanas vacuum family	certified local predecessor	The local BH0B certificate derives the complete static spherical Bach-flat family in the declared conformal gauge.
Weak-field circular-orbit law	certified local predecessor	The local BH0C certificate derives the leading weak-field beta/r + gamma r/2 - k r^2 circular-speed law and records its exact-family correction.
Luminous exponential-disk prediction	published model transcription	Mannheim–O'Brien Eqs. (5) and (20) integrate the Newtonian and linear kernels over thin stellar and gas disks and add universal linear and quadratic terms.
Published NGC 3198 parameters	content pinned transcription	The model uses the paper's distance, scale length, stellar and HI masses, fitted M/L, endpoint radius, and fixed universal constants without refitting.
Published endpoint reproduction	coarse numerical reproduction	Independent evaluation of the displayed equations predicts the endpoint velocity within the declared five-percent coarse gate of the velocity reconstructed from the paper's endpoint acceleration.
Independent SPARC curve comparison	mixed random error gate failed	Without refitting, the curve passes the declared five km/s RMS shape gate but fails the reduced-chi-squared gate based on SPARC random errors alone.

Table 12: The seven typed stages of the Mannheim conformal-gravity NGC 3198 assembly. A failed uncertainty-sensitive comparison remains visible rather than being averaged with the coarser passing checks.

Independent evaluation predicts 153.9 km/s at the paper's endpoint, compared with 149.6 km/s reconstructed from its displayed endpoint acceleration. The relative residual is 2.893%, which passes the declared 5% coarse audit gate. Across 39 later SPARC points inside that radius, the no-refit RMS residual is 4.538 km/s and passes the declared 5 km/s coarse shape gate, while the reduced χ^2 from SPARC random errors alone is 5.592 and fails the declared gate $\chi^2_\nu \leq 2$.

No parameter is refitted. SPARC is a later 3.6-micrometre photometric reduction, not the original heterogeneous blue-band dataset, and its random-error column omits inclination and other systematics. The result is therefore a no-refit external stress test with a mixed disposition, not a reproduction of the original likelihood and not evidence sufficient to promote the model to empirical support. The massive-tracer matter-coupling assumption is recorded but not resolved.

Common-protocol NGC 3198 control. The same 39 velocities, random-error-only objective, distance rescaling, and analytic stellar/gas geometry are used for all three families. The stellar scale is fitted in every family; the NFW family additionally fits halo speed and concentration.

Family	Fitted parameters		RMS (km/s)	Reduced χ^2	AICc	Gate
Newtonian baryons only	q*=1.6321		23.896	128.722	4893.56	FAIL
GR plus NFW dark halo	q*=0.8254, c200=6.802	V200=116.32,	5.148	0.965	41.44	PASS
Mannheim conformal gravity	q*=1.0645		4.694	3.202	123.78	FAIL

Table 13: Common-protocol NGC 3198 fit. AICc penalizes the two extra NFW parameters. The ranking is scoped to this one-galaxy, random-error-only analytic-disk comparison.

Mannheim has a lower unweighted RMS than NFW, while NFW has the lower uncertainty-weighted chi-squared and AICc. NFW is the only family passing the declared reduced-chi-squared gate. This does not include distance, inclination, photometric, gas-profile, or other systematic marginalization; use the full SPARC numerical mass model; impose a halo concentration–mass prior; generalize beyond NGC 3198; or select a complete physical theory.

Prototype	Direct obligations	Certified joins	Coverage rail	Composition rail
Mainstream GR and quantum-field-theory reference	16/16	2/7	SATISFIED	PARTIALLY_CERTIFIED
Algebraic QFT and local-covariance tradition	12/16	0/7	OPEN	NOT_ASSESSED
Finite, discrete, and exactly checkable models	10/16	0/7	OPEN	NOT_ASSESSED
Bateman–Turok hidden-ghost-parity programme	8/16	0/7	OPEN	NOT_ASSESSED
Reverse mathematics and weak-foundation programme	9/16	0/7	OPEN	NOT_ASSESSED
Mannheim conformal-gravity programme	15/16	1/7	OPEN	PARTIALLY_CERTIFIED
Pure-Weyl BV–BFV and causal programme	15/16	2/7	OPEN	PARTIALLY_CERTIFIED
Constructive and computable physics tradition	12/16	0/7	OPEN	NOT_ASSESSED
Topos and internal quantum-foundations tradition	9/16	0/7	OPEN	NOT_ASSESSED

Table 14: Cube-selected coverage and composition maturity. The classical-standard mixed-carrier reference has complete direct coverage; this does not certify the unregistered joins.

Cube-selected prototype envelopes. The BT positive Euclidean lattice programme is the only prototype with a numerical-reproducibility record. Its status is COARSE REPRODUCTION ONLY: two independent algorithms pass the declared four-standard-error gate but not the two-standard-error precision gate. The empirical and out-of-sample rails remain empty. The separate Euclidean-to-Krein carrier record is incompatible only with identification as the same full nonperturbative measure; it does not refute every conditional bridge.

External positive control. Standard general relativity: observational-domain control. A calibration control showing that the atlas can distinguish registered predictions and successful empirical comparisons from missing records. It is not selected from the foundations cube and is not a candidate Weyl-gravity assembly.

Table 15: Registered standard-GR control comparisons.

Benchmark	Primary source	Registered finding and boundary
Solar-system dynamics	B. Bertotti, L. Iess, and P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, Nature 425, 374-376 (2003). https://doi.org/10.1038/nature01997	The reported estimate is consistent with the standard-GR prediction within the experiment's stated uncertainty. Boundary: Does not test strong-field dynamics, gravitational radiation, galactic phenomenology, cosmology, quantum gravity, or a foundations-cube composition map.
Compact binaries	M. Kramer et al., Strong-Field Gravity Tests with the Double Pulsar, Physical Review X 11, 041050 (2021). https://doi.org/10.1103/PhysRevX.11.041050	The tested post-Keplerian and radiative predictions are mutually consistent with GR at the reported precision. Boundary: A high-precision strong-field binary test, not a proof of GR in arbitrary compact objects, cosmology, or the quantum domain.
Gravitational waves	LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, Tests of General Relativity with GWTC-3, arXiv:2112.06861 (2021). https://arxiv.org/abs/2112.06861	The reported tests find no evidence for deviations from GR in the analysed GWTC-3 signals. Boundary: Consistency within the analysed catalogue and waveform systematics is not proof of GR for every source, propagation environment, polarization model, or cosmological history.
Gravitational waves	B. P. Abbott et al., Gravitational Waves and Gamma-rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A, Astrophysical Journal Letters 848, L13 (2017). https://arxiv.org/abs/1710.05834	The observed delay yields stringent bounds compatible with standard luminal GR propagation in the tested setting. Boundary: Does not isolate all modified-gravity effects, determine waveform generation by itself, or provide a universal propagation theorem.

The control populates three of six benchmark families with four records. It is outside the cube, does not make GR complete, and transfers no observational support to a Weyl-gravity prototype.

The complete normalized dataset is `foundations/site/data.json` (SHA-256 `735063c38d7f53964a53631df718bbf09e658e9944b28497a13fba37a0c7ab8c`; canonical digest `c83ff6cddba8d587b4e647c7c533bb9fc5d020dcea27c1b23ad59f96bcd2dcad`). These tables are generated from that file and contain no hand-copied coverage counts, relation edges, evidence roles, boundaries, or citations. The assembly and calibration tables are generated from `foundations/site/assemblies.json` (SHA-256 `ca8e11e33adb9a40b0e5e2cb05062fa6124dca7b59c16350b268538d42d233c5`; canonical digest `678065ba94d98a76e92f326dbae80cca1c2695377c6ae2a51e3945c0df17db90`).

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